



Tropical college of medicine



Obstetrics and gynecology Nursing II
for 3rd year Bsc nursing student

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Debremarkos 2016 EC

Multiple pregnancy

- When more than one fetus simultaneously develops in the uterus
 - *Two fetuses (twins) is the commonest
 - *Three fetuses (triplets)
 - *Four Fetuses (quadruplets)
 - *Five fetuses (quintuplets)
 - *Six fetuses (sextuplets) may also occur.
- Occurs in 1% of all pregnancies

Type of multiple pregnancy

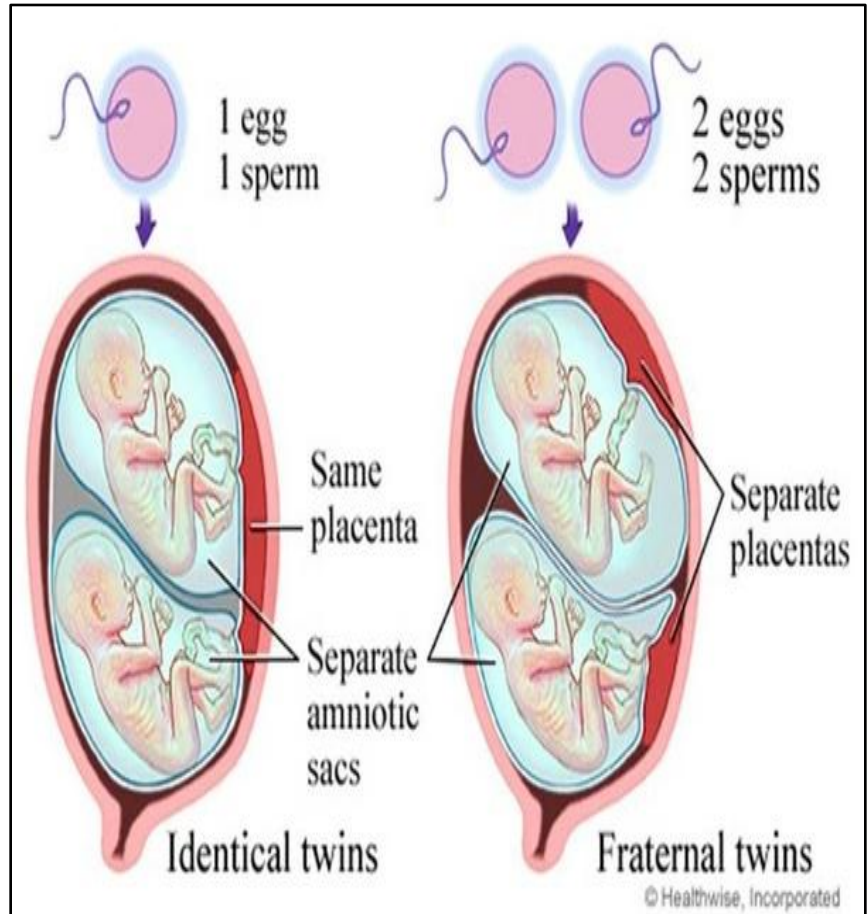
- Occurs in 1% of all pregnancies.

1. Dizygotic

- One ovum with two sperm
- Two ovum with two sperm

2. Monozygotic

- A single fertilized ovum divides early $\Rightarrow$ two identical embryos



Risk factor

Increasing parity and age (37 peak age)

Discontinuation of OCPs

Assisted reproductive technologies

Ovulation induction

Taller, heavier, and older mothers

Race(high in blacks)

Family history

Diagnosis

- History

- ⇒ Family history of twins on the mother's side
- ⇒ Higher incidence of hyper emesis gravidarum
- ⇒ Past history of multifetal gestations
- ⇒ Increased abdominal size and increased fetal movements
- ⇒ History of ovulation induction or assisted reproductive technologies
- ⇒ History of OCP discontinuation before conception

Diagnosis....

- Abdominal examination
 - ⇒ Accurate measurement of the fundal height
 - ⇒ Positive discrepancy
 - ⇒ Usually reveals multiple fetal parts
 - ⇒ Two FHB auscultated at a time varying by more than 10bpm
 - ⇒ Three or more fetal poles felt
 - ⇒ Polyhydramnios

Diagnostic work up

- Sonography:
 - ⇒ Routinely at 16-18 weeks for all to diagnose early
 - ⇒ Number of fetuses
 - ⇒ Sex of fetuses
 - ⇒ Well being of fetuses
 - ⇒ Careful search for congenital anomalies
 - ⇒ Gestational age and estimated fetal weight
 - ⇒ Fetal presentations of both twins
 - ⇒ Localization of placenta

Differential

Wrong date

Polyhydramnios

Big baby

Any space occupied lesion

Ascites in pregnancy

Complications

Maternal	Placental/fetal
• Hyperemesis • Anemia • Miscarriage • Pregnancy-induced hypertension • Polyhydramnios (excess amniotic fluid) • Poor contractions during labor • Post-partum hemorrhage	• Placenta previa • Abruptio placentae • Placental insufficiency • Preterm delivery • Low birth weight; • Malpresentations • Cord prolapse • Congenital anomalies (especially in monozygotic)

Conjoined twins[Siamese twins]

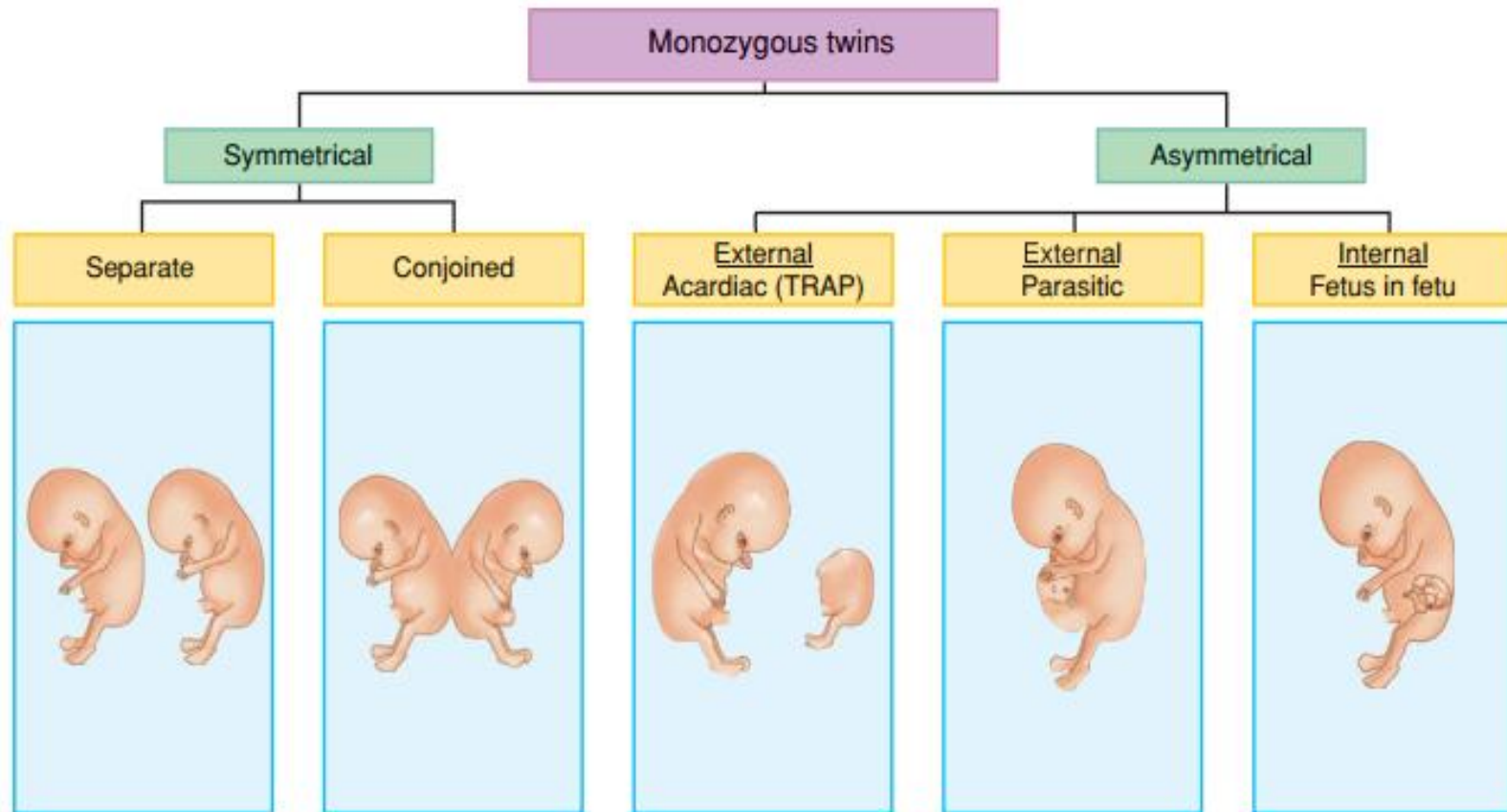


FIGURE 45-14 Possible outcomes of monozygotic twinning. The asymmetrical category contains twinning types in which one twin complement is substantially smaller and incompletely formed.

Conjoined twins[Siamese twins]...

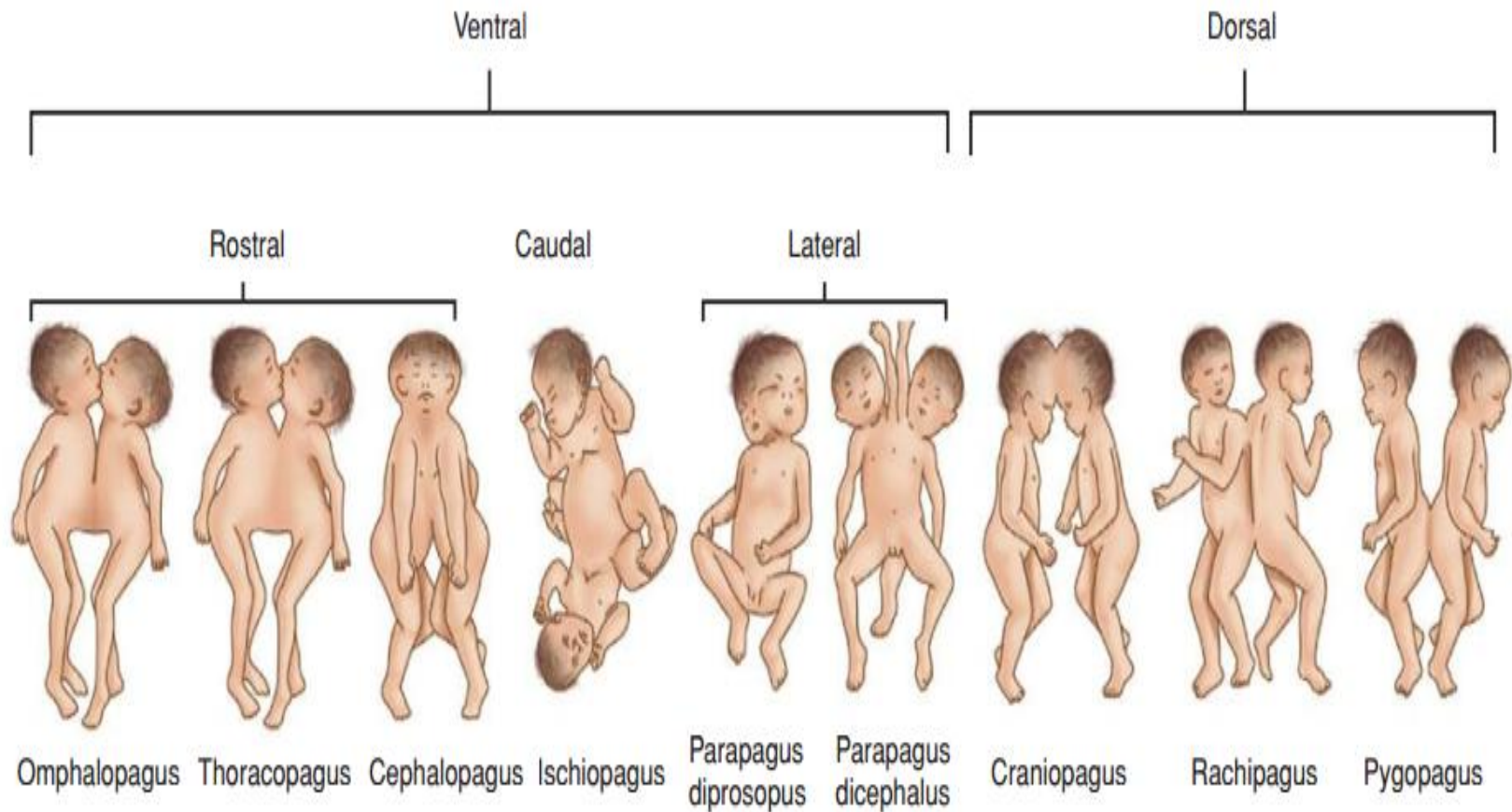


FIGURE 45-15 Types of conjoined twins. (Redrawn from Spencer, 2000.)

Management

- The purposes of management are to:
 1. Prevent preterm birth
 2. Identify growth alterations in one or both twins
 3. Determine the timing and type of delivery.
 4. Referral to tertiary centers

Management...

- Advice

- ⇒ Energy increased by 300kcal from singleton
- ⇒ Increased rest
- ⇒ Supplement –iron 60-120mg ,folic acid 1mg /day
- ⇒ More frequent ANC follow up
- ⇒ Prevention of preteterm labor-bed rest

Duration of pregnancy

- The mean duration of gestation for twins is 37 weeks.
- The mean duration of gestation for triplets 35 weeks
- Post term is at 40 weeks (2 weeks earlier than singleton)
- Mode of delivery is CS if the first twin is non-vertex and Triplets and higher order pregnancies
- Mode of delivery is spontaneous vertex delivery

Route of delivery

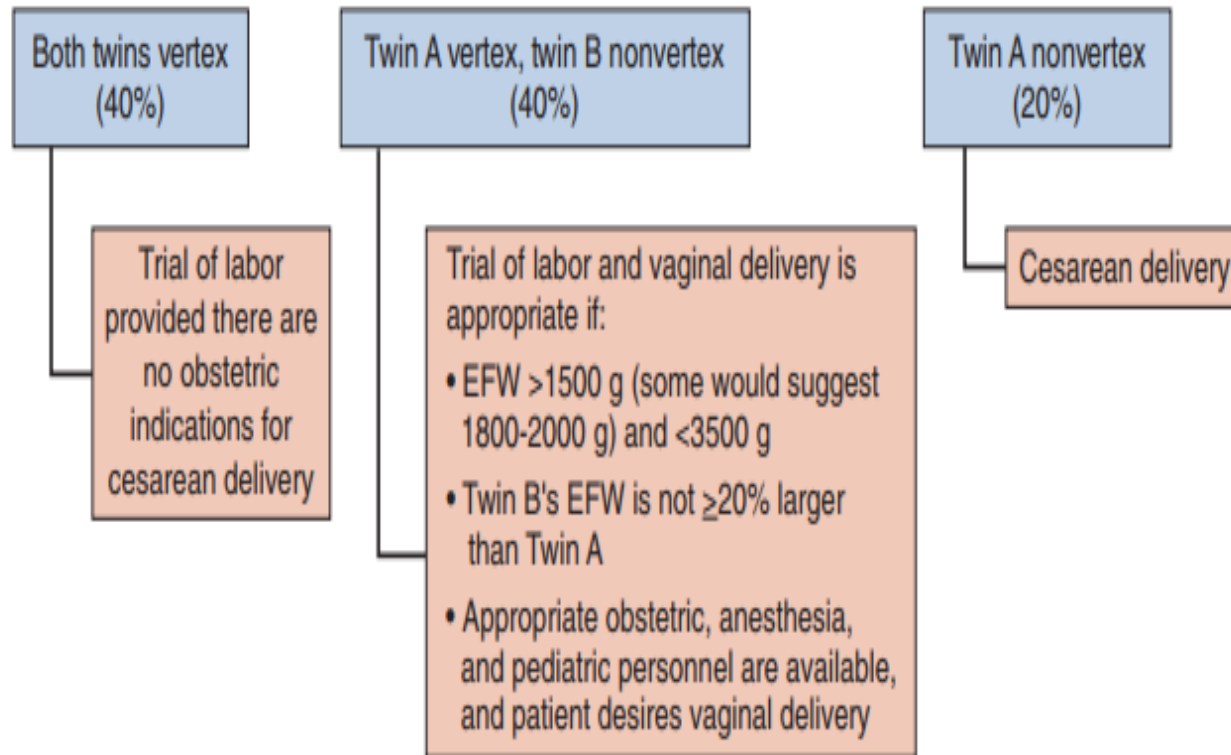


FIG 32-14 Algorithm for mode of delivery in diamniotic twins. (Monoamniotic twins should have cesarean delivery.) EFW, estimated fetal weight.

Unit-VI: Disease associated with pregnancy

- Objective

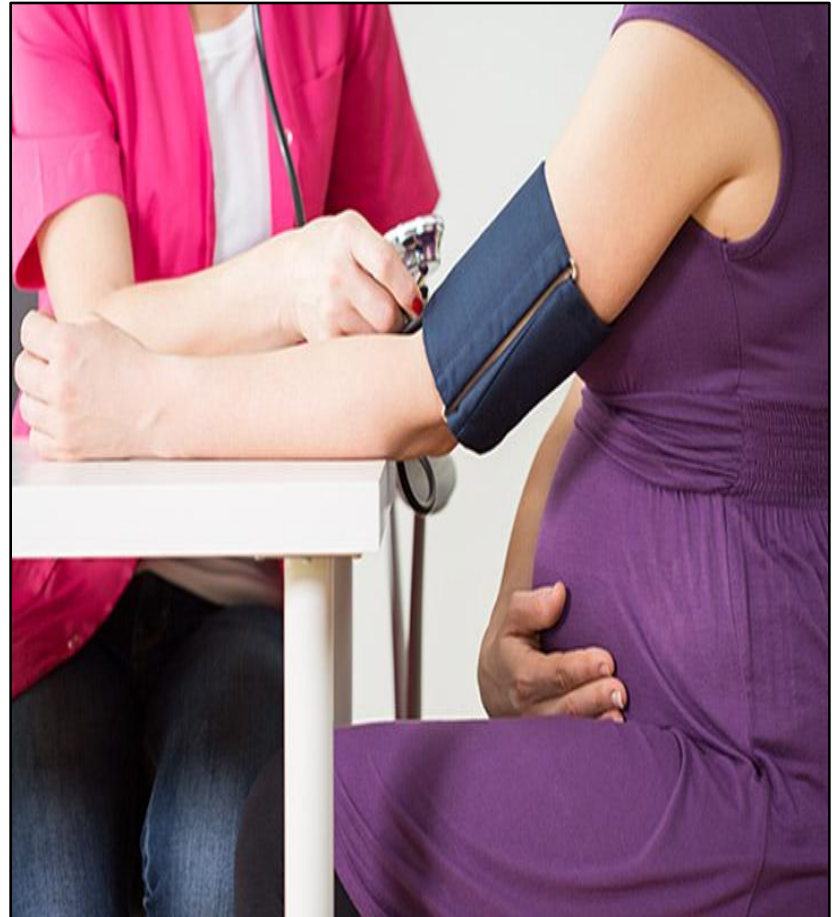
- ⇒ Define anemia and hypertension during pregnancy
- ⇒ Explain risk and cause of anemia and hypertension
- ⇒ Discuss on clinical presentation of anemia and hypertension
- ⇒ Explain diagnosis of anemia and hypertension
- ⇒ Explain management and complication of anemia and hypertension

Hypertensive disorders

- The most common medical complications of pregnancy.
- Incidence is between 5% and 10%.
- Major cause of maternal and perinatal mortality and morbidity
- One of deadly triad:
 - 1.Hemorrhage
 - 2.Infection
 - 3.Hypertension

Hypertension in pregnancy

- A systolic blood pressure ≥ 140 mmHg, diastolic blood pressure ≥ 90 mmHg or both in two occasions taken 4 hours or more apart; or a single blood pressure recording of $\geq 160/110$ mmHg.



Hypertension in pregnancy classification

Gestational hypertension

Preeclampsia eclampsia syndrome

Chronic hypertension

Superimposed preeclampsia

Hypertension in pregnancy classification...

1. Gestational hypertension

- ⇒ Hypertension without proteinuria
- ⇒ Hypertension without other signs of preeclampsia
- ⇒ Developing after 20 weeks of gestation in a previously normotensive woman.

Hypertension in pregnancy classification...

2. Preeclampsia eclampsia syndrome

- Preeclampsia

- ⇒ New onset of hypertension and proteinuria after 20 weeks of gestation in a previously normotensive woman.

- Eclampsia

- ⇒ Grand mal seizure or coma in a woman with preeclampsia.

Hypertension in pregnancy classification...

3. Chronic hypertension

- Hypertension that antedates pregnancy
- Is present before 20 weeks of gestation; or persists after 12 weeks postpartum.

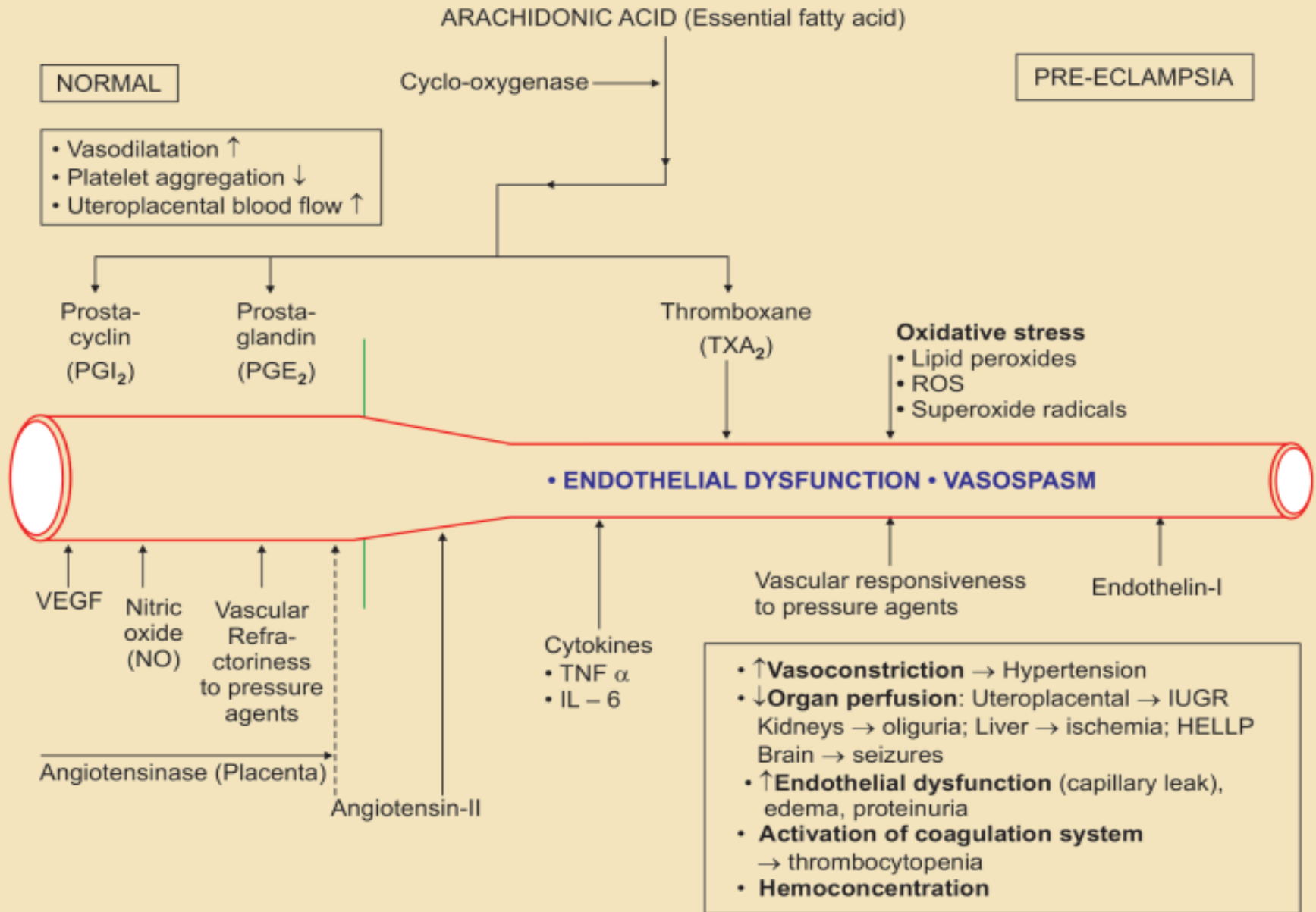
4. Superimposed preeclampsia

- Superimposed pre-eclampsia without severe features
- Superimposed pre-eclampsia with severe features

Risk factors for preeclampsia

- First pregnancy
- Young or old age
- Multiple gestation
- History of hypertension
- Renal disease
- Diabetes
- Obesity
- Family history of pre-eclampsia

ETIOPATHOLOGY OF PRE-ECLAMPSIA (See table on p. 220)



Classification of preeclampsia

Pre-eclampsia with severe features	Pre-eclampsia without severe features
• Blood pressure $\geq 160/110$ • Headache, blurred vision • Oliguria (<400 ml/24 hours) • Epigastric pain or pain in RUQ • Difficulty breathing • Low platelet count ($<100,000/\mu\text{l}$). • $\uparrow$ Liver enzymes more than 2x of upper limit of normal. • Serum creatinine higher than 1.1mg/dl	• Blood pressure $\geq 140/90$ • Proteinuria • No sign severe features

Clinical features of hypertension in pregnancy

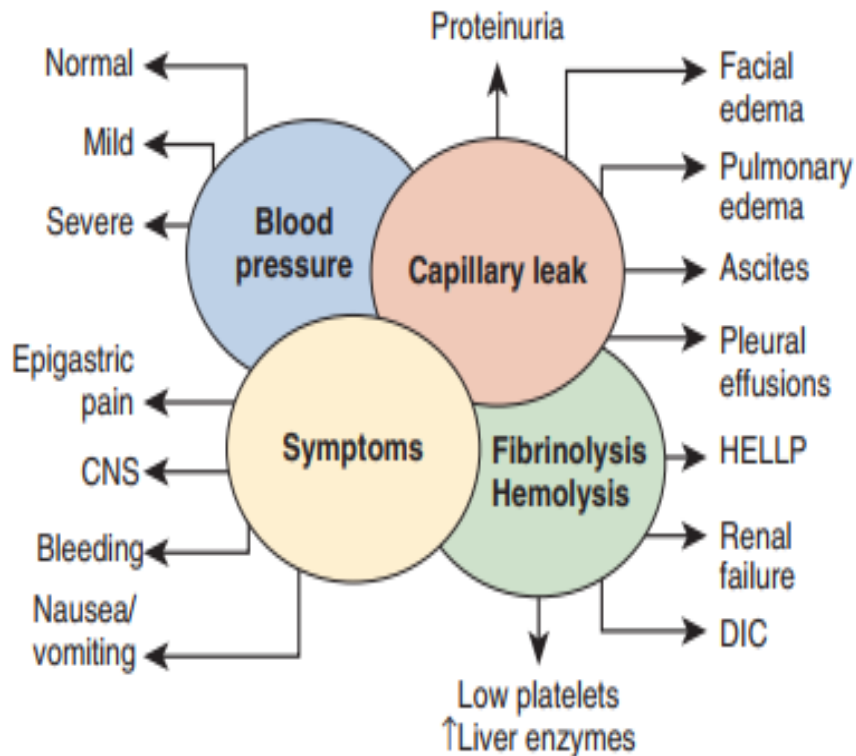


FIG 31-2 Maternal manifestations in preeclampsia. CNS, central nervous system; DIC, disseminated intravascular coagulation; HELLP, hemolysis, elevated liver enzymes, and low platelets.

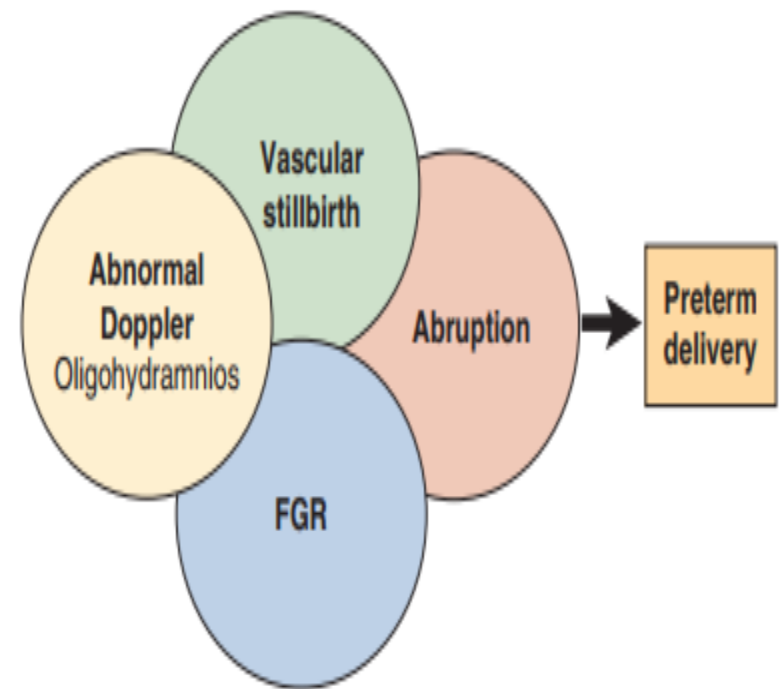


FIG 31-3 Fetal manifestations of preeclampsia. FGR, fetal growth restriction.

Clinical features of hypertension in pregnancy...



Fig. 17.1A: Marked pedal edema



Fig. 17.1B: Marked vulval edema in pre-eclampsia

Diagnosis

- Hypertension and proteinuria are the hallmark
- Severity features of preeclampsia
- Urine protein, CBC, liver enzymes, LDH and renal function test
- Ultrasound is used to monitor fetal growth and to assess fetal wellbeing.

Management of HDP

- Objectives
 - ⇒ Prevent eclampsia
 - ⇒ Preserve health of mother
 - ⇒ Deliver alive ,mature baby

Pre eclampsia with out severe features

- Gestational age less than 37weeks
 - ⇒ Monitor BP, fetal condition, CBC, OFT twice weekly(outpatient)
 - ⇒ Counsel danger signs associated with features of severe PE.
 - ⇒ Encourage the woman to eat a normal diet.
 - ⇒ Orient on fetal movement counting (kick chart) daily.
 - ⇒ Do not give anticonvulsants or antihypertensives
 - ⇒ Delivery at 37 completed weeks

Preeclampsia with severe features

General measures

Prevent convulsion

Control hypertension.

Delivery / expectant management

Preeclampsia with severe features...

- General Measures

- ⇒ Admit the patient urgently, preferably to the labor ward
- ⇒ Manage in left lateral position
- ⇒ Setup IV line & infuse maintenance fluids
- ⇒ Monitor urine output and maintain urine output at >30 ml/hr.
- ⇒ Monitor vital signs, FHB & reflexes
- ⇒ Auscultate the lung bases for fine crepitation.

Preeclampsia with severe features...

- Anticonvulsant therapy (seizure prophylaxis)
 - ⇒ Give Magnesium sulfate as shown in the box below.
 - ⇒ In all severe preeclamptic mothers during admission & continued during period of evaluation & observation for 24 hours.
 - ⇒ Diazepam: may be used as alternative, if MgSO₄ is not available

Preeclampsia with severe features...

- Control hypertension

- ⇒ Start antihypertensives if the BP > 160/110 (one of it)

- ⇒ Hydralazine or labetalol is the drug of choice for acute control.

- ⇒ Maintain BP above the lower limits of normal (90-110 mmHg)

- ⇒ Maintaining

- Systolic BP between 140 and 155 mm Hg

- Diastolic BP between 90 and 105 mm Hg

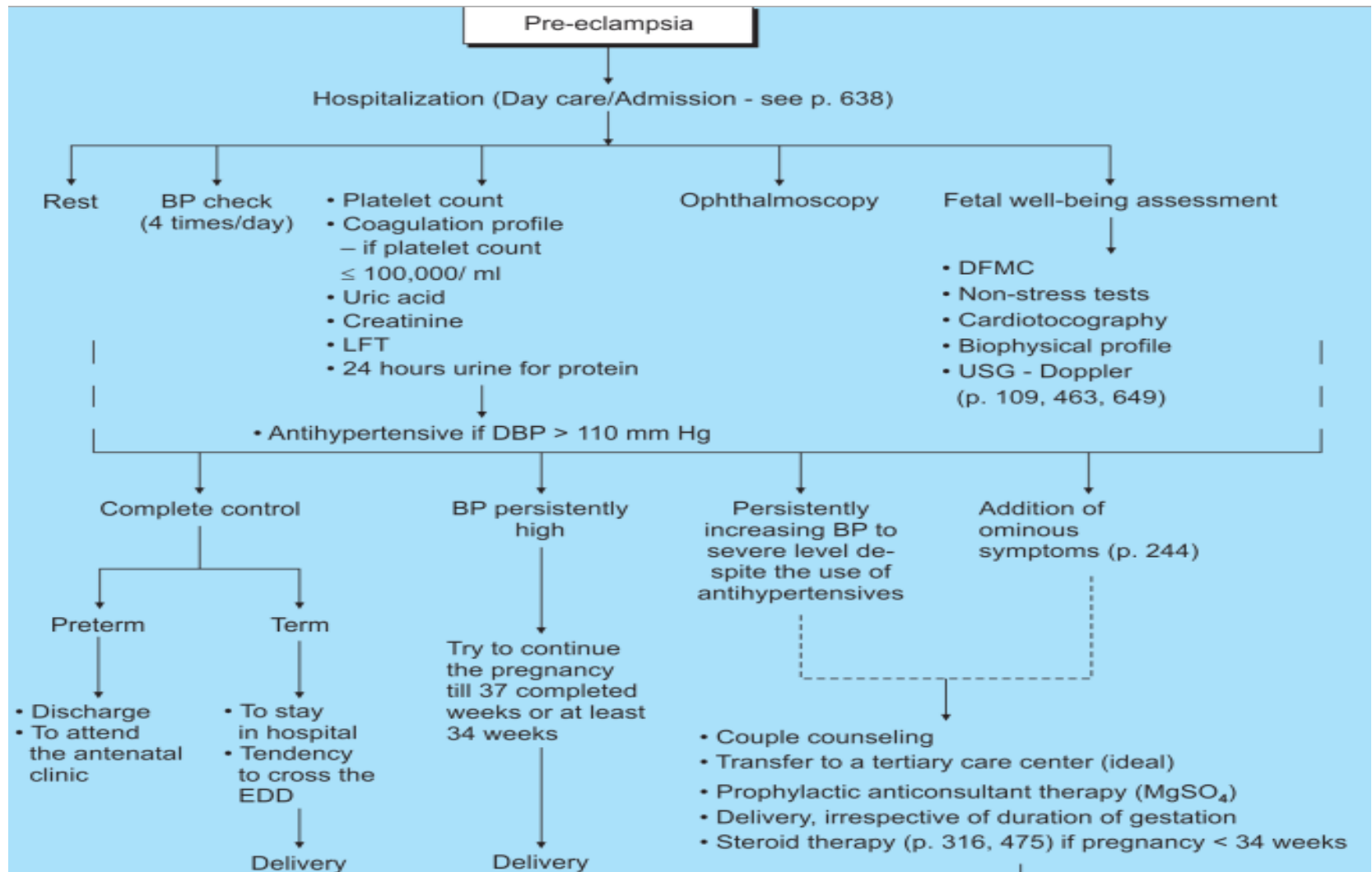
Preeclampsia with severe features...

Hydralazine	• Give 5 mg IV slowly every 20 minutes until DBP<110 mmhg. • The maximum dose is 20 mg per 24
Labetalol	• PO 200 mg ⇒ Repeat dose after one hour until the treatment goal is achieved. ⇒ The maximum dose is 1200 mg in 24 hours. • 10 mg IV ⇒ If response is inadequate after 10 minutes, administer 20 mg IV. ⇒ The dose can be doubled to 40 mg and then 80 mg with 10-minute intervals until blood pressure is lowered below threshold. ⇒ The maximum total dose is 300 mg; then switch to oral treatment.
Nifedipine	• 10 mg orally. • Repeat dose after 30 minutes if response is inadequate until optimal BP. • The maximum total dose is 30 mg in the acute treatment setting. • For maintenance therapy 10-20 mg PO bid is given.
Alpha methyldopa	• Administer 250-750 mg every six to eight hours. • The maximum dose is 3000 mg per 24 hours.

Planning delivery

- Gestational age < 28 weeks $\Rightarrow$ Termination of pregnancy
- Gestational age $\geq$ 28 weeks and <34 weeks: expectant management if no contraindication
- Gestation after 37 Completed Weeks ,giving birth is recommended

Preeclampsia management summery



Eclampsia

- Derived from a Greek word, meaning “like a flash of lightening”
- Pre-eclampsia when complicated with generalized tonic–clonic convulsions and/or coma.
- May occur quite abruptly, without any warning manifestations.
- 80% preceded by features of severe pre-eclampsia.

Eclampsia management

- General measures(IV fluid,position,suction,O2,spatula...)
- Control of convulsions(Mg₂So₄)
- BP control, stabilization of the condition of the mother & fetus
- Fluid balance.
- Delivery should take place within 12 hours of onset of convulsion
- Delivery & intrapartum/postpartum care

Anticonvulsant

MAGNESIUM SULFATE SCHEDULES FOR SEVERE PRE-ECLAMPSIA AND ECLAMPSIA

Loading dose

- Magnesium sulfate 20% solution, 4g IV over 5 minutes.
- Follow promptly with 10 gm of 50% magnesium sulfate solution, 5 gm in each buttock as deep IM injection with 1mL of 2% lidocaine in the same syringe. Ensure that aseptic technique is practiced when giving magnesium sulfate deep IM injection. Warn the woman that a feeling of warmth will be felt when magnesium sulfate is given.
- If convulsion recurs after 15 minutes, give 2 gm magnesium sulfate (20% solution) IV over 5 minutes.

Maintenance dose

- 5 gm magnesium sulfate (50% solution) + 1 mL lidocaine 2% IM every 4 hours into alternate buttocks.
- Continue treatment with magnesium sulfate for 24 hours after delivery or the last convulsion, whichever occurs last.

Complications hypertension in pregnancy

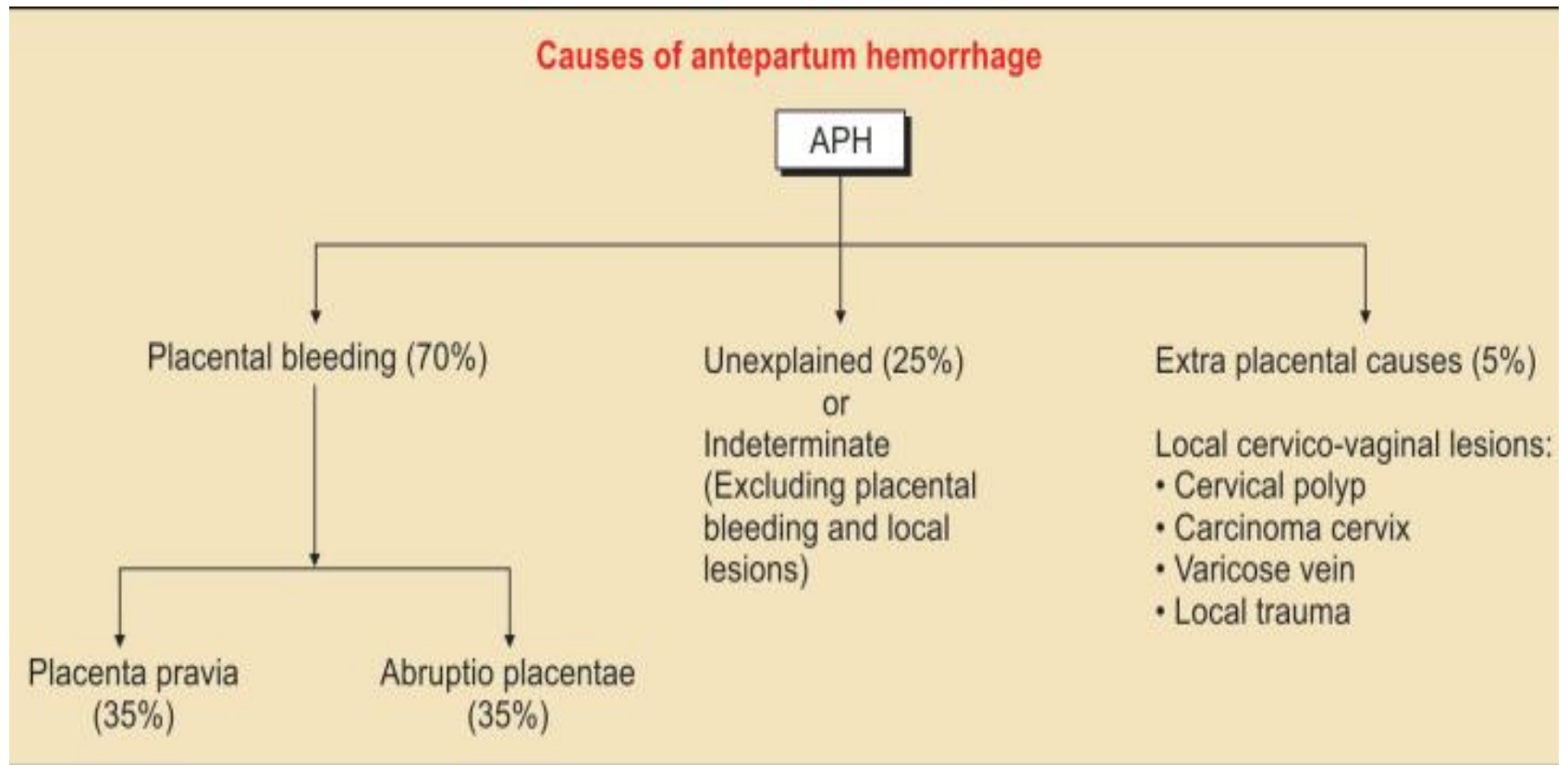
Maternal complications	Fetal complications
• Abruptio placentae • DIC • Acute renal failure • Hepatocellular injury, liver rupture • Brain damage • Transient blindness • Cardiorespiratory arrest • Aspiration pneumonitis • Acute pulmonary edema • Postpartum hemorrhage	• Fetal-growth restriction • Preterm birth • Fetal death

Ante partum hemorrhage (APH)

- Is the occurrence of vaginal bleeding (VB) after 28th weeks of gestation and before delivery of the last fetus.
- Incidence is 1-4 % of pregnancy.



Causes APH



Placenta previa

- The implantation of the placenta over or near the internal os of the cervix (within 2cm) which diagnosed in 3rd TMP.
- 0.3 % to 0.4 % of all term pregnancies

Risk factors of placenta previa

1	Scarring or poorly vascularized endometrium	• Grandmultiparity • Advanced age(>35years) • Abortion • Uterine fibroids • Procedure(curettage, C/S, myomectomy)
2	Need for increased placental surface area (large placenta)	• DM • Multiple gestation • Smoking • Male fetus
3	Prior placenta previa	• Increase 8x

Classification of placenta previa

1. Placenta previa(marginal PP, minor PP)

- Internal cervical OS is covered by placental tissue.

2. Low-lying placenta(marginal PP, minor PP)

- Placenta lies within 2 cm of the cervical OS
- Does not cover the cervical OS

Classification of placenta previa...



Low lying
placenta



Marginal
placenta



Partial
previa



Complete
previa

Clinical feature

- Painless, bright red, causeless, recurrent bleeding
- Uterus corresponds to the period of amenorrhea
- Uterus is soft and relaxed
- FHB is normal



Fig. 18.4: Labor in a woman with placenta previa complicated with prolapse of umbilical cord and footling. *By Courtesy: Dr S Bhattacharya, Silchar*

Investigations

- HCT
- Blood group and Rh
- Ultrasound: Fetal assessment, retroplacental clot and for localization of placenta

Treatment

- Resuscitate and stabilize on arrival and admit the patient.
- Assess maternal and fetal wellbeing.
- Prepare cross matched blood (at least 2 units).

Treatment...

- Expectant management indication
 - ⇒ <37 weeks
 - ⇒ Patient in stable condition
 - ⇒ Reassuring fetal condition.

Treatment...

- Expectant management
 - ⇒ Give dexamethasone if GA < 37 weeks.
 - ⇒ Anti D 300µg IM if Rh negative and not sensitized.
 - ⇒ Closely monitor maternal and fetal conditions with APH chart.
 - ⇒ Prevent and treat anemia.

Treatment

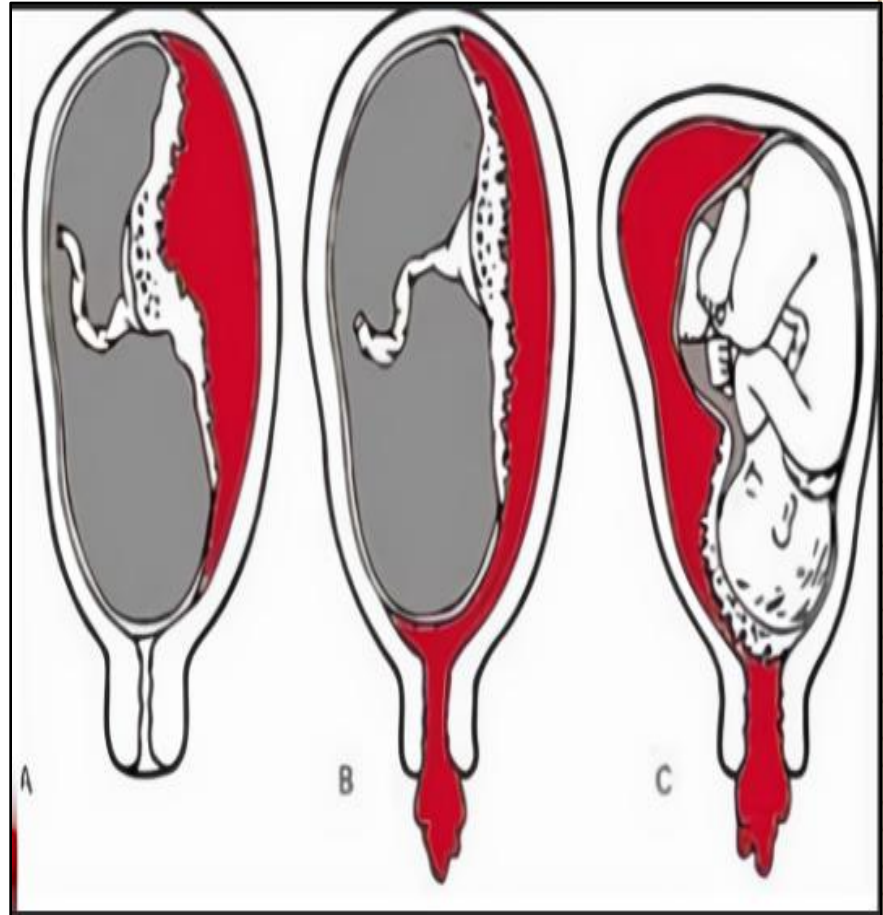
- Immediate delivery indication
 - ⇒ Gestational age is >37 weeks
 - ⇒ Deranged vital signs
 - ⇒ Heavy bleeding,
 - ⇒ NRFHRP, IUFD
 - ⇒ Lethal congenital anomaly of the fetus
 - ⇒ Established labor.

Treatment...

- Mode of delivery
 - ⇒ Vaginal delivery can be allowed for marginal placenta.
 - ⇒ Cesarean delivery:
 - Placenta previa(partial and complet)
 - Excessive bleeding
 - NRFHR

Abruptio placenta

- Premature separation of a normally implanted placenta after 28 weeks from the uterus before delivery of the last fetus.
- Classification.
 - A. Revealed(85%)
 - B. Concealed(15%)
 - C. Mixed



Risk factors

- Hypertensive disorders(30%)
- Multiparity
- Advanced age(>35yr)
- Metabolic abnormalities(eg, DM)
- Vascular disease(e.g, Systemic lupus erythematosus)
- Placental insufficiency
- Asthma

Classification of abruptio placenta.

Clinical presentation	Mild (Grade1)	Moderate (Grade 2)	Severe (Grade 3)
Degree of separation	<1/6 th	1/6 th -2/3 rd	>2/3 rd
Amount of blood	<400 mL	400-1000mL	>1000mL
Uterine irritability (pain/tenderness)	Normal / slightly increased	Increased	Tetanic, reactive, board like, tender on palpation
Fetal condition	No fetal heart abnormality	Abnormal / rarely	Fetal distress or death
Shock	Absent	Mild, postural hypotension	Severe and always present
BP	Normal	Postural hypotension	Reduced, recordable
PR	Normal	Increased	Feeble, fast
Fibrinogen level	Normal	Slightly decreased	Decreased
DIC	Absent	Absent	present

Clinical presentation

- Vaginal bleeding: menstrual-like (dark), totally concealed or the amount is less than the degree of the shock.
- Abdominal pain/ (uterine) tenderness.
- NRFHRP or absent fetal heart beat.
- Coagulation defect: bleeding (epistaxis, ecchymosis, petechiae).

Investigations

- HCT
- Blood group and Rh
- Coagulation profile: Platelet count, PT, PTT, fibrinogen or bedside clotting and bleeding tests
- Ultrasound: Fetal assessment, retroplacental clot and for exclusion of placenta previa

Treatment

- Resuscitate and stabilize on arrival and admit the patient.
- Assess maternal and fetal wellbeing.
- Prepare cross matched blood (at least 2 units).

Treatment...

- Expectant management indication

⇒ <37 weeks

⇒ Patient in stable condition

⇒ Reassuring fetal condition.

Treatment...

- Expectant management indication
 - ⇒ Give dexamethasone if GA < 37 weeks.
 - ⇒ Anti D 300µg IM if Rh negative and not sensitized.
 - ⇒ Closely monitor maternal and fetal conditions with APH chart.
 - ⇒ Prevent and treat anemia.

Treatment...

- Immediate delivery indication
 - ⇒ Gestational age is >37 weeks
 - ⇒ Estimated fetal weight is >2.5 Kg
 - ⇒ Deranged vital signs
 - ⇒ Heavy bleeding
 - ⇒ NRFHRP, IUFD, malformed fetus
 - ⇒ Established labor

Treatment...

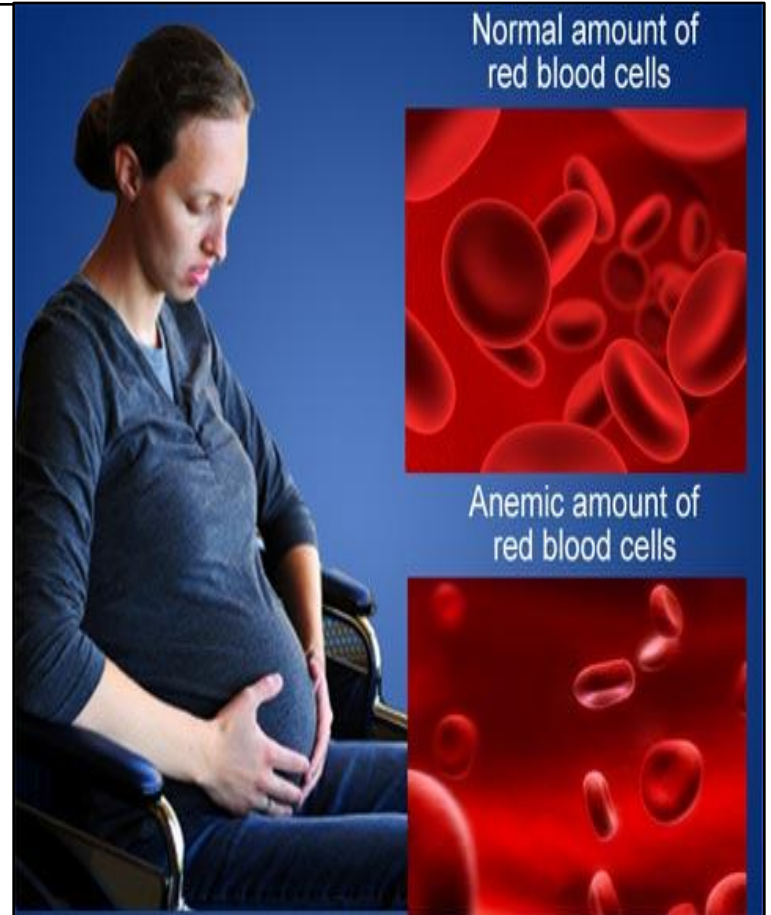
- Immediate delivery
 - ⇒ Mode of delivery: Vaginal delivery is preferred.
 - ⇒ Cervical ripening and induction of labor, amniotomy.
 - ⇒ Emergency cesarean section:
 - For severe bleeding endangering maternal life
 - NRFHR

Complications

- Post partum hemorrhage
- Hemorrhagic shock
- Organ failure (acute kidney injury, congestive heart failure).
- Disseminated coagulopathy
- Utero-placental insufficiency $\Rightarrow$ IUGR, fetal distress or IUFD

Anemia in pregnancy

- Is decrease of RBC in blood
- The commonest disorder
- Anemia in pregnancy is a hemoglobin level of
 - ⇒ $<11\text{g/dl}$ in 1st TMP and 3rd TMP
 - ⇒ $<10.5\text{ g/dl}$ in 2nd TMP



Incidence

- The commonest medical complication of pregnancy.
- 50-65% of pregnant women develop anemia in low-resource countries
- Anemia is responsible for 20% of maternal deaths in the third world countries

Risk mother

- Multi-fetal gestations
- Poor socio-economic status
- Malnourished
- High parity
- Adolescent mothers
- Inter- pregnancy intervals less than two years
- Women with obstetric hemorrhage during current pregnancy

Cause of anemia during pregnancy

Common Causes—85% of Anemia

Physiologic anemia
Iron deficiency

Uncommon Causes

Folic acid deficiency
Vitamin B₁₂ deficiency (due to the rapid increase in bariatric surgery)
Hemoglobinopathies

- Sickle cell disease
- Hemoglobin SC
- β -Thalassemia minor

Bariatric surgery
Gastrointestinal bleeding

Rare Causes

Hemoglobinopathies

- β -Thalassemia major
- α -Thalassemia

Syndromes of chronic hemolysis

- Hereditary spherocytosis
- Paroxysmal nocturnal hemoglobinuria

Hematologic malignancy

Symptoms of Anemia in Pregnancy



Excessive tiredness



Weakness



Headaches



Dizziness



Chest pain



Shortness of breath



A rapid or irregular heartbeat



Pale skin



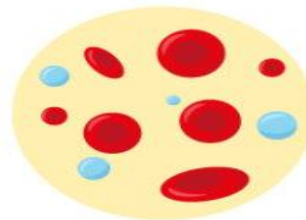
A numb or cold feeling in hands and feet



A low body temperature



Irritability



Diagnostic approach to anemia in pregnancy

Nurse role in the assessment of women with anemia in pregnancy

- **History taking:** hx of any of the predisposing factors.
- **symptoms of anemia :**
- C.V.S→ palpitation.
- C.N.S→ headache , visual disturbance , weakness , fatigability , drowsiness.
- R.S→ breathlessness.
- G.I.S→ anorexia , nausea , vomiting.
- G.U.S→ loss of libido.

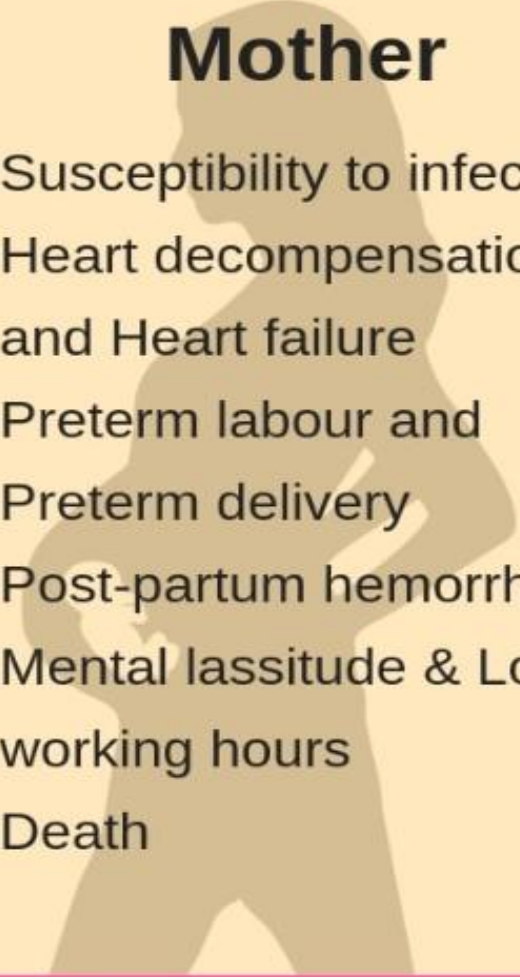
- History and examination
- Routine Hemoglobin check at first ANC visit
- If anemia is diagnosed
 - ⇒ Intestinal parasites stool exam
 - ⇒ Malaria by blood film

Complications of anemia during pregnancy

Fetus

- 
- Neural Tube defects (esp. folate def.)
 - Miscarriage
 - OIUGR/ Low birth weight
 - Prematurity
 - Anemia in infancy
 - IUFD

Mother

- 
- Susceptibility to infection
 - Heart decompensation and Heart failure
 - Preterm labour and Preterm delivery
 - Post-partum hemorrhage
 - Mental lassitude & Loss of working hours
 - Death

Management of anemia during pregnancy

- Depends on the specific cause
- Iron rich diet nutritional anemia
- Treat the cause (Hookworm, malaria ...)
- Elemental Iron from 120-200mg daily
- Blood transfusion for severe anemia (Hgb <7gm/dl)

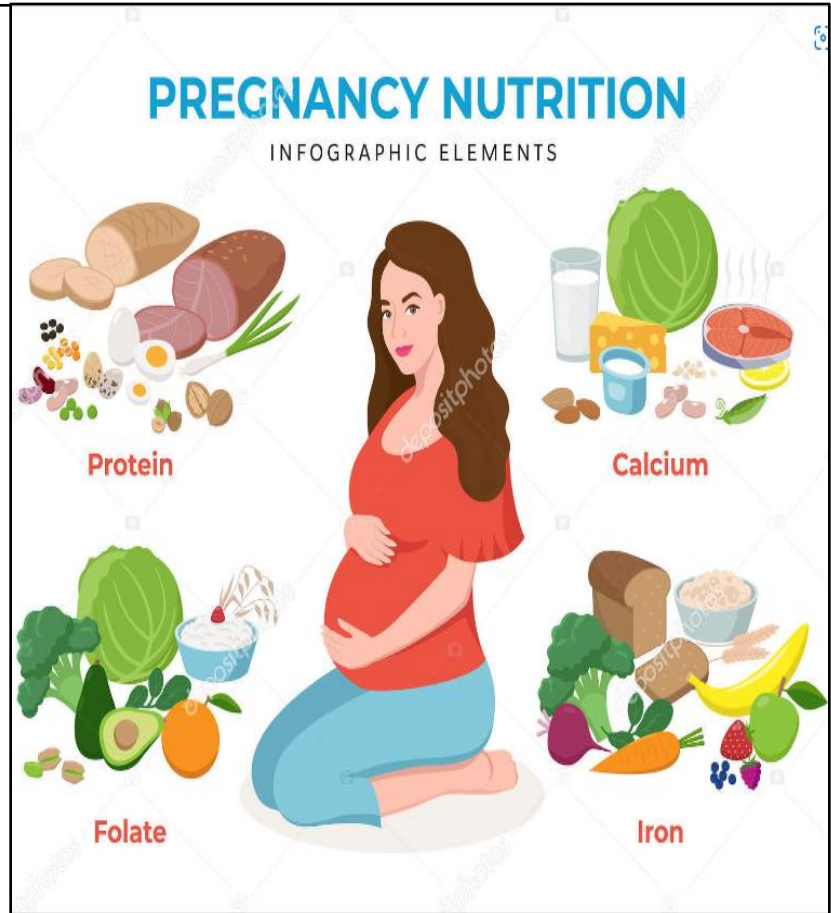
TABLE 42-2

ELEMENTAL IRON AVAILABLE FROM COMMON
GENERIC IRON PREPARATIONS

PREPARATION	ELEMENTAL IRON (mg)
Ferrous gluconate 325 mg	37-39
Ferrous sulfate 325 mg	60-65
Ferrous fumarate 325 mg	107

Prevention

- Adequate iron rich diet
 - Prevent infection
 - Prophylaxis iron
- ⇒ One 325 mg tablet of ferrous sulfate daily
- Wear shoes
 - Interpregnancy interval(2years)



Content

Unit I

- Abnormal labor

Unit II

- Operative delivery

Unit III

- Family planning

Unit IV

- Abnormal puerperium

Unit v

- Abnormalities of early pregnancy

Unit VI

- Benign and malignant tumors of genital tract

Unit I: Abnormal labor

Mal position

Mal presentation

Uterine inertia

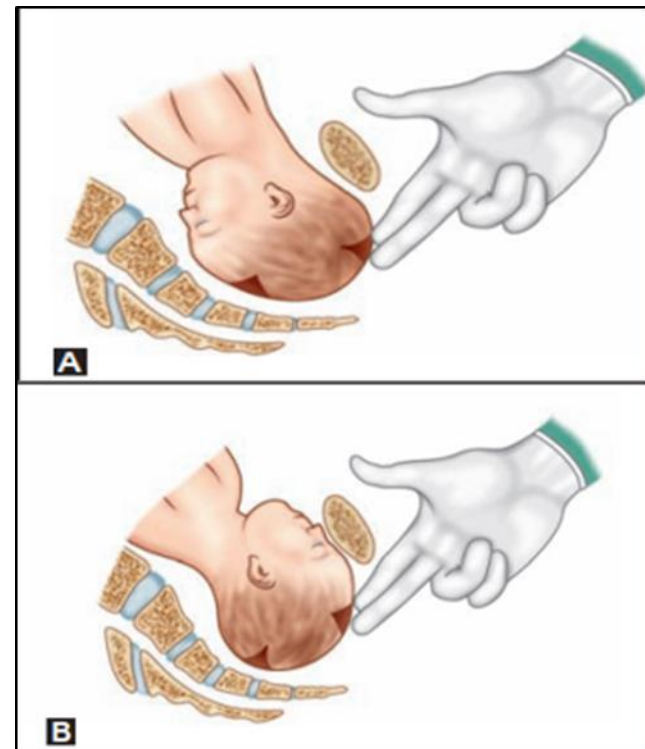
Cephalopelvic disproportion (CPD) and obstructed labor

Malposition and malpresentation

- Denominator

⇒ Part of the fetus which determines the presenting part

1. Vertex-Occiput
2. Breech-Sacrum
3. Face– mentum
4. Brow-sinciput



Fetal position

- Is the relationship between the denominator of the presentation to the six areas of the mother pelvis.



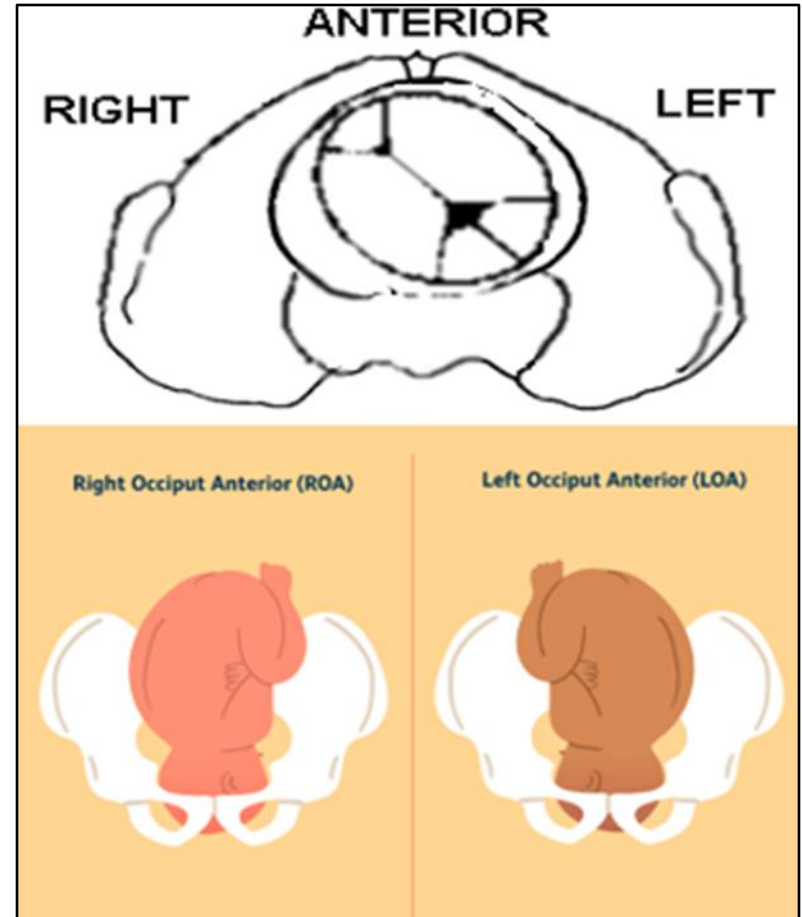
Fetal position...

- The normal fetal position is occiputo anterior

⇒ Direct

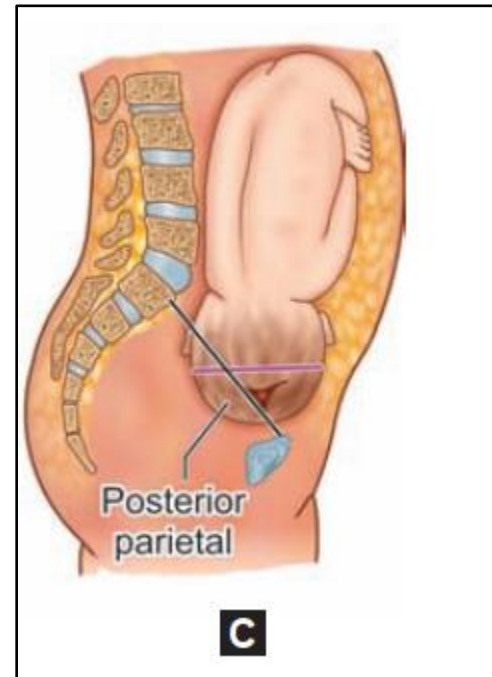
⇒ Right

⇒ Left

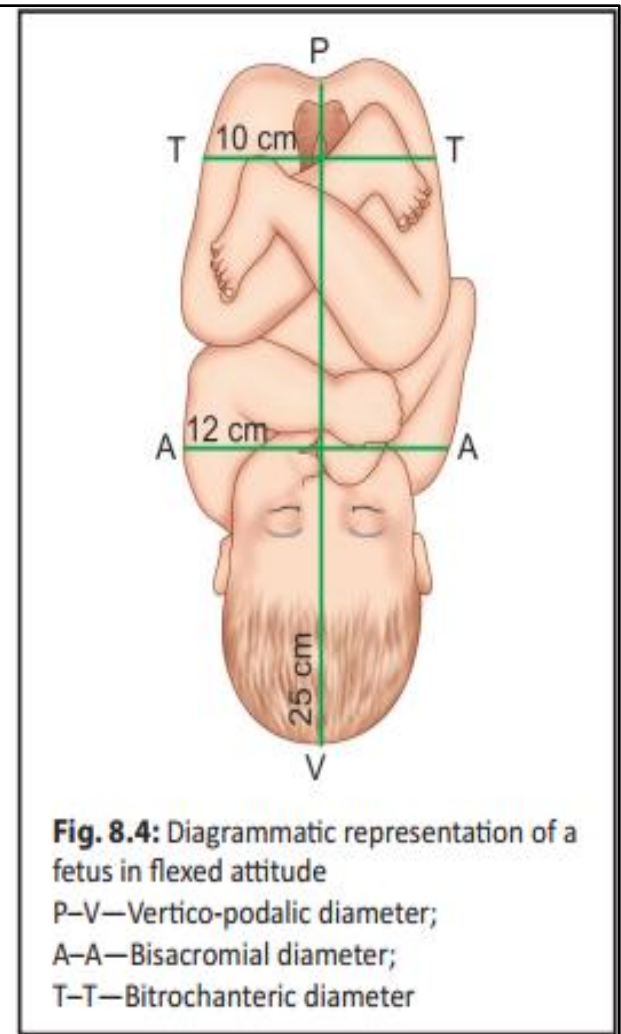
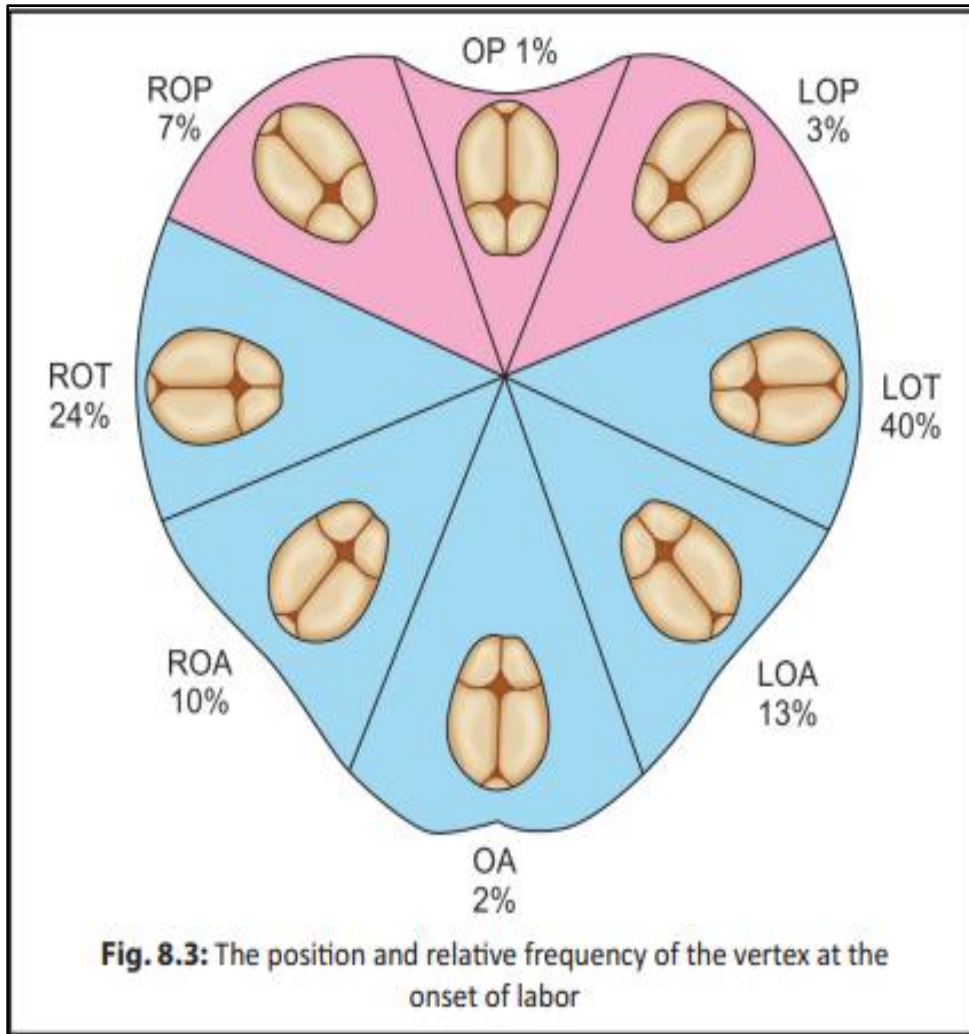


Malposition

- Malposition are any position of the vertex other than flexed occipito-anterior one
- Include
 - ⇒ Occiput posterior
 - ⇒ Persistent occiput transverse



Malposition...

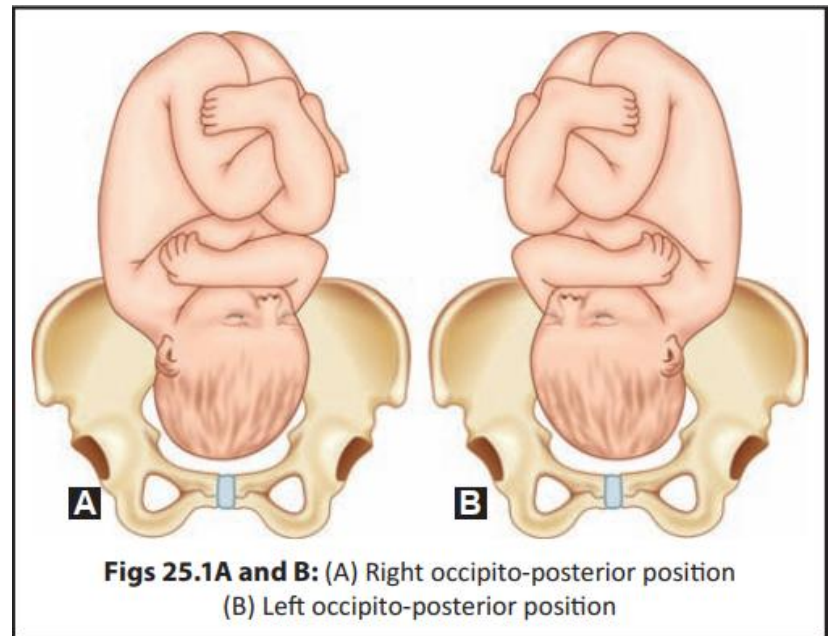


Predisposing factors

Maternal	Fetal and placental
• Contracted pelvis • Pelvic tumors: uterine myomas, ovarian tumors etc • Uterine anomalies: bicornuate uterus, uterine septum etc • High parity	• Prematurity • Fetal anomaly (e.g. hydrocephalus, anencephalus) • Polyhydramnios / oligohydramnios • Multiple pregnancy • Placenta previa

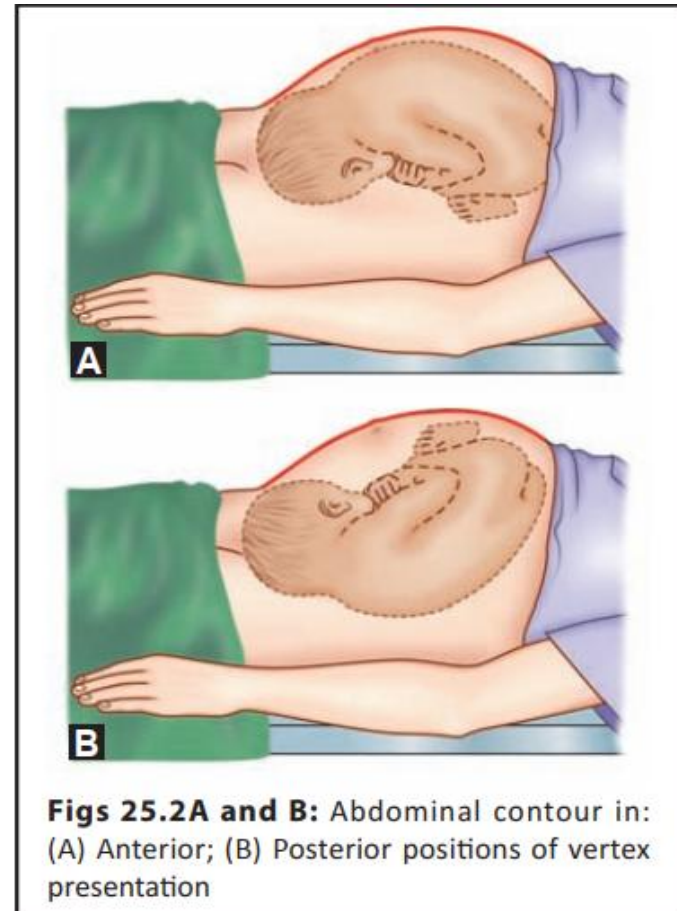
Occiput posterior position

- Is when the fetal occiput is posterior in relation to the maternal pelvis



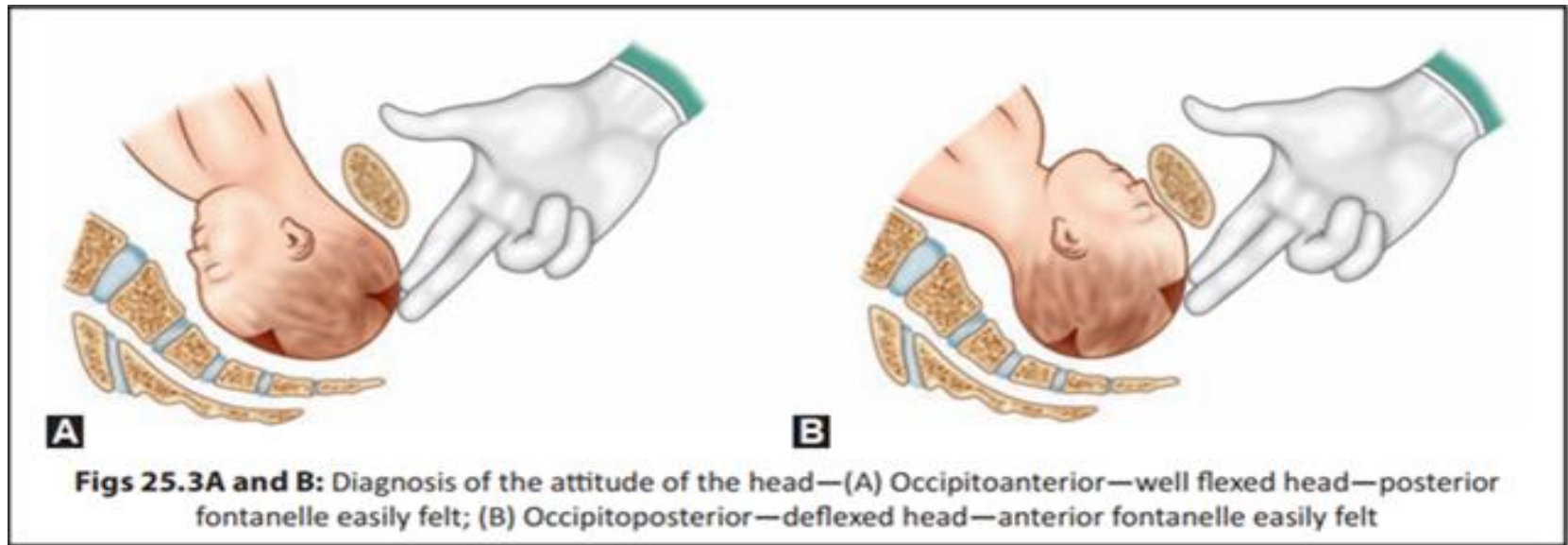
Diagnosis

- Suggestive abdominal findings
 - ⇒ Flattened lower part of abdomen.
 - ⇒ Anteriorly palpable fetal limbs.
 - ⇒ Fetal heart heard in the flank.



Diagnosis

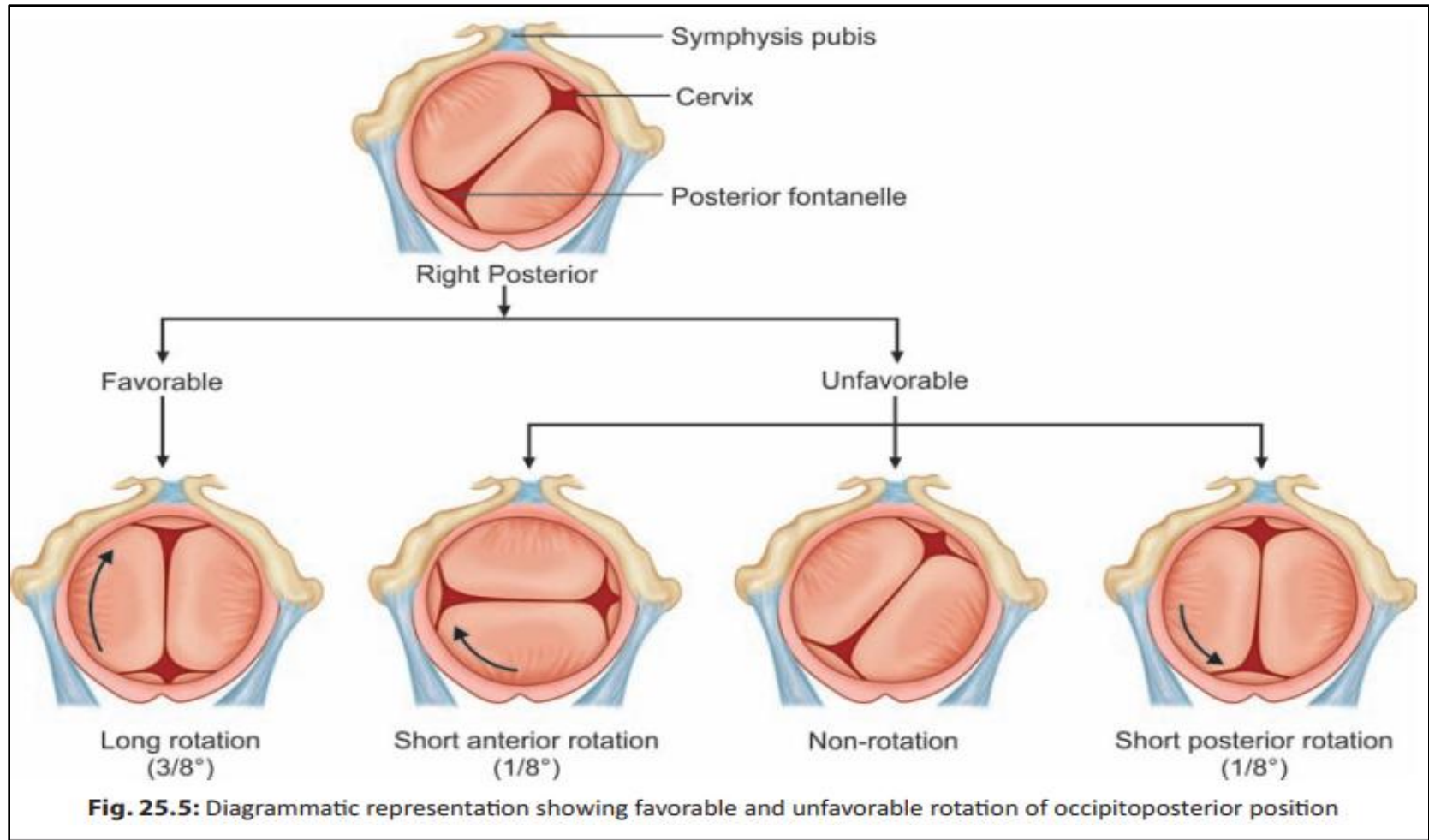
- On vaginal examination
 - ⇒ Posterior fontanelle towards the sacrum.
 - ⇒ Anterior fontanelle felt anteriorly if neck is flexed.



Management

- Rotation to occiput anterior, expect vaginal delivery
- Incomplete rotation leading to occipito-transverse position:
 - ⇒ Expect vaginal delivery
 - ⇒ Vacuum delivery(below station +2)
 - ⇒ Deliver by cesarean section

Management

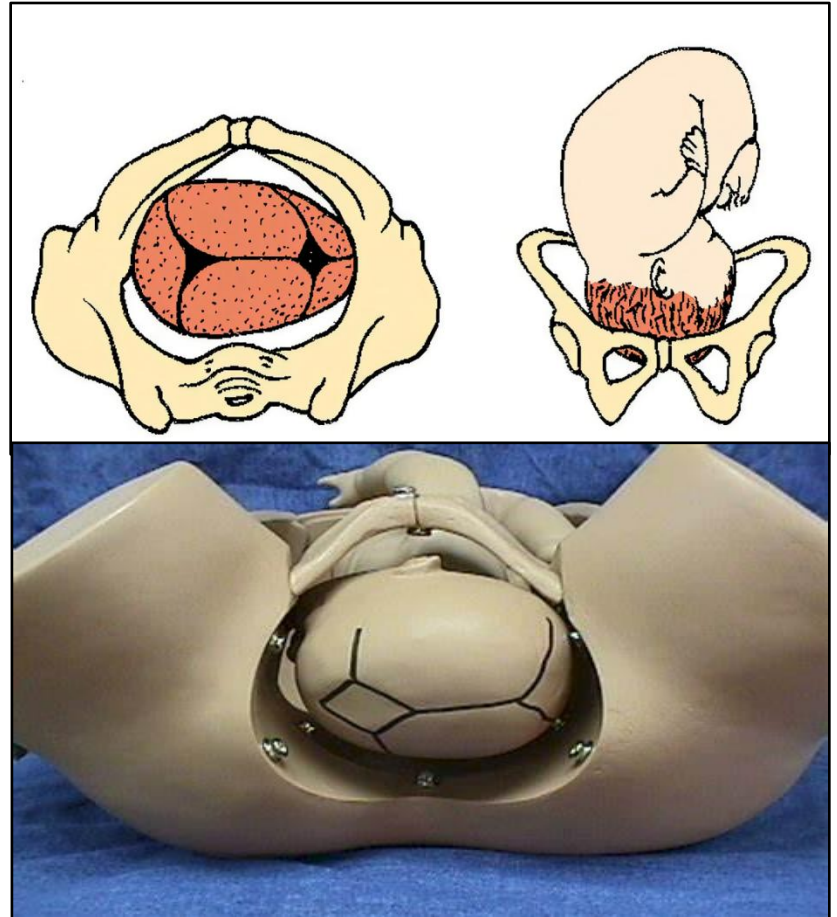


Management

- If no rotation
 - ⇒ Expect vaginal delivery
 - ⇒ Augment with oxytocin
 - ⇒ Vacuum delivery
 - ⇒ Deliver by cesarean section

Persistent occiput transverse position

- Is an occiput transverse position that is maintained for an hour or more in the second stage of labor.



Causes and risk factors

- Inadequate power (contraction and poor pushing)
- Platypelloid and android pelvis.
- Fetal head in long occipito-frontal diameter

Diagnosis

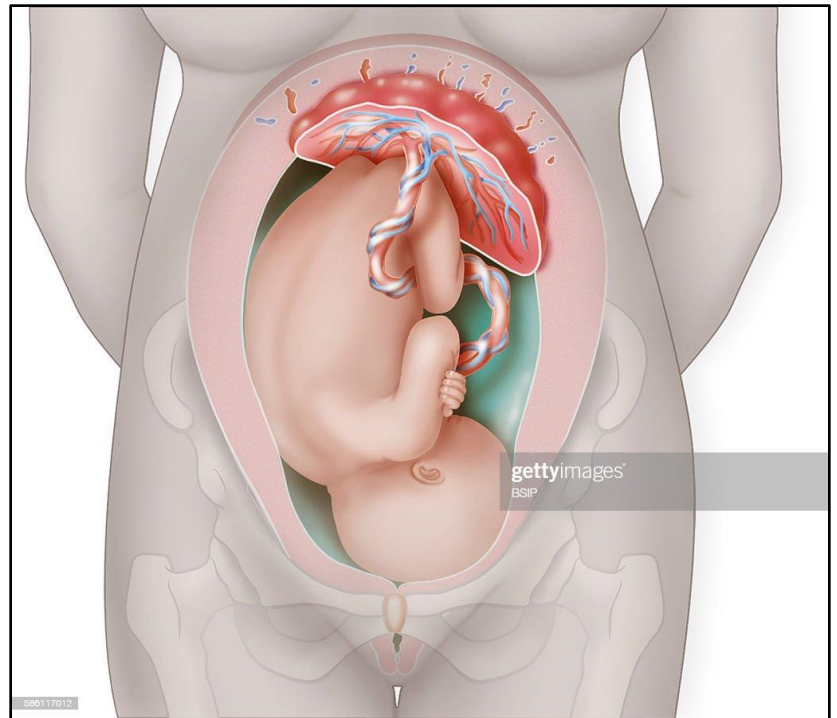
- Suspect when fetal descent is protracted or arrested.
- On vaginal examination
 - ⇒ Suture and fontanelles are palpable in the transverse diameter of the pelvis
 - ⇒ Fetal ears can be palpated superiorly under the symphysis and inferiorly above the sacrum/coccyx.

Management

- Expectant management:
 - ⇒ Small fetuses can usually be delivered
 - ⇒ If there is progress in descent and FHR is reassuring
 - ⇒ Partial or complete rotation may still occur spontaneously.
- Augmentation.
- Cesarean delivery(High transverse arrest,no maternal effort)

Fetal presentation

- Is the fetal body part which occupies the lower pole of the uterus and thereby first enters the pelvic passage
- The normal presentation is vertex .



Fetal presentation....

- Malpresentations
- ⇒ Are all fetal presentations other than vertex.

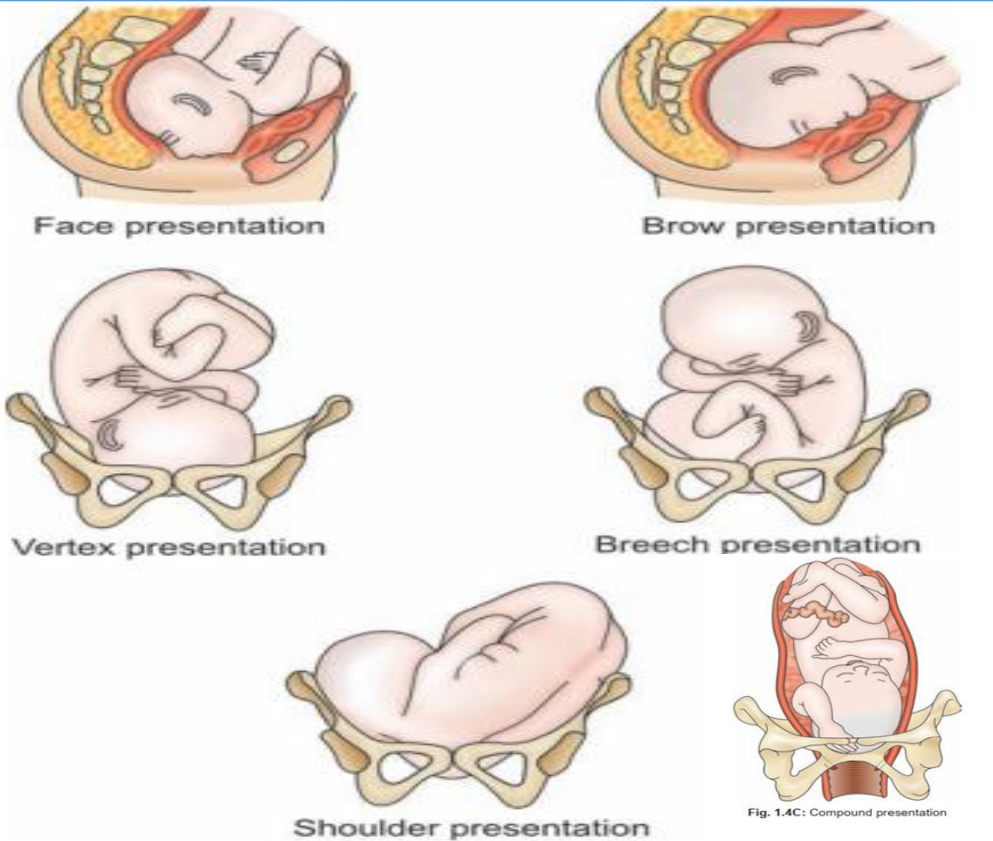
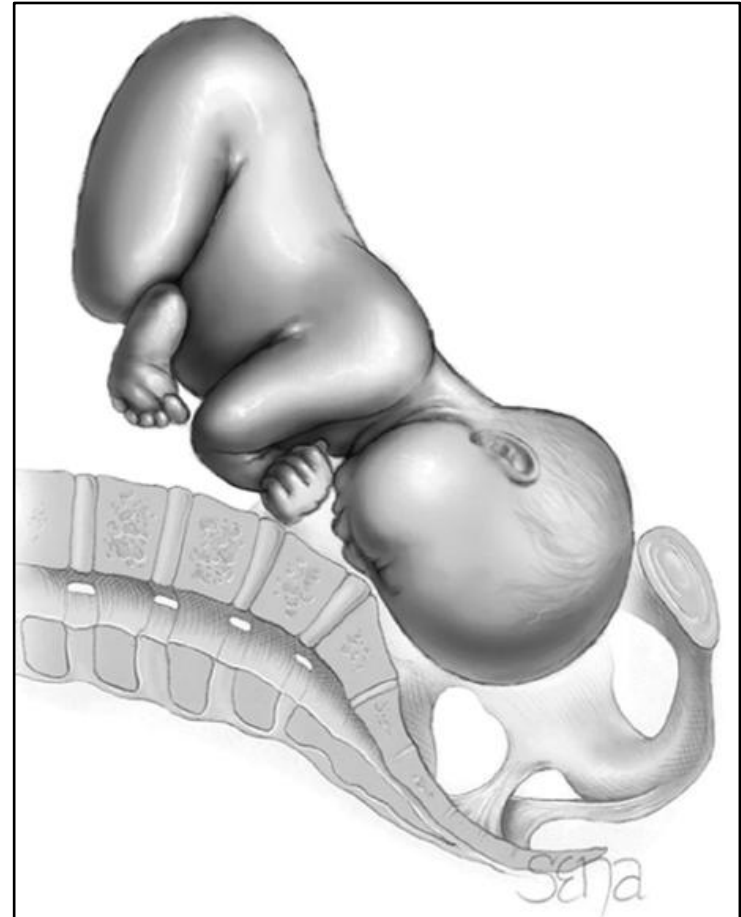


Fig. 1.4A: Various types of fetal presentation

Brow presentation

- Partial extension of the fetal head making the occiput higher than the sinciput
- Presenting part are frontal bone, orbital ridges and anterior fontanel



Diagnosis

- Suggestive abdominal finding:
- Occiput felt above sinciput.
- On vaginal examination
- Anterior fontanelle and orbits are felt



Management of persistent brow

- Persistent brow presentation is brow presentation in the later part of the first stage (after 6 cm of cervical dilatation) and second stages of labor.
- Deliver by cesarean section.



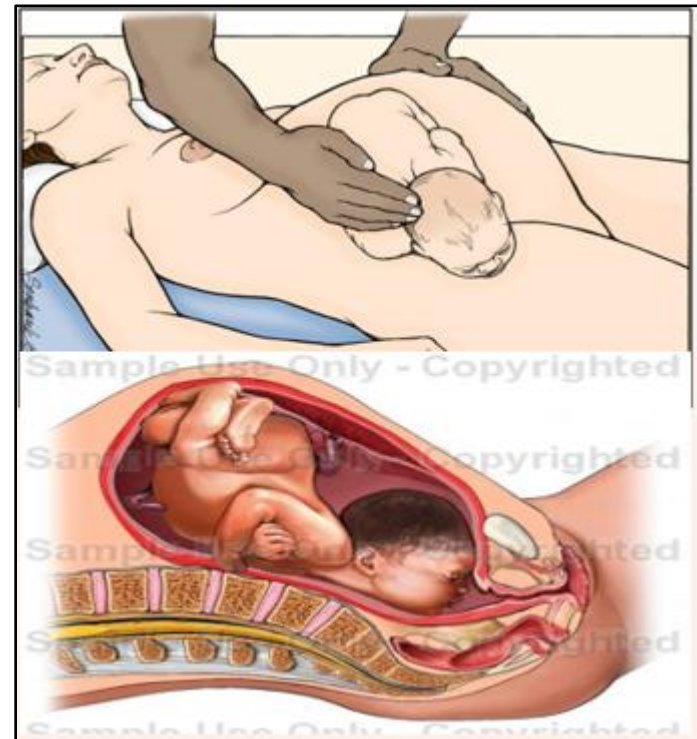
Face presentation

- Face presentation is hyperextension of the fetal head with the face being the leading part (figure 10).
- Fetal chin (mentum) is used as a reference point.



Diagnosis

- Suggestive abdominal finding
- ⇒ Groove may be felt between the occiput and the back (Leopold III).



Diagnosis...

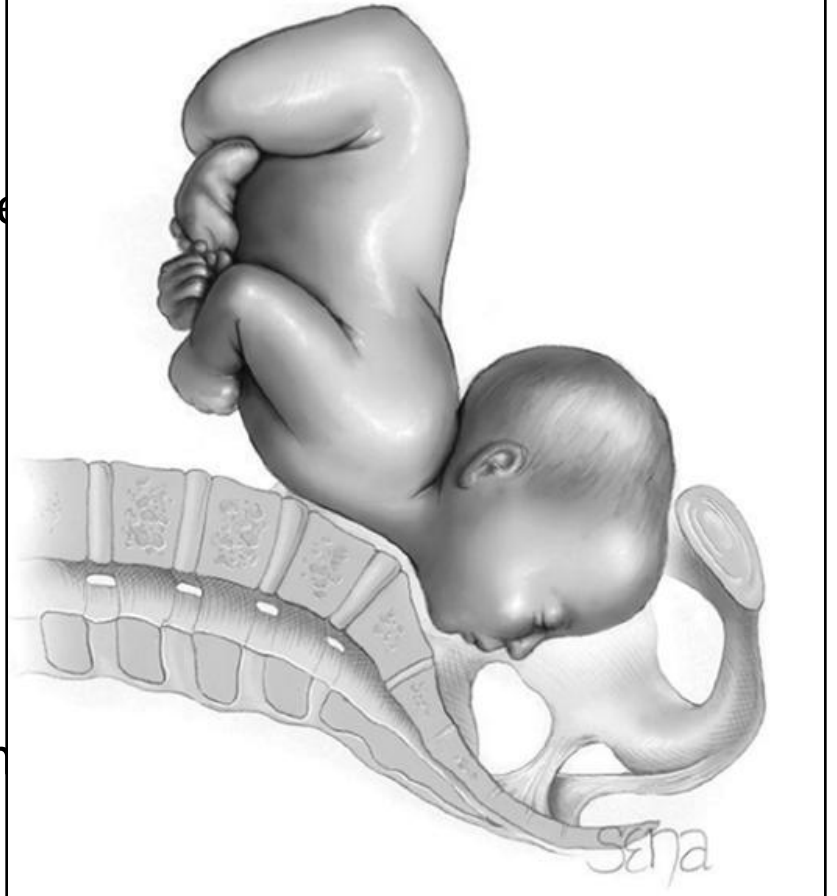
- On vaginal examination
 - ⇒ Fetal chin, mouth and nose palpated.
 - ⇒ The mouth with the two malar bone prominences make a triangle
 - ⇒ Mento-anterior: chin anterior position.
 - ⇒ Mento-posterior: chin posterior position



Fig. 25.25: Vaginal touch picture showing landmarks in mento-anterior position

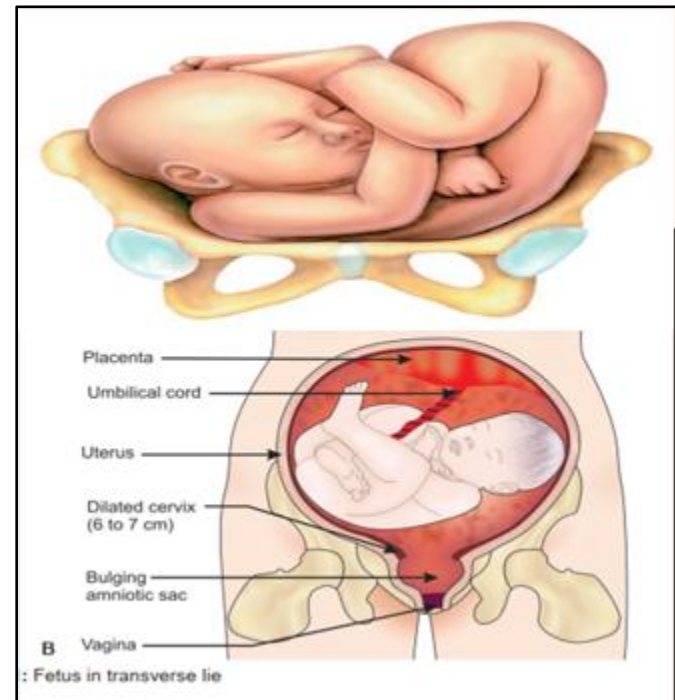
Management

- Mento-anterior:
 - ⇒ Adequate pelvis follow labor
 - ⇒ Forceps delivery can also be used
- Mento-posterior may rotate and follow labor
- Persistent mento-posterior
 - ⇒ If fetus is alive → CS
 - ⇒ If the fetus is dead → Craniotomy



Transverse lie (shoulder presentation)

- Transverse lie is when the long axis of the fetus is perpendicular to the longitudinal axis of the mother/uterus.
- Shoulder presentation is when the shoulder is the presenting part in a transverse lie.



Transverse lie....

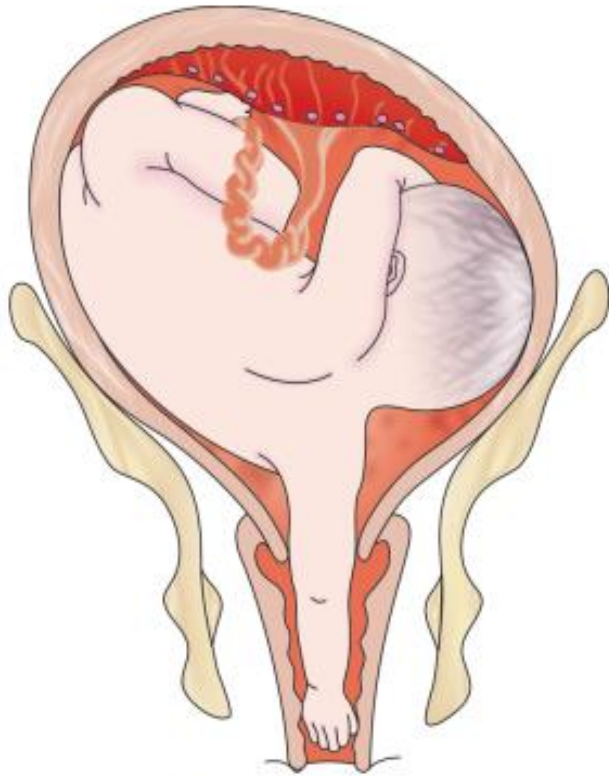


Fig. 3.6: Arm prolapse

Dorso-anterior position

Dorso-posterior position

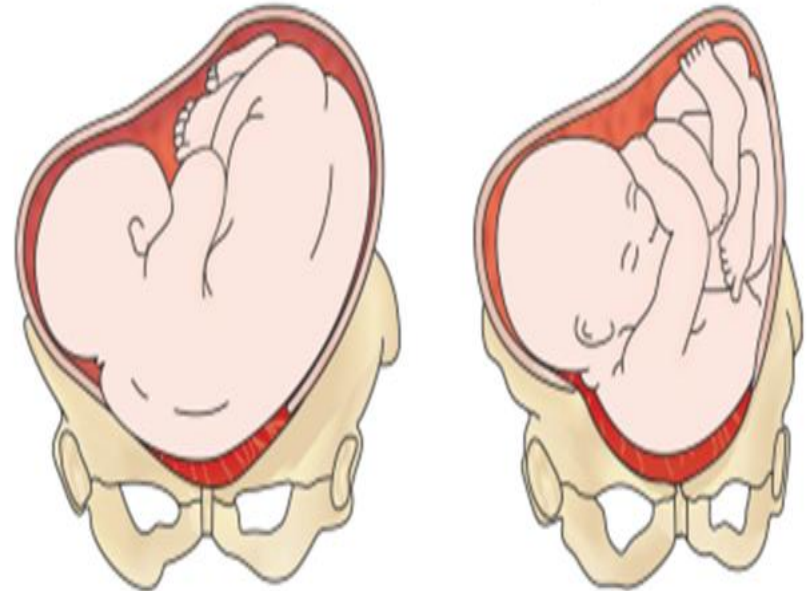


Fig. 3.2: Different positions of transverse lie

Diagnosis

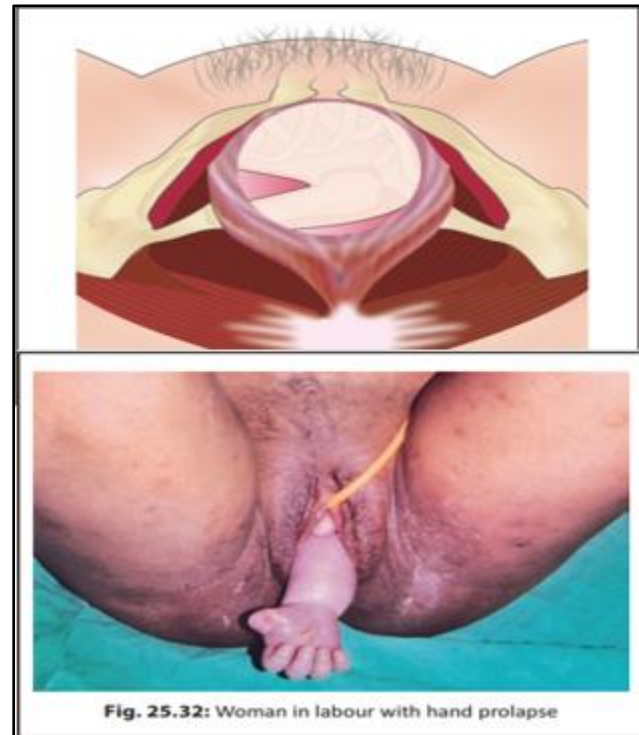
- Abdominal findings:

- ⇒ Neither the fetal head nor the breech is felt in the upper and lower part of the uterus.
- ⇒ The abdomen is transversely elongated that longitudinally.
- ⇒ Fundal height is less than gestation age.



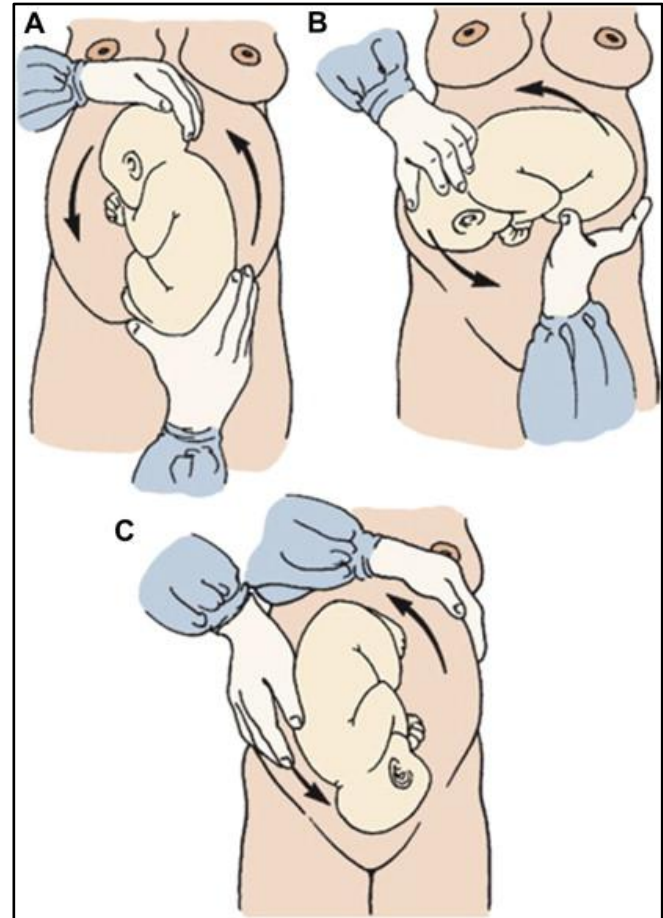
Diagnosis...

- Vaginal finding:
⇒ The shoulder or the prolapsed arm is felt.



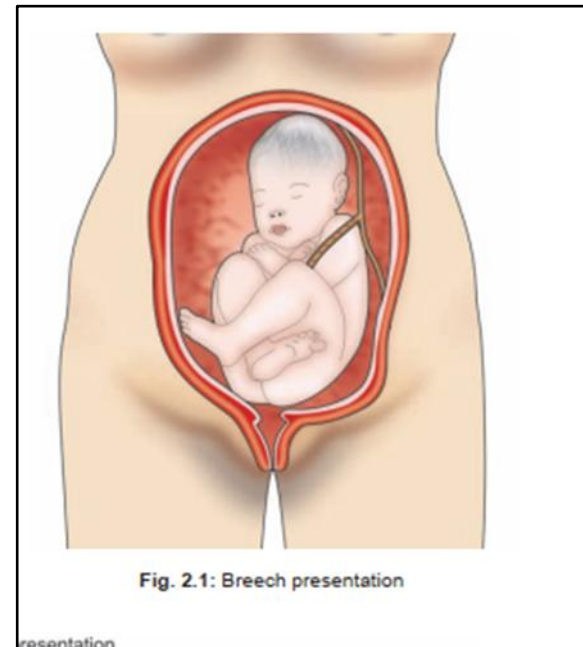
Management

- External cephalic version
- Cesarean section

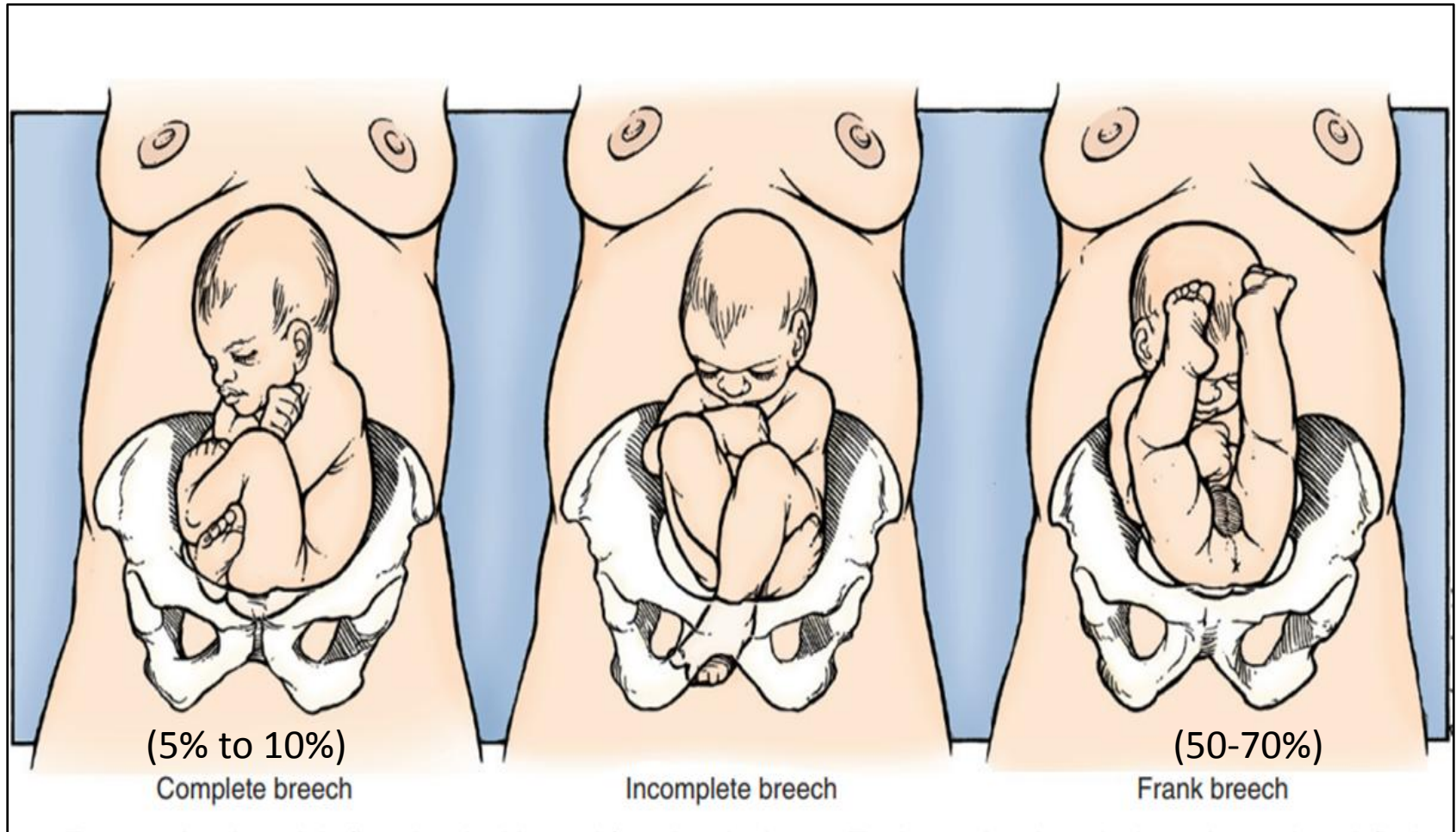


Breech presentation

- Is when the fetal buttock and/ or feet are the presenting part occupying the lower pole.
- Incidence
 - ⇒ 20% at 28th week
 - ⇒ 5% at 34th week
 - ⇒ 3–4% at term



Classification breech presentation



Predisposing factors

Maternal factor	Fetal Factors
• Cephalo-pelvic disproportion • Contracted maternal pelvis • Ppolyhydramnios, oligohydramnios) • Uterine anomalies (bicornuate or septate) • Placental abnormalities (placenta previa) • Multiparity (especially grand multiparas) • Previous history of breech delivery • Presence of pelvic tumor • Contracted pelvis.	• Prematurity • Fetal anomalies (e.g. neurological abnormalities,hydrocephalus, anencephaly and meningomyelocele) • Intrauterine fetal death

Diagnosis

- History
 - ⇒ Discomfort under the rib (due to the hard head)
 - ⇒ Feeling of more fetal movement in the lower segment



Diagnosis...

- Abdominal examination
 - ⇒ A round hard and smooth mass (head) at the fundus
 - ⇒ A soft, broad, and non ballotable mass (the breech) occupying lower pole of the uterus
 - ⇒ FHB loudest just above the umbilicus



Diagnosis...

- Vaginal examination
 - ⇒ Ischial tuberosities, sacrum and its spines, genitals and the anus



Diagnosis...

- Ultrasound
 - ⇒ Confirm clinical diagnosis
 - ⇒ Assess fetal attitude
 - ⇒ Assess amniotic fluid volume
 - ⇒ To estimate fetal weight, and to investigate for fetal anomalies and other predisposing factors



Management

- First stage of labor
 - ⇒ Secure IV line.
 - ⇒ Analgesics as labor pain management.
 - ⇒ Vaginal examination at rupture of membranes
 - ⇒ Avoid ARM.
 - ⇒ Instructed mother to push until the cervix is fully dilated
 - ⇒ Augmentation of labor is contraindicated

Management...

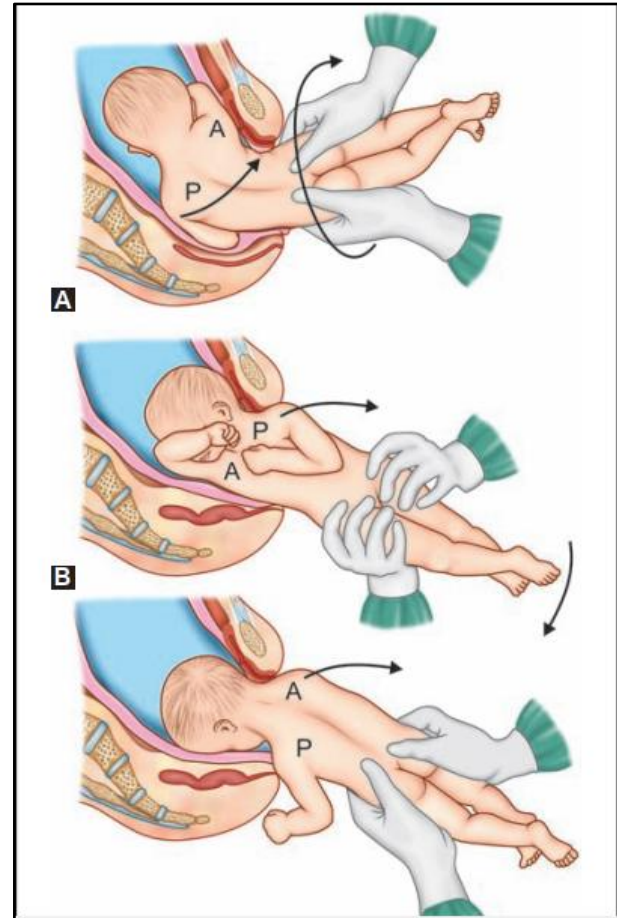
- Second stage of labor
 - ⇒ Wait buttocks till body is born to the level of the umbilicus
 - ⇒ If the legs do not deliver spontaneously
 - Pinard's maneuver



Management...

- Delivery of arm

⇒ Lovset's maneuver



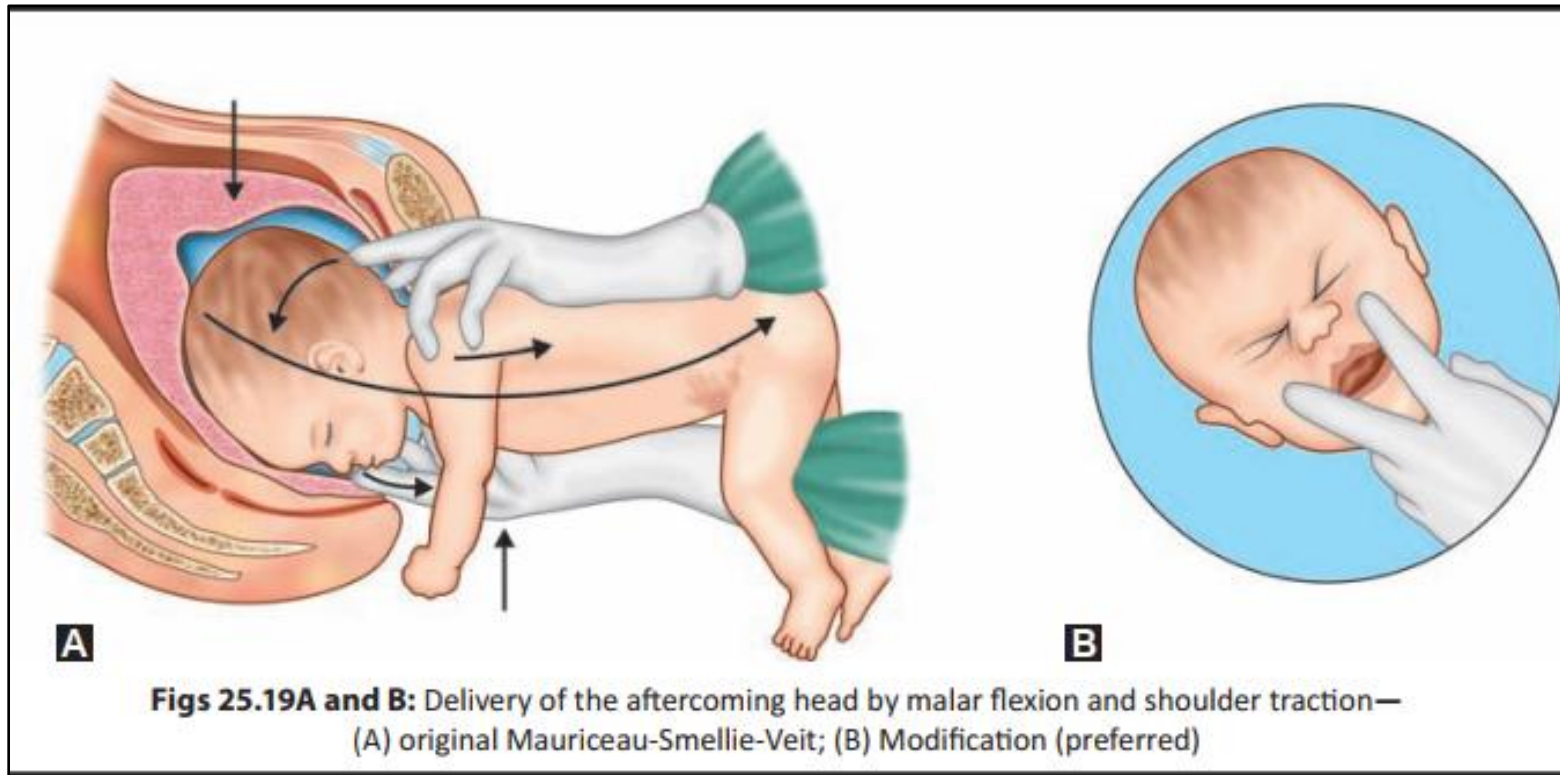
Management...



Figure 15. *Delivery of Arms and Shoulders.*

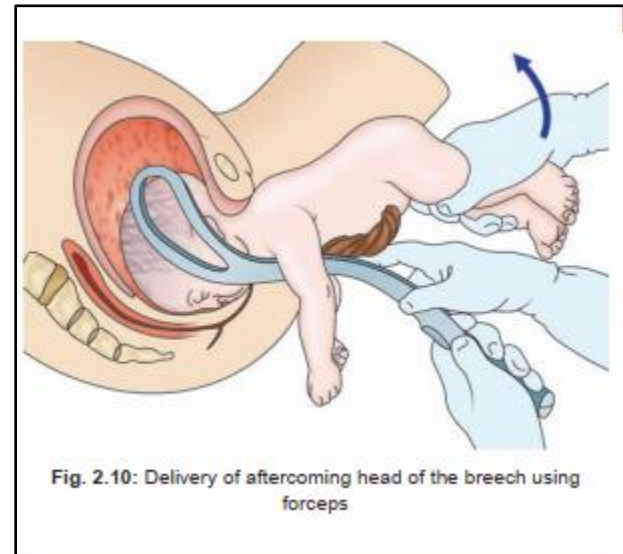
Management...

- Delivery of head :Mauriceau Smellie Veit Maneuver (MSV)



Management...

- Forceps delivery:
 - ⇒ Specialized type of obstetric forceps (Piper forceps) can be used to deliver after-coming head when the MSV cannot be accomplished easily.



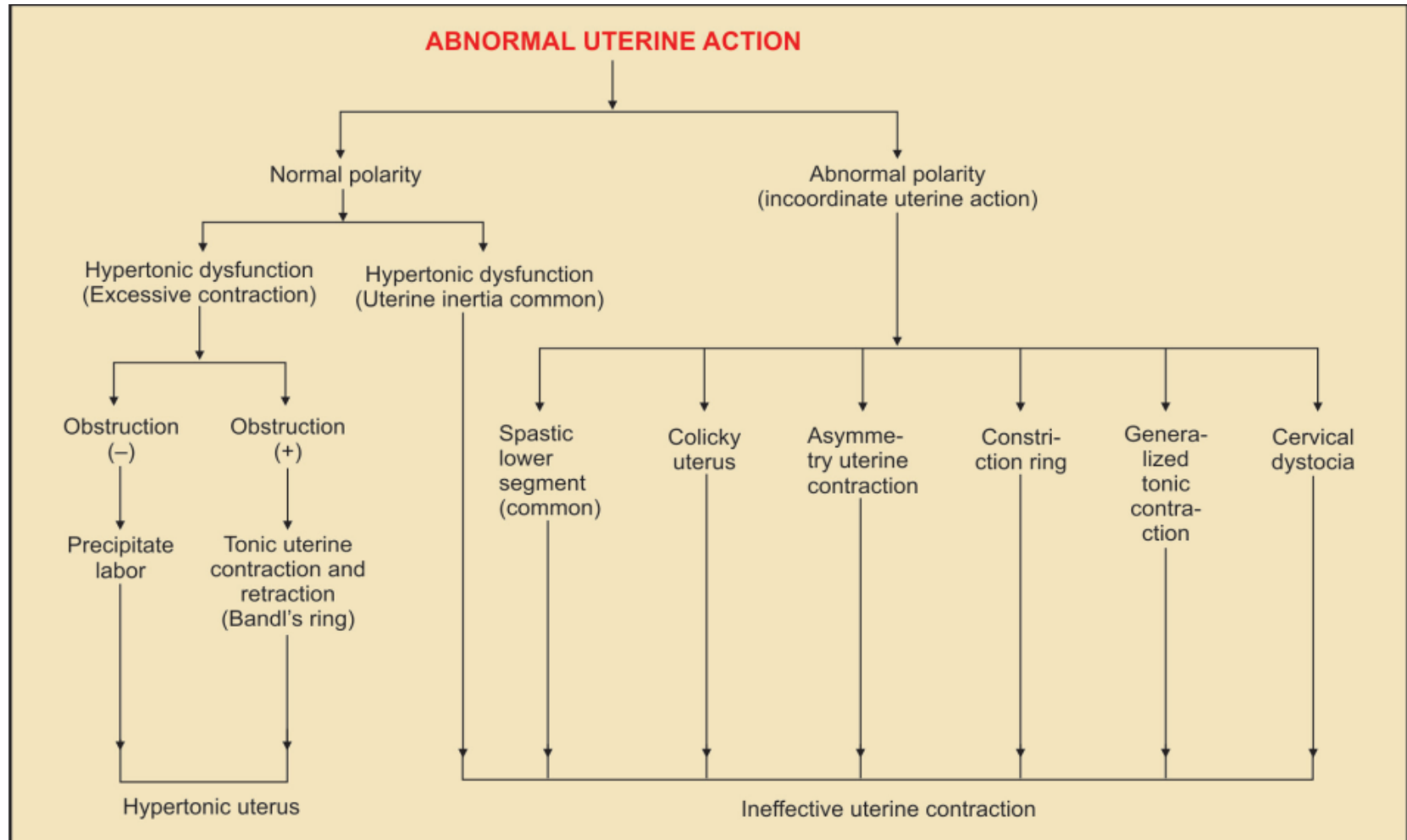
Abnormal uterine action

- Normal labor
 - ⇒ Coordinated uterine contraction
 - ⇒ Progressive dilatation of the cervix(> 1 cm/h in a nulliparous)
 - ⇒ Descent of the fetal head.
- Overall labor abnormalities occur in about 25% of the nulliparous women and 10% of multiparous women.

Abnormal uterine action

- Any deviation of the normal pattern of uterine contractions affecting the course of labor is designated as disordered or abnormal uterine action.

Uterine inertia....



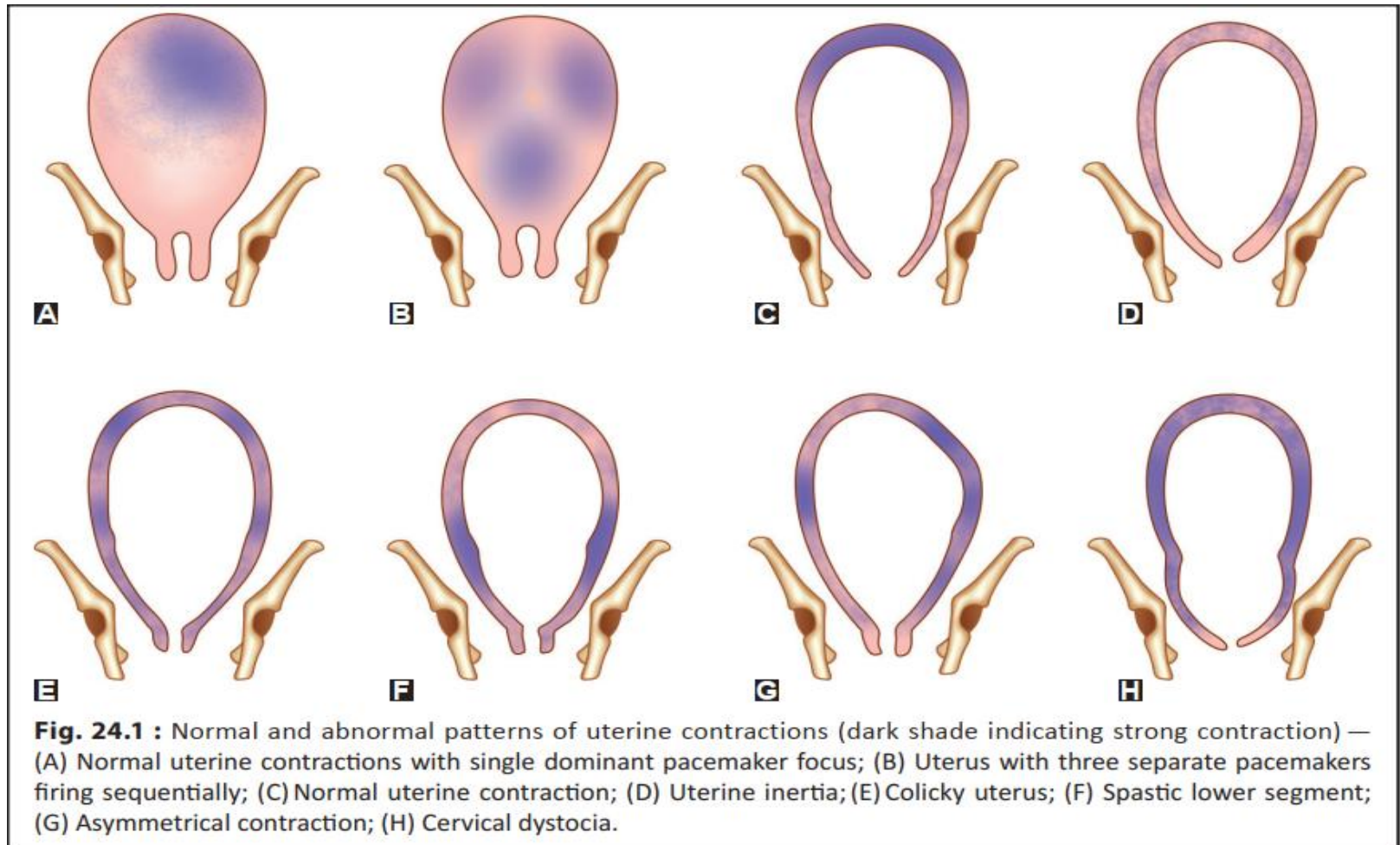
Uterine inertia

- Is abnormal uterine action.
- Is the common type of disordered uterine contraction
- Comparatively less serious.
- It may complicate any stage of labor.
- It may be present from the beginning of labor or may develop subsequently after a variable period of effective contractions.

Uterine inertia...

- Uterine contraction:
 - ⇒ The intensity is diminished
 - ⇒ Duration is shortened
 - ⇒ Good relaxation in between contractions
 - ⇒ The intervals are increased.
 - ⇒ Intrauterine pressure during contraction is <25 mm of Hg.

Uterine inertia....



Etiology

- Elderly primigravida women
- Prolonged pregnancy
- Over distension of the uterus (twins and fibroids)
- Emotional factor (anxiety, stress)
- Constitutional factor (obesity)
- Contracted pelvis and malpresentation
- Use of sedatives, analgesics and oxytocics
- Premature attempt at vaginal delivery
- Attempted instrumental vaginal delivery under light anesthesia.

Diagnosis

- Patient feels less pain during uterine contraction
- Hand placed over the uterus during uterine contraction reveals less hardening of the uterus
- Uterine wall is easily indentable at the time of a pain
- Uterus becomes relaxed after the contraction
 - ⇒ Fetal parts are well palpable
 - ⇒ Fetal heart rate remains normal

Diagnosis....

- Internal examination reveals
 - ⇒ Poor dilatation of the cervix
 - ⇒ Contracted pelvis
 - ⇒ Malposition or malpresentation
 - ⇒ Membranes usually remain intact.

Effects on the mother and fetus

- Maternal exhaustion
- Fetal distress are unusual and appear late
- High rate of cesarean section

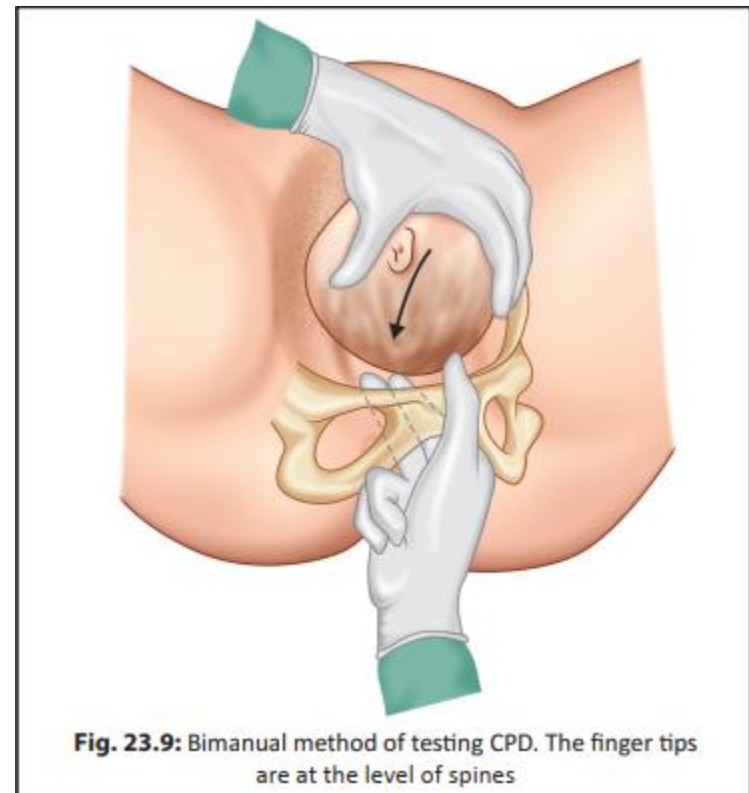
Management

- Exclude cephalopelvic disproportion or malpresentation.

1. Vaginal delivery	A. General measures	• Emotion support • None supine posture of the woman is changed • Empty the bladder, catheterization is made • Maintain hydration by infusion of LV fluid • Adequate pain relief
	B. Active measures	• Acceleration of uterine contraction ⇒ Rupture of the membranes ⇒ Augmentation with oxytocin drip.
2. Place of cesarean section		⇒ Presence of contracted pelvis ⇒ Malpresentation ⇒ Evidences of fetal or maternal distress.

Cephalopelvic disproportion (CPD)

- Is disparity between the fetal head and maternal pelvis which leads to inability of the fetal head to pass through the maternal pelvis for mechanical reasons.
- Failure of the fetal head to enter the pelvic inlet.



CPD classification

1. Absolute

- The fetus is simply too large to negotiate the pelvis.
- True mechanical obstruction as a result of:
 - ⇒ Maternal(permanent)
 - ⇒ Temporary(fetal factors)

CPD classification...

Maternal(permanent)	Fetal(temporary)
• Contracted pelvis (commonest) (TC/OC/BTD < 8cm) • Pelvic exostoses • Spondylolisthesis • Anterior sacrococcygeal tumors	• Large fetus (macrosomia). • Shoulder dystocia. • Congenital abnormalities e.g., Hydrocephalus, ascites or tumor • Fetal monsters • Locked twins

CPD classification...

2.Relative

- Where the fetus may be delivered vaginally if a favorable combination of other factors can be achieved.
 - ⇒ Maternal-border line contracted pelvis (TC= 8-10cm).
 - ⇒ Fetal-mal presentation or position(Face,brow,OP).

Diagnosis

- Difficult before labor(ligaments & joints relax, fetal skull bones overlap)
- Trial of labor
- History
- Physical examination
- Radiological tests

Diagnosis....

- Antepartum examination, measurement and imaging cannot reliably diagnose CPD to preclude a trial of labor.

Diagnosis of CPD

History	Examination	Imaging
• Primigravid • Prolonged labor • Previous prolonged labor • Previous perinatal death • History of obstetric trauma • Properly documented obstetric record (e.g. intraoperative direct measurement of the obstetric conjugate)	• Arrest and protraction disorders of cervical dilatation. • Failure of head descent • High station of the head • Failure of progress of labor after correction of inadequate uterine activity (augmentation) • Fetal macrosomia • Fetal distress • Molding +2 and above • Caput • Dry vagina • Abnormal pelvimetry	• Ultrasound examination may reveal macrosomia or congenital anomalies e.g.hydrocephalus.

Diagnosis of CPD

True conjugate(TC)< 10 cm

Flat sacrum

Narrow sacrosciatic notch

Prominent ischial spine

Diverging/converging pelvic side walls

Bituberous diameter(BTD) < 8 cm

Narrow subpubic angle

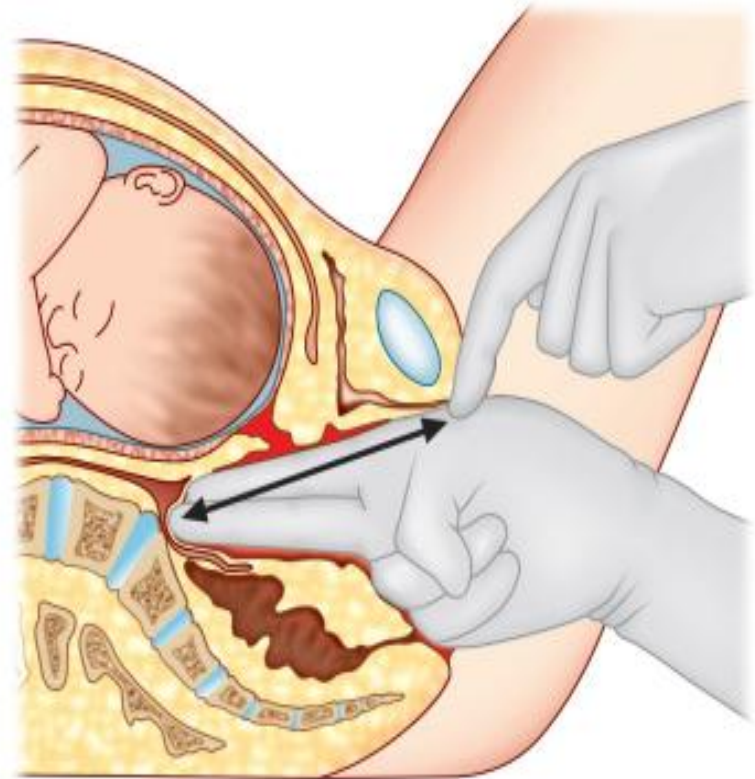


Fig. 9.10: Measurement of diagonal conjugate

Sign of contracted pelvis

Dr Abreham T

Management of CPD...

- Route of delivery:
 - ⇒ Presence of CPD during labor is an indication for caesarean delivery.
 - ⇒ Craniotomy is indicated if the fetus is dead

Complications

Maternal	Fetal / neonatal
• Prolonged / obstructed labor: If CPD is not diagnosed & properly managed the end result is obstructed labor and its associated complications. • PPH • Maternal sepsis	• Fetal distress • Perinatal asphyxia • Neonatal infections • Perinatal death

Discharge counseling and education

- Explained about the indication (CPD) and the need for repeat CS in future.
- Feedback to referring health centers.
- Previous CS for CPD can be followed at a nearby health center and referred after 36 - 37 weeks of gestation.

Obstructed labor

- Is failure of descent of the fetus in the birth canal for mechanical reasons in spite of good uterine contraction.
- It is an outcome of neglected and mismanaged labor.

Cause of obstructed labor

Maternal	Fetal
• Contracted pelvis / cephalopelvic disproportion (commonest) • Soft tissue abnormalities ⇒ Tumor previa ⇒ Vaginal septum ⇒ Tight perineum ⇒ Uterine congenital anomalies	• Macrosomia • Malpresentations- e.g. shoulder presentation • Malposition – persistent occipito posterior or occipito transverse positions • Locked twins, conjoined twins • Fetal anomalies e.g. hydrocephalus • Shoulder dystocia

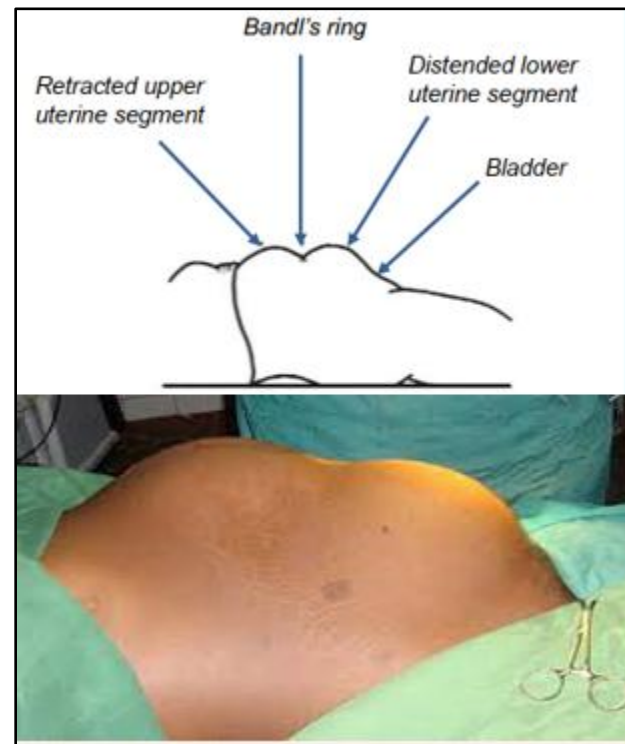
Clinical features

- History
 - ⇒ Abnormally prolonged labor
 - ⇒ Early ROM or prolonged rupture of the membranes
 - ⇒ Most do not have antenatal care
 - ⇒ Painful contractions
 - ⇒ Fever
 - ⇒ Previous prolonged labor, stillbirth or early neonatal death
 - ⇒ Previous operative deliveries
 - ⇒ Medical history;rickets, osteomalacia, or pelvic injury

Clinical features...

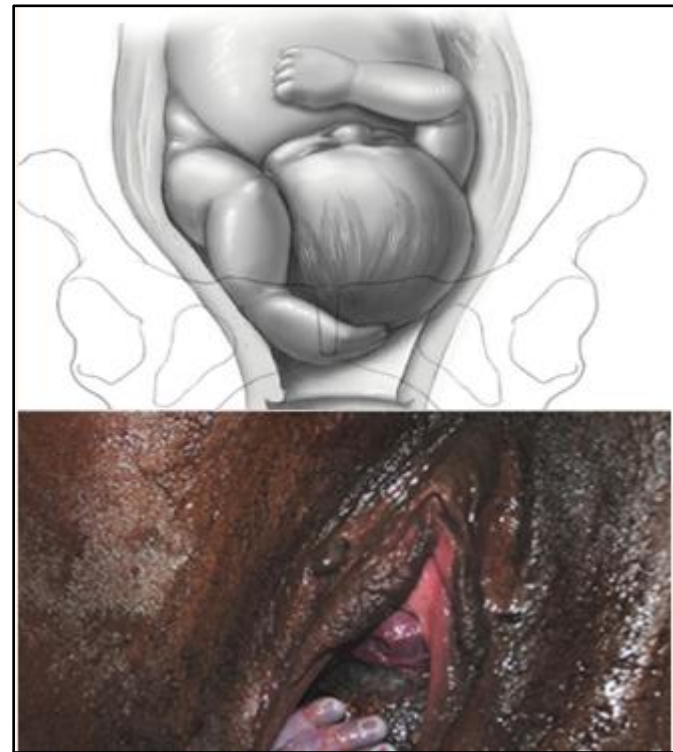
- Abdominal findings

- ⇒ Distention of the bowel as a result of acidosis and hypokalemia.
- ⇒ Two/ Three tumor abdomen
- ⇒ Abnormal FHB
- ⇒ No fetal heart tone if the fetus dies from anoxia



Clinical features...

- Vaginal examination finding
 - ⇒ Edema of lower vagina and vulva (canula sign).
 - ⇒ Meconium- stained and foul-smelling discharge.
 - ⇒ Edema of the cervix
 - ⇒ Cervix poorly applied to the head.
 - ⇒ Little or no descent of the presenting part.
 - ⇒ Increasing molding & Caput



Management

- Secure good venous access
- Give crystalloids (ringers lactate is the fluid of choice)
- Give dextrose solution IV but never give orally.
- Monitor closely
- Blood group and Rh, cross-match, complete blood count
- Give broad spectrum antibiotics
- Catheterize
- Empty the stomach(NGT)
- Prophylactic dose of TAT: 1500 units
- Alive fetus: do emergency cesarean delivery.
- Dead fetus do destructive delivery if prerequisite fulfill

Complications

Early complications	Late complications:
• Atonic PPH • Uterine rupture • Peripartum infection • Tetanus • Maternal death • Fetal distress • Fetal & neonatal infections • Fetal and neonatal death.	• Fistula vesico-vaginal, rectovaginal) • Vaginal stenosis & stricture • Foot drop (sciatic, common peroneal nerve damage • Infertility following postpartum PID or hysterectomy • Psychological trauma due to the painful labor experience • Loss of the baby • Social isolation

Unit-II:Operative delivery

1

- Forceps delivery

2

- Vacuum delivery

3

- Cesarean section

4

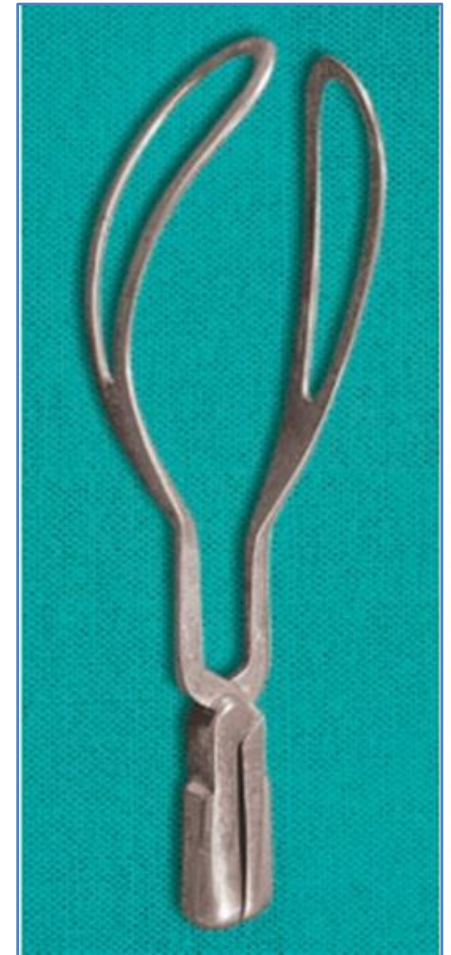
- Induction

Objective

- Define each operative delivery
- Describe types of operative deliveries.
- Identify indications & contraindication for operative deliveries.
- Explain prerequisite condition operative deliveries.
- Describe complication of operative deliveries.
- Describe definition, indication method and complication of induction

Obstetric forceps

- Is a pair of instruments specially designed to assist extraction of fetal head and thereby accomplishing delivery of the fetus.
- Forceps functions
 - ⇒ Traction
 - ⇒ Rotation.



Anatomy of obstetric forceps

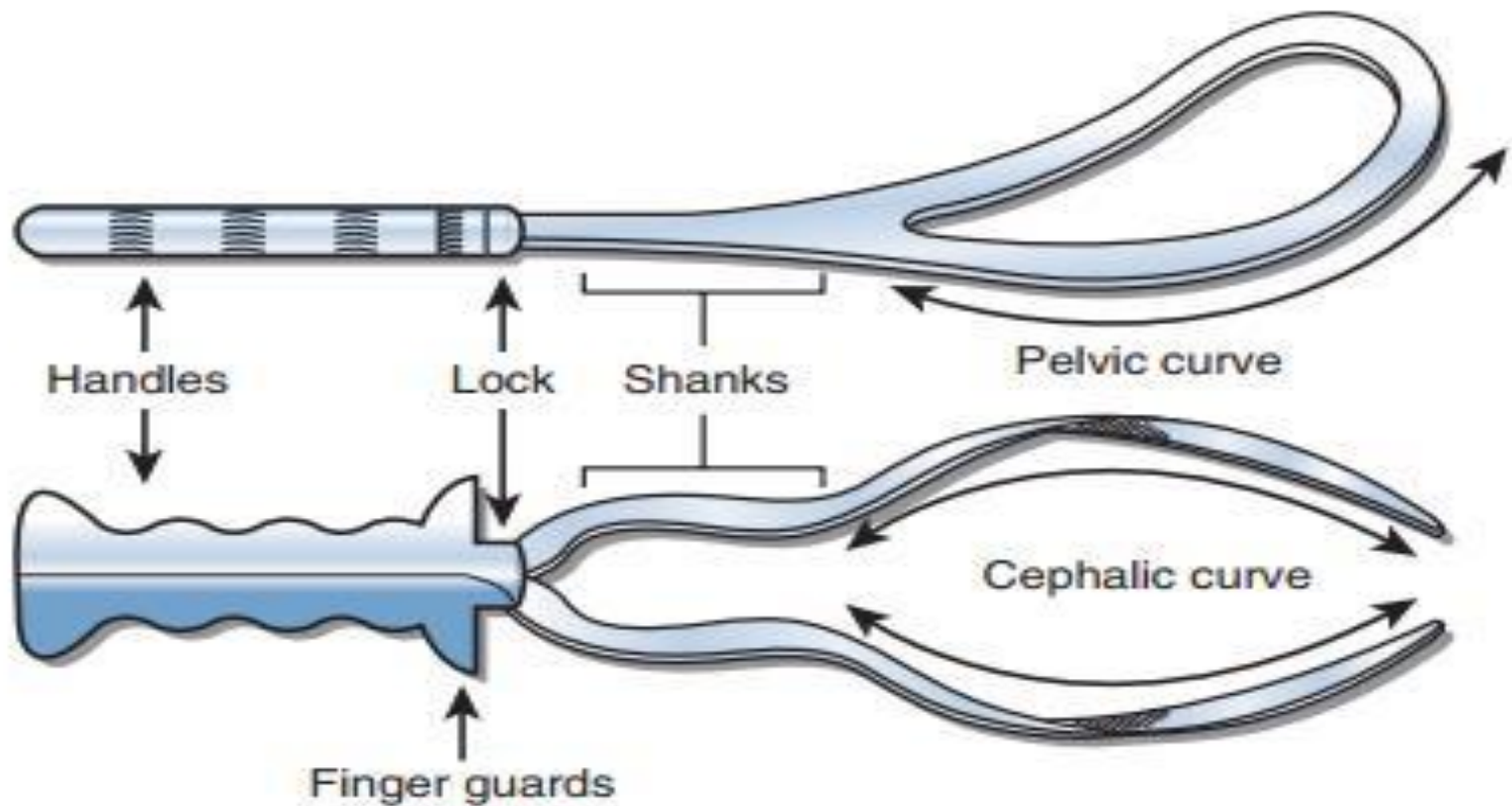


FIG 14-1 Anatomy of the forceps.

Type of forceps

Based on function

Classic
forceps

Rotational
forceps

Specialized
forceps

Based on application level

Outlet
forceps

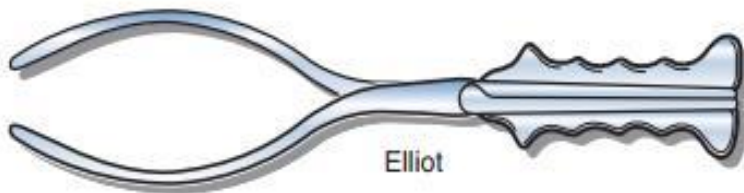
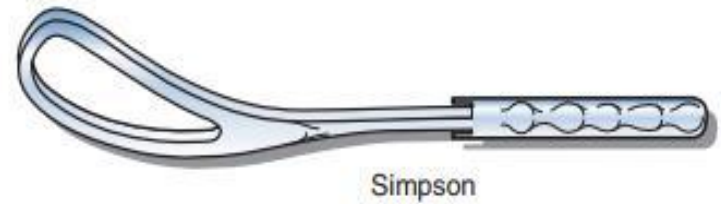
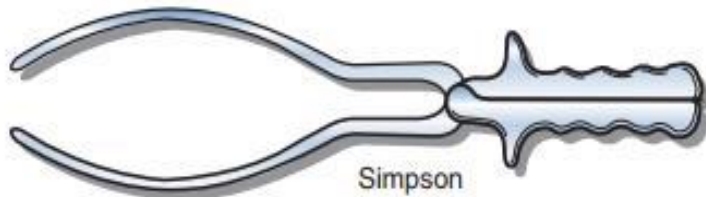
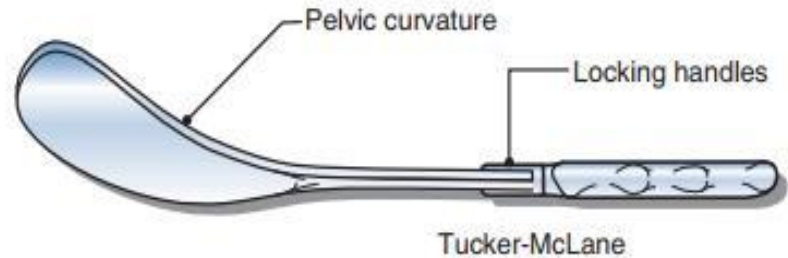
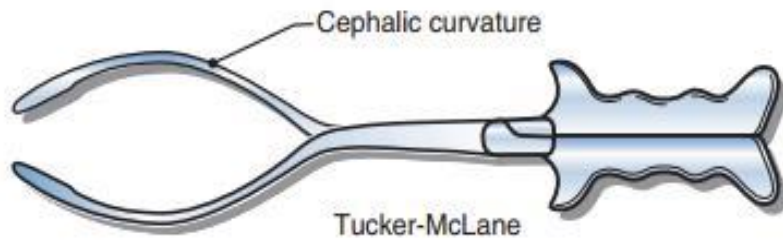
Low
forceps

Mid
forceps

Classic forceps

- Is conventional traction forceps
- Are typically used when rotation of the vertex is not required.
 - ⇒ Simpson forceps (has parallel shanks)
 - ⇒ Elliot forceps (has overlapping shanks)
 - ⇒ Tucker-McLane forceps (has solid blades)

Classic forceps....



Rotational forceps

- Used for rotation
- Absence of a pelvic curve facilitates rotation of the vertex.
- Cannot be depressed sufficiently low against the perineum.
 - ⇒ Tucker-McLane
 - ⇒ Kielland

Rotational forceps.....

② Rotational forceps



Fig. 36.15 : Kielland's Forceps

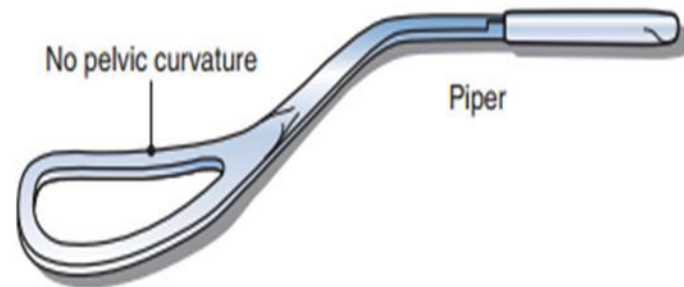
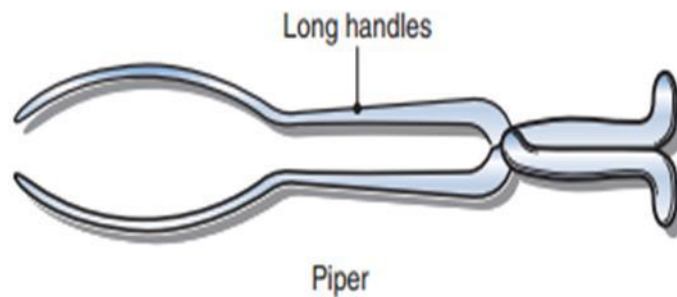
Piper forceps is a specialized forceps, used to assist the delivery of the after-coming head of breech. It has a cephalic curve, reverse pelvic curve, long parallel shanks that permit the baby's body to rest against it during head delivery.

Specialized forceps

- Forceps to assist with delivery of the after coming head during vaginal breech delivery.

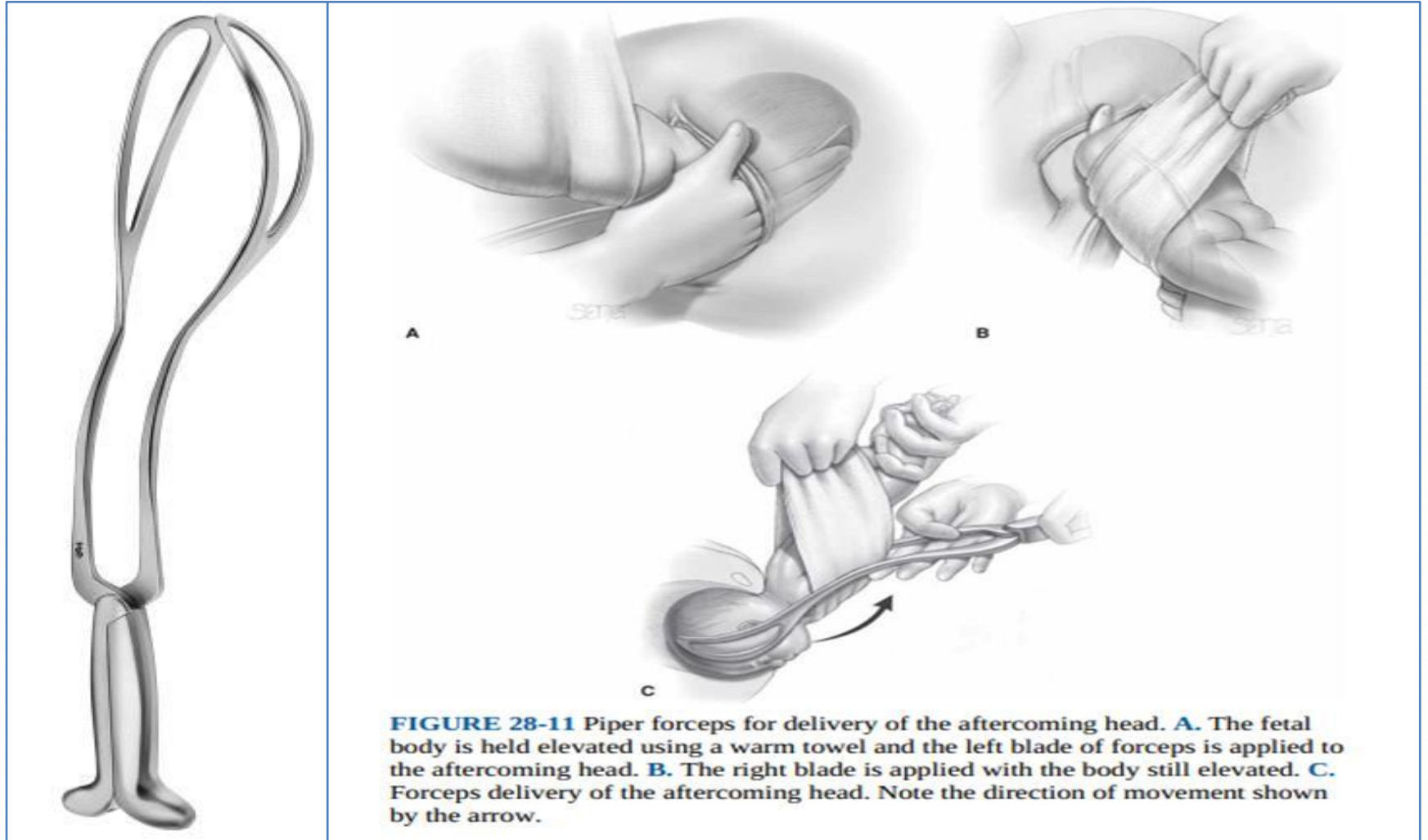
⇒ Piper forceps

③ Forceps for delivery of aftercoming head of the breech



JWKOI *M. Cooley*

Specialized forceps...



Forceps classification

1	Outlet	• Applied when the fetal head is at station +3 (at the pelvic floor). • Scalp is visible at the introitus without separating the labia. • Sagittal suture is in AP diameter or right or left OA or OP. • Fetal head is at or on perineum. • Rotation does not exceed 45 degrees.
2	Low	• Applied when the station is +2 or below not on pelvic floor • Rotation ≤ 45 degrees to LOA/ROA to OA or. • Rotation ≥ 45 degrees to LOP/ ROP to OP.
3	Mid forceps	• Station is above +2 cm but head is engaged. • Station is between 0 and till 2cm. • Done $\Rightarrow$ Only with skilled operator. $\Rightarrow$ When assessment suggests a high chance of success.

Indication and contraindications of forceps

Indication	Contraindications
• Prolonged second stage • Fetal compromise in 2nd stage of labor ⇒ Non reassuring fetal HR ⇒ Cord prolapse ⇒ Abruptio placenta • After coming head of breech • Shortening of the 2nd stage of labor ⇒ Maternal exhaustion ⇒ Maternal cardiopulmonary ⇒ Cerebrovascular disease ⇒ Proliferative retinopathy ⇒ Severe anemia ⇒ Pregnancy induced hypertension ⇒ Eclampsia	• Absence of full dilatation of cervix. • Cephalopelvic disproportion. • High station of fetal head • If uterine contraction cease • Lack of experience of operator. • Mentum posterior face presentation. • Hydrocephalic infant • Brow presentation.

Prerequisites for forceps application

- Presentation & position(Vertex $\Rightarrow$ OA, Face $\Rightarrow$ mento-anterior (MA)).
- Station of +2 or below
- Fully dilated cervix
- Ruptured membranes
- Adequate pelvis
- Provider should be skilled in performing forceps delivery
- No contraindication to vaginal delivery

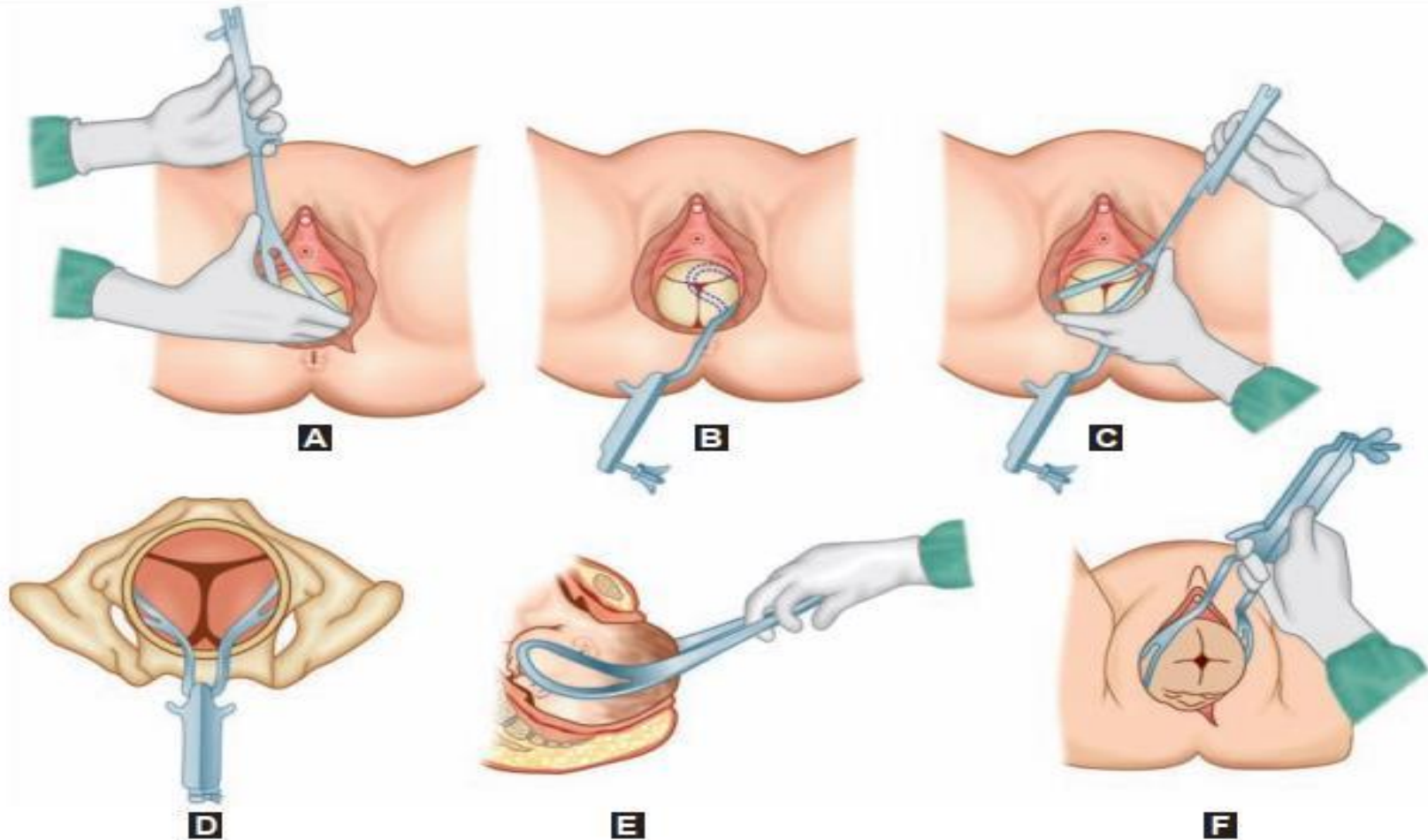
Preparation

- Counsel and get written consent.
- Empty bladder.
- Local anesthesia infiltration for episiotomy (if required).
- Prepare all necessary equipment and asses functionality

Procedure

1. Completely locked forceps in front of the perineum to orient and identify the right and left blades.
2. Insert two fingers of the right hand into the vagina
3. Slide the left blade gently between the head and fingers to rest on the side of the head(repeat the same maneuver on right side)
4. Check that application and no maternal tissue is entrapped.
5. After locking, apply steady traction

Procedure



Figs 36.12A to F: Steps of low forceps operation — (A) Introduction of the left blade; (B) The handle lying flat on the perineum after introduction; (C) Introduction of the right blade; (D) Showing perfect apposition and locking of the blades; (E) Bimalar, biparietal placement of blades and position of the fingers during traction; (F) Change in the grip in the final stage of delivery

Procedure...

6. After locking, apply steady traction
 - Between contractions check FHB and application of forceps.
 - When the head crowns, make an episiotomy if necessary.



Fig. 36.13: The blades are equidistant from the lambdoid suture indicating correct application

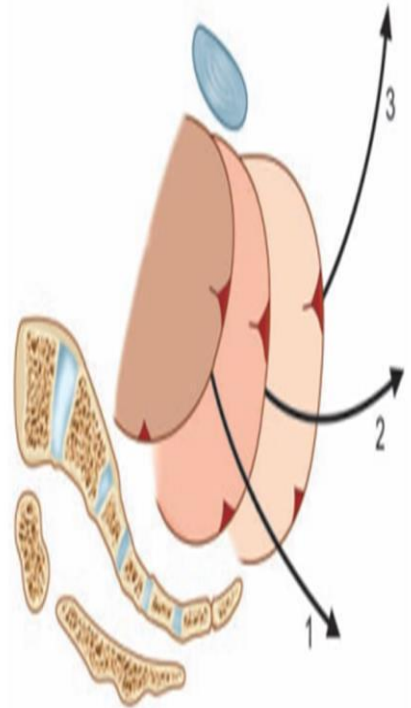


Fig. 36.14: Showing direction of traction along the axis of the birth canal — (1) Downwards and backwards; (2) Straight horizontal pull (3) Upwards and forwards

Procedure...

7. Remove first the right forceps followed by the left.

- Further care

⇒ After delivery inspect the vagina and cervix; and repair any tear or episiotomy.

⇒ Examine the newborn .

⇒

Forceps

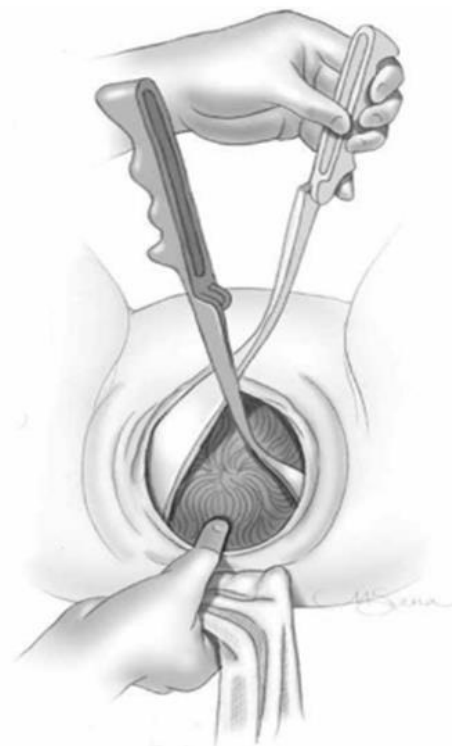


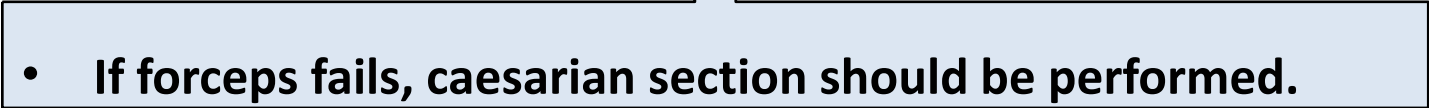
FIGURE 29-11 Branches are removed in the opposite order from that in which they were originally placed. The fingers of the right hand, covered by a sterile towel, bolster the perineum. The thumb is placed directly on the head to prevent sudden egress. (Reproduced with permission from Yeomans ER: Operative vaginal delivery. In Yeomans ER, Hoffman BL, Gilstrap LC III, et al (eds): Cunningham and Gilstrap's Operative Obstetrics, 3rd ed. New York, McGraw-Hill Education, 2017.)

Trial of forceps

- It is a tentative attempt of forceps delivery in case of suspected midpelvic contraction with a declaration of abandoning it in favor of caesarean section if moderate traction fails to overcome the resistance.
- It should be attempted in double set up.
- If there is difficulty of introduction of blades, locking or resistance to gentle traction, withdrawn forceps and do caesarian section

Failed forceps

- When a deliberate attempt in vaginal delivery with forceps has failed to expedite the process, it is called failed forceps
- Forceps Failed If
 1. Locking of forceps is difficult
 2. Fetal head does not advance with each pull.
 3. Fetus is undelivered after three pulls with no descent
 4. Fetus is undelivered after 30minute

- 
- **If forceps fails, caesarian section should be performed.**

Advantages over vacuum

- Fast
- Active rotation possible
- Traction independent of contraction
- Also used in face (MA) & after coming head of breech
- Can be used for preterm(≤ 34 weeks)
- Less failure to effect delivery
- Less fetal trauma

Complication of forceps delivery

Fetal complications	Maternal complications:
• Injury to facial nerves • Lacerations of the face and scalp • Cephalhematoma • Intracranial hemorrhage. • Fracture of the facial bones or skull	• Tear of cervix, vagina, or vulva. • Uterine rupture • Postpartum hemorrhage (traumatic) • Shock • Infection • Delayed or long-term sequel ⇒ Chronic low backache ⇒ Genital prolapsed ⇒ Incontinence.

Complication of forceps delivery...



FIGURE 33-4 Left facial nerve injury. This was almost completely resolved two days after delivery.

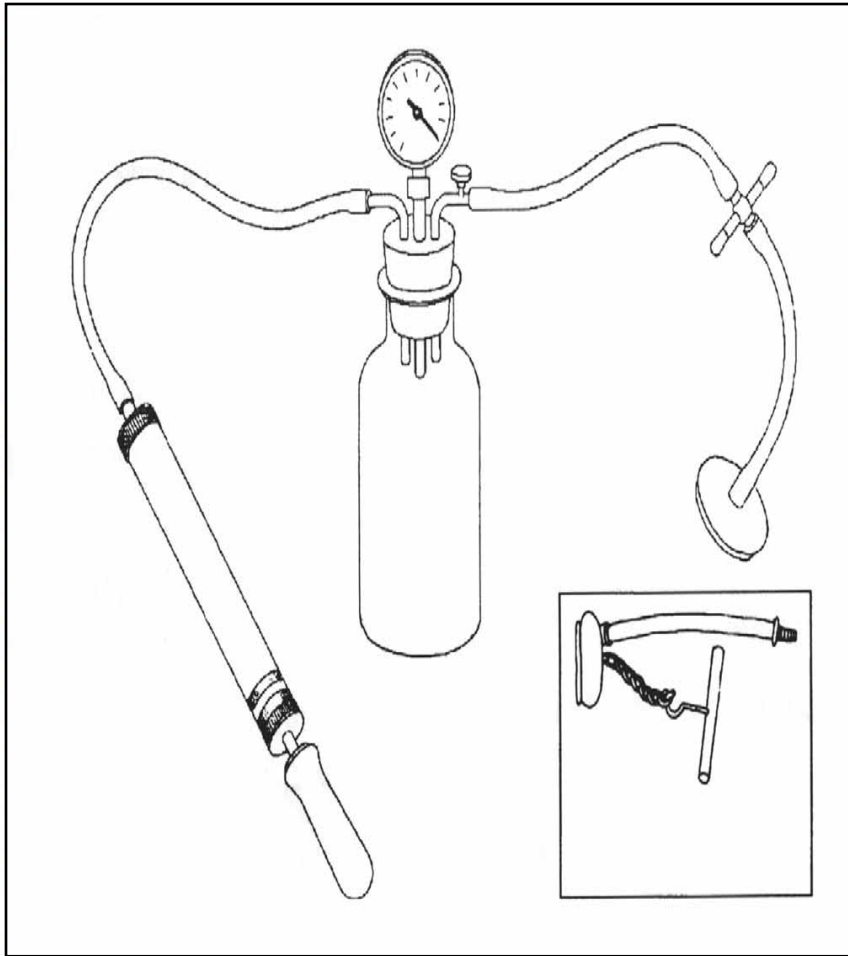


Vacuum extraction (ventouse)

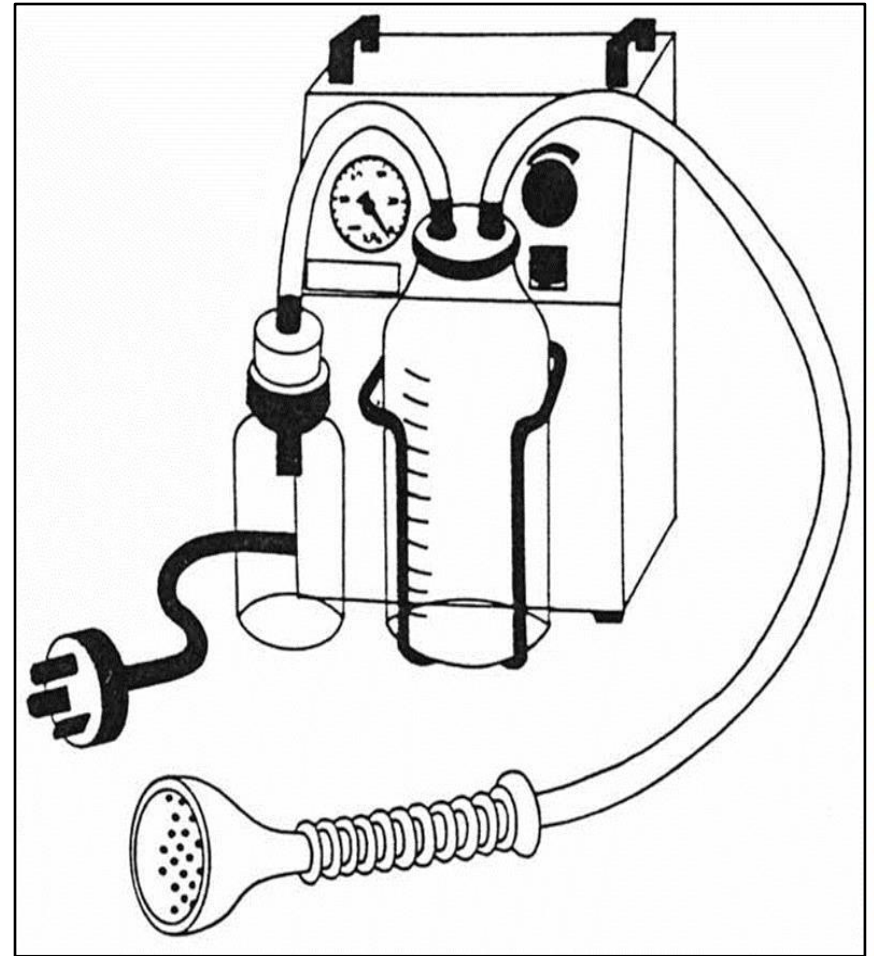
- It is an instrumental device designed to assist delivery by creating a vacuum between it and the fetal scalp.
- Function of the cup
 - ⇒ Flexion
 - ⇒ Descent
 - ⇒ Autorotation



Types of vacuum extractors



Vacuum hand pump



Electrical vacuum pump with soft cup

Types of vacuum extractors...

Cup type

Metal cup/M-style

- Rigid
- Has more traction force
- Size: 40, 50, & 60 mm
- Can be autoclaved
- Is a mushroom-shaped
- Have a higher success rate
- Easier cup placement in OPP
- ↑ risk of fetal scalp injuries.
- More difficult to apply

Soft cup/Pliable

- Vacuum attained more quickly
- Are bell, mushroom or funnel shaped.
- Fewer neonatal scalp injuries.
- Disposable and reusable
- Faster and easy application
- Less pronounced chignon
- Fewer superficial scalp injuries
- Less intracranial hemorrhage
- Autorotation
- Has higher failure rate
- Easy to apply.

Types of vacuum extractors...

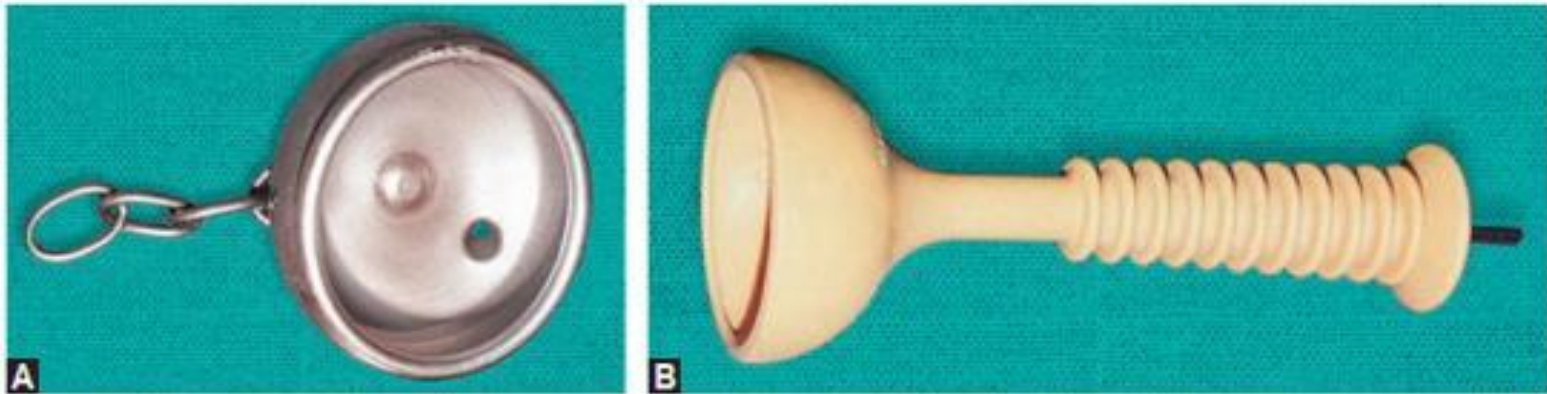
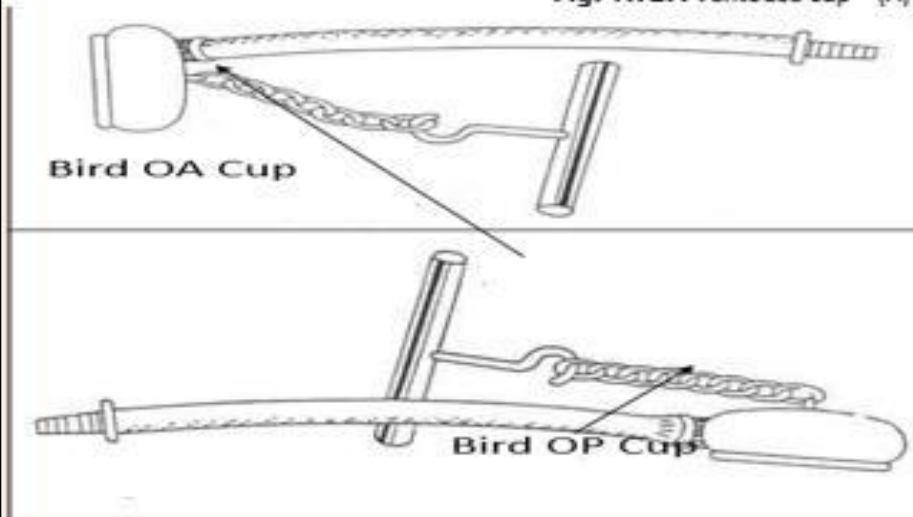


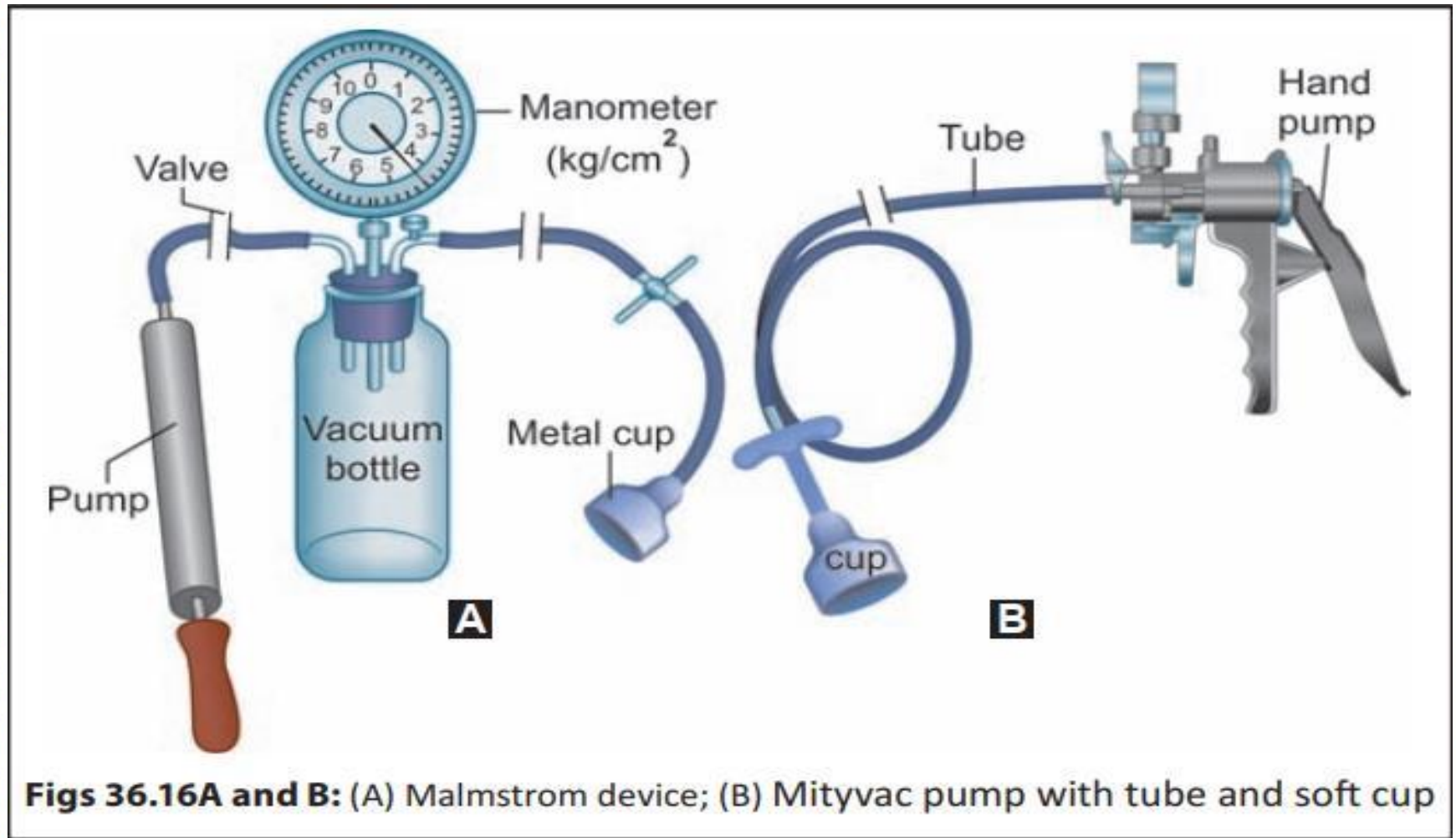
Fig. 41.27: Ventouse cup—(A) Metal (B) Silastic



Components of vacuum

1. Cup - diameter of 3, 4, 5 or 6 cm.
2. Rubber tube -attaching the cup to a glass bottle
3. A manometer - declare the negative pressure
4. Another rubber tube - connecting the bottle to a suction piece which may be manual or electronic creating a negative pressure that should not exceed - 0.8 kg per cm².

Components of vacuum....



Indications of vacuum extraction

- Indications and pre-requisites are the same as for forceps.
- Not indicated for fetuses under 34 weeks gestational age.
- Special use-before the cervix is fully dilated ($> 8\text{cm}$)

Contraindications

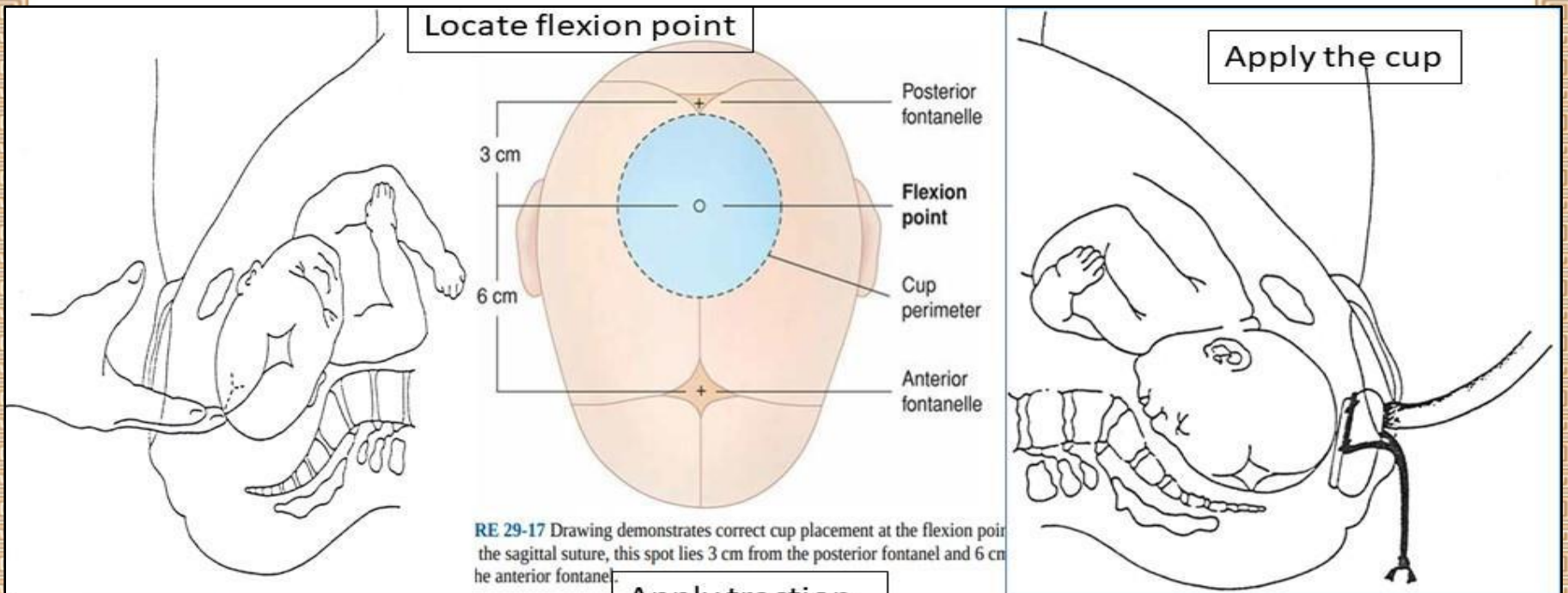
- Operator inexperience
- Inability to assess fetal position
- High station(above 0 station)
- Suspicion of cephalopelvic disproportion
- Other presentations than vertex.
- Premature fetus(<34 weeks).
- Intact membranes

Vaccume Prerequisites	• Only vertex presentation • Fully dilated cervix • Engaged head: station at 0 and below • Ruptured membranes • Gestational age 34 weeks and above • Adequate pelvis • No contraindication to vaginal delivery
Preparation	• Counsel and get written consent. • Empty bladder • Local anesthesia infiltration for episiotomy if required. • Assemble, check all connections and test the vacuum on a gloved hand

Procedure

Step	Procedure	
1	Locate flexion point	
2	Apply the cup	
3	Create a vacuum.	• Create a vacuum of 0.2 kg/cm² negative pressure and check the application of the cup. • Increase the vacuum to 0.8 kg/cm² and check the application of the cup again.
4	Apply traction.	• Applied at the onset of each contraction • First pull for right direction for descent and causes flexion. • Between contractions check: ⇒ Fetal heart rate ⇒ Application of cup ⇒ Progress of descent
5	Encourage the woman to assist descent with expulsive efforts	
6	Removal of cup.	

Procedure



A



B



C

Figs 36.17A to C: Application of vacuum extractor; A, B, C, indicating the cup placement at different stations of the fetal head. Flexion or pivot point is an imaginary site located midsagittally about 2 cm in front of the posterior fontanellae or about 2 cm behind the center of the anterior fontanellae. Traction over this site either by the vacuum or forceps promotes flexion and presents the fetal head with a smaller diameter to the pelvis (Fig. 36.16A).

Vacuum

Procedure....

Tips

- Never use the cup to actively rotate the baby's head.
- Rotation of the baby's head will occur with traction.
- Pulls for a maximum of 30 minutes(with progress, no fetal distress)

Failure of vacuum

- Vacuum extraction is considered failed if:
 - ⇒ The head does not advance with each pull.
 - ⇒ The fetus is undelivered after three pulls.
 - ⇒ The fetus is not delivered within 30 minutes.
 - ⇒ The cup that is applied appropriately and pulled in the proper direction with maximum negative pressure slips off the head twice.



NB: If vacuum delivery fails, the fetus should be delivered by Cesarean section.

Advantages of vacuum over forceps

- Regional anesthesia is not required.
- Cup doesn't increase diameter beside the head unlike forceps.
- Less compression force($0.77 \text{ Vs } = 1.3 \text{ kg/cm}^2$ forceps) to head.
- Less genital tract lacerations.
- Can be applied before full cervical dilatation($\geq 8\text{cm}$)

Complications vacuum delivery

Maternal	Fetal
• Perineal, vaginal ,labial, periurethral and cervical lacerations. • Annular detachment of the cervix (incompletely dilated) • Cervical incompetence • Future prolapse if used with incompletely dilated cervix.	• Cephalohematoma. • Scalp lacerations • Subgaleal hematomas • Intracranial hemorrhage. • Neonatal jaundice • Subconjunctival hemorrhage • Retinal hemorrhage

Complications vacuum delivery.....



Figure 24: Bicornous cephalohematoma-looks like a horn

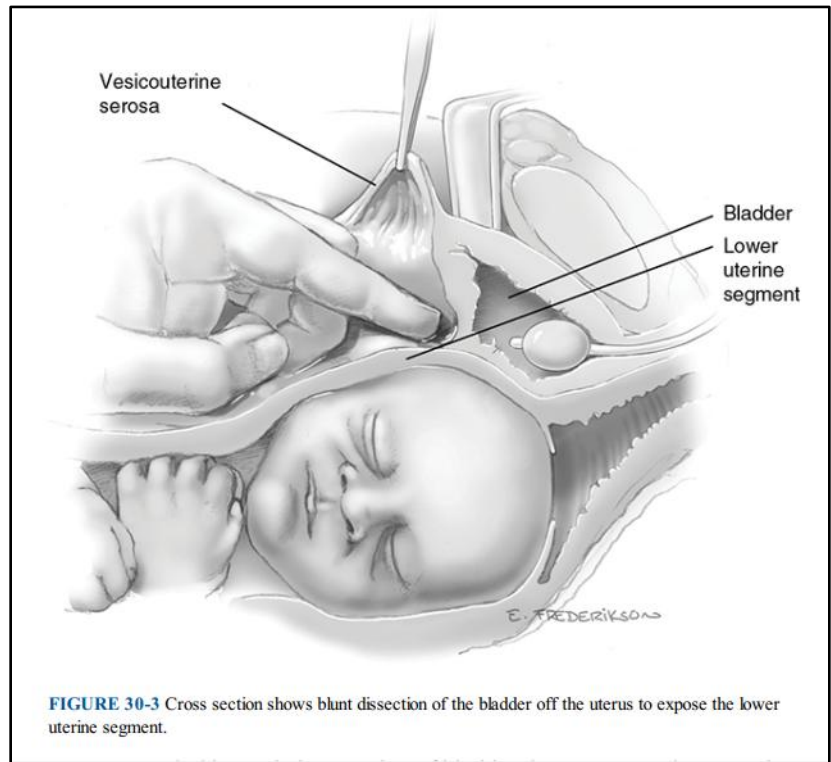


Operative vaginal delivery and PMTCT

- As much as possible operative vaginal delivery should be avoided in mother with ARVI.
- When instrumental delivery is indicated or mandatory .
 - ⇒ Limit use of vacuum extraction
 - ⇒ Prefer obstetric forceps whenever

Cesarean section(CS) /delivery

- Is the delivery of the fetus, placenta and membranes through an incision on the abdominal and uterine wall at or after 28 weeks of gestation.



Type of CS

Elective CS	• Is a planned cesarean delivery • Performed before the onset of labor or the appearance of any complication that might constitute an urgent indication.
Emergency CS	• Is when the CS is done in labor or due to any complication that necessitates immediate delivery.

Indications

ABSOLUTE INDICATIONS	RELATIVE INDICATIONS
Vaginal delivery is not possible. Cesarean is needed even with a dead fetus Indications are few: 1. Central placenta previa 2. Contracted pelvis or cephalopelvic disproportion (absolute) 3. Pelvic mass causing obstruction (cervical or broad ligament fibroid) 4. Advanced carcinoma cervix	Vaginal delivery may be possible but risks to the mother and/or to the baby are high More often multiple factors may be responsible 1. Cephalopelvic disproportion (relative) see p. 352 2. Previous cesarean delivery (p. 330)—(a) when primary CS was due to recurrent indication (contracted pelvis). (b) Previous two CS (c) Features of scar dehiscence. (d) Previous classical CS 3. Non-reassuring FHR (fetal distress) 4. Dystocia may be due to (three Ps) relatively large fetus (passenger), small pelvis (passage) or inefficient uterine contractions (Power)

Investigations

- Hemoglobin / hematocrit.
- Blood group (ABO) and Rh.
- HIV
- HBsAg
- At least 2 units of cross matched blood

Prerequisites

- Appropriate indication.
- Competent team of providers.
- Appropriate facility and equipments.

Preoperative preparation

- Elective CS
 - ⇒ Plan on days
 - ⇒ Is done at 39 weeks
 - ⇒ NPO for 8 hours for regular meal and 2 hours for clear fluid
- Emergency CS
 - ⇒ Cannot be planned
 - ⇒ Limit feeding to fluid diet in laboring women

Preoperative preparation....

- Obtain informed written consent.
- Preparation of blood (if required).
- Secure IV (16 or more gauge cannula).
- Catheterization.
- Prophylactic antibiotic (ampicillin, cephalixin, cefazolin...)
- Administer antacid solution PO (or through NG tube).
- Make sure the necessary drugs and equipment are in place.
- Make sure neonatal resuscitation set and personnel are in place.
- Check fetal heart beat before proceeding to the CS.

Intra-operative care

- Record maternal vital signs before anesthesia.
- Position on operation table(tilt to left)
- Monitoring of vital signs during the CS.
- After delivery of the baby administer 20 IU oxytocin in 1000 ml of N/S or R/L at 60 drops per-minute for two hours.

Types of skin during CS

- Types of skin during CS
 1. Midline vertical
 2. Suprapubic transverse

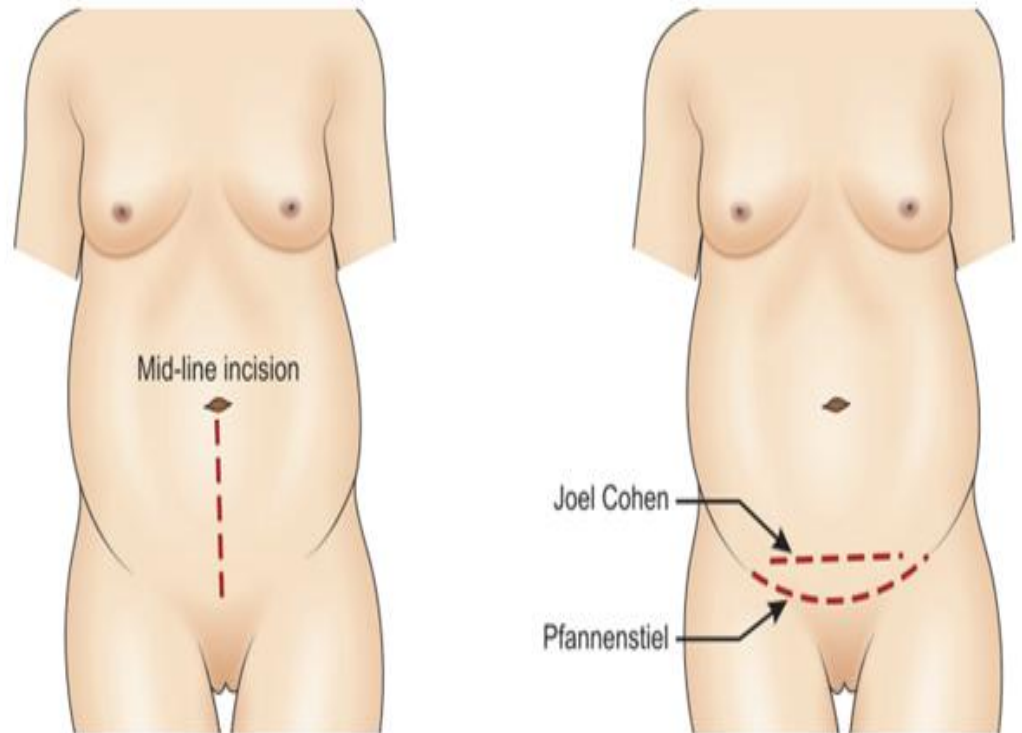


Fig. 7.2: Type of abdominal skin incision

Types of uterine incision for CS

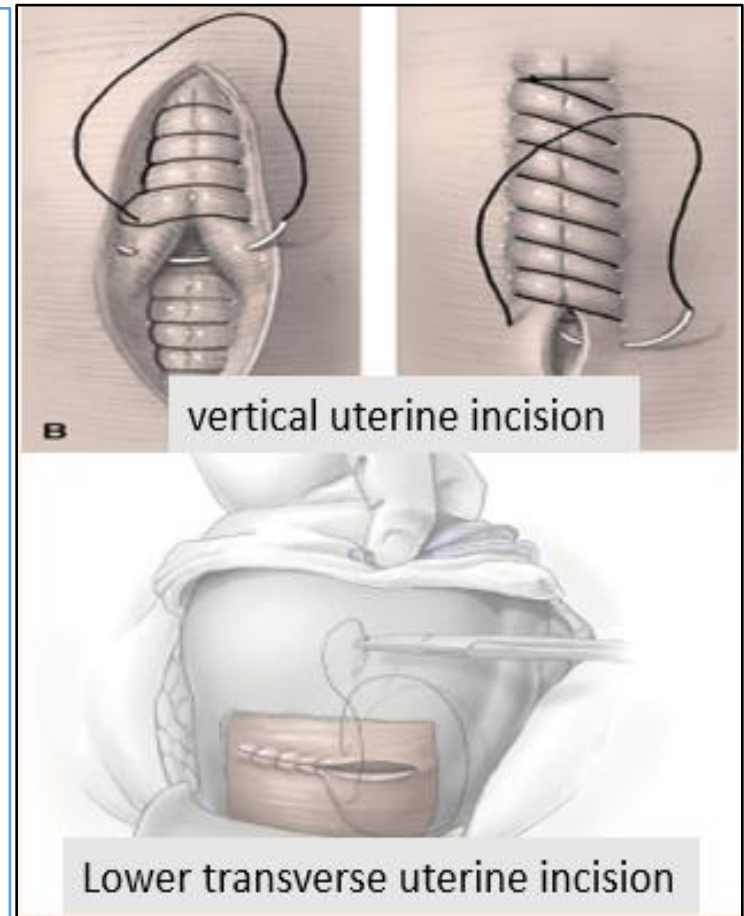
1. Lower transverse uterine incision

⇒ Commonest

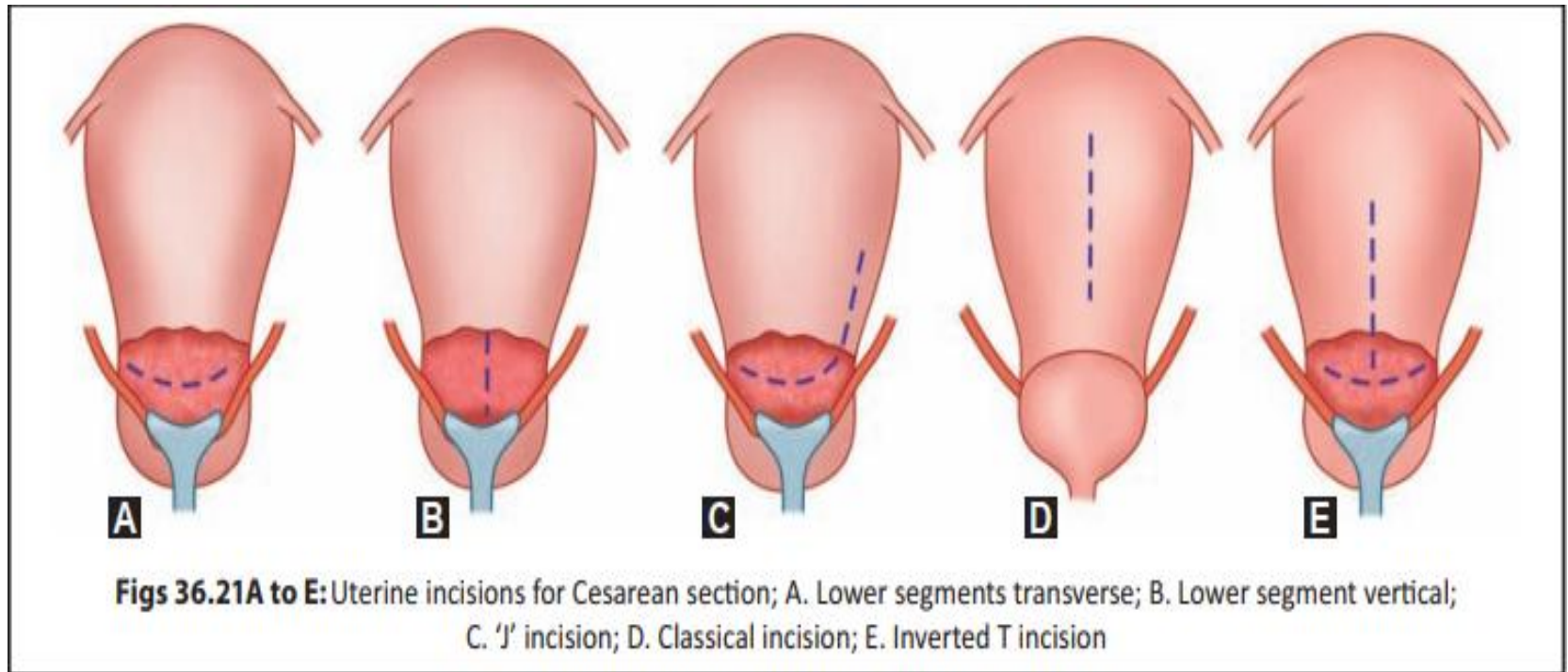
⇒ Preferable type

2. Lower vertical uterine (De Lee) incision.

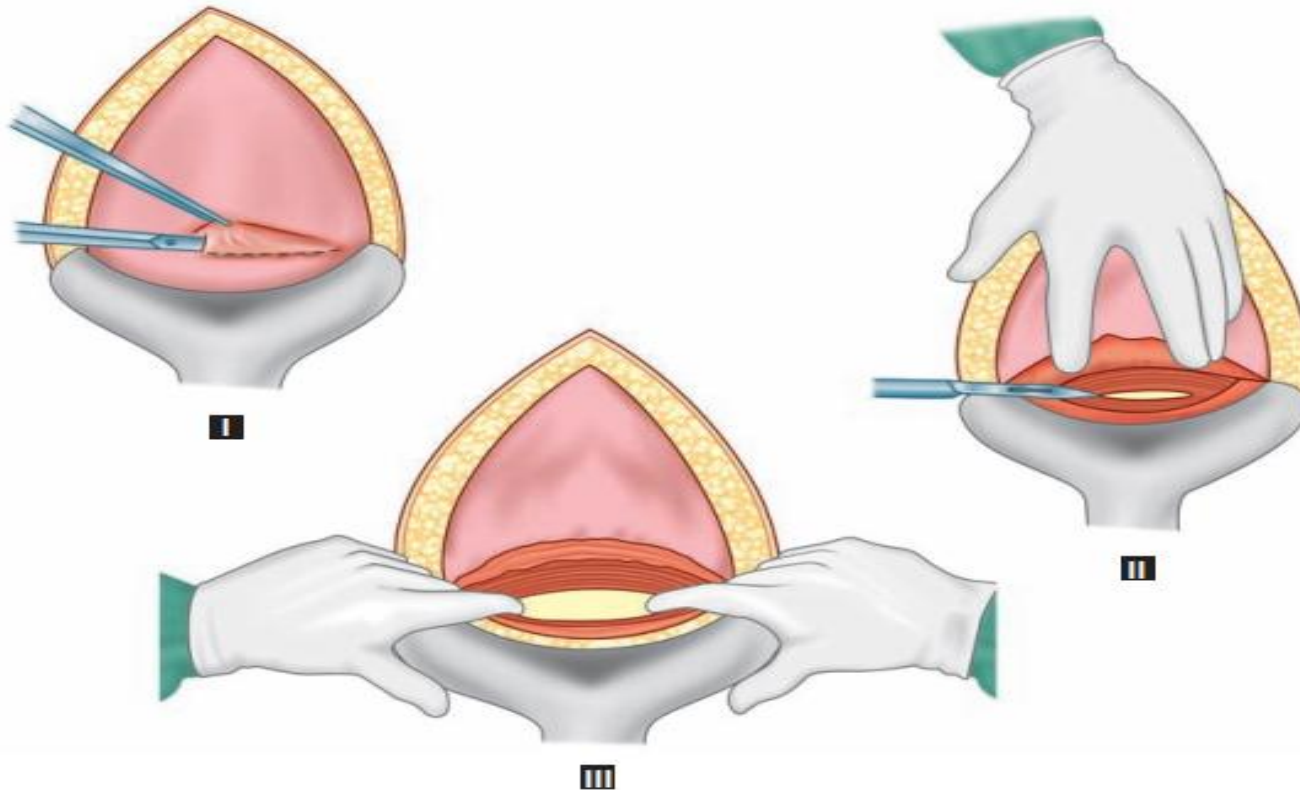
3. Upper vertical uterine incision (classical incision).



Types of uterine incision for CS...



Types of uterine incision for CS...



Figs 36.22I to III: Steps of LSCS : (I) The loose peritoneum on the lower segment is cut transversely; (II) A short incision is made in the midline down to the membranes; (III) The incision of the lower segment is being enlarged using index finger of both hands.

Steps contd.

Types of uterine incision for CS...

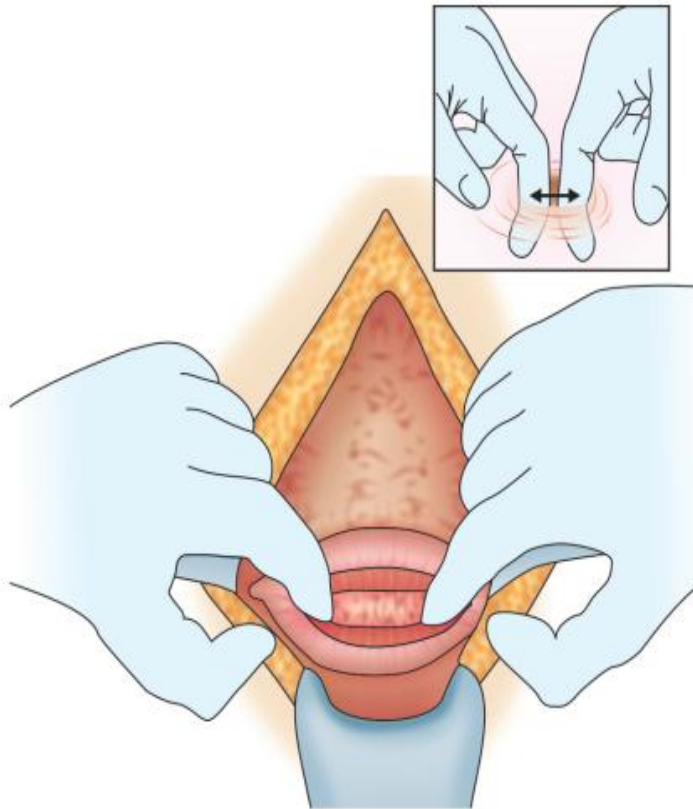


Fig. 7.3A: Extension of the uterine incision by manual stretching using the index fingers

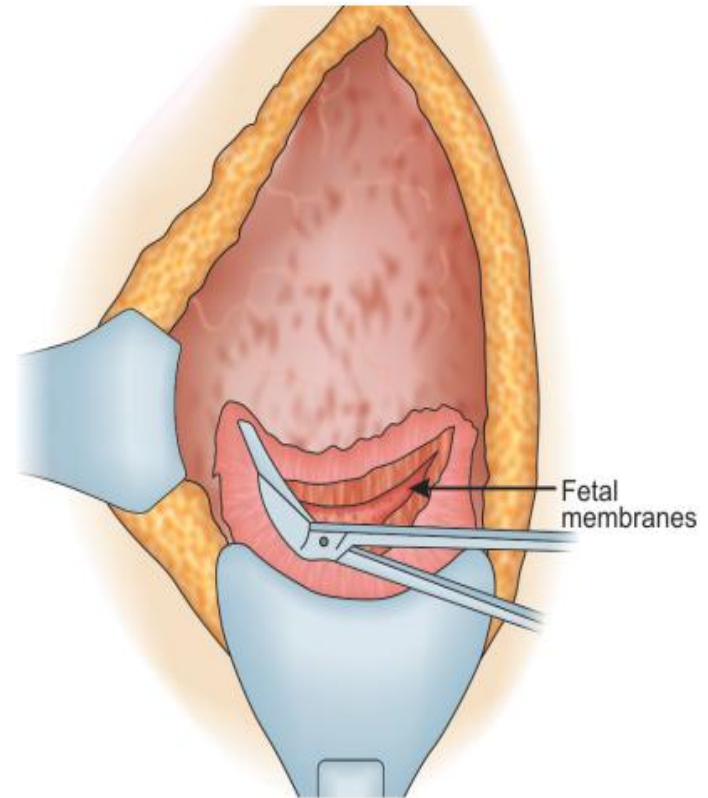
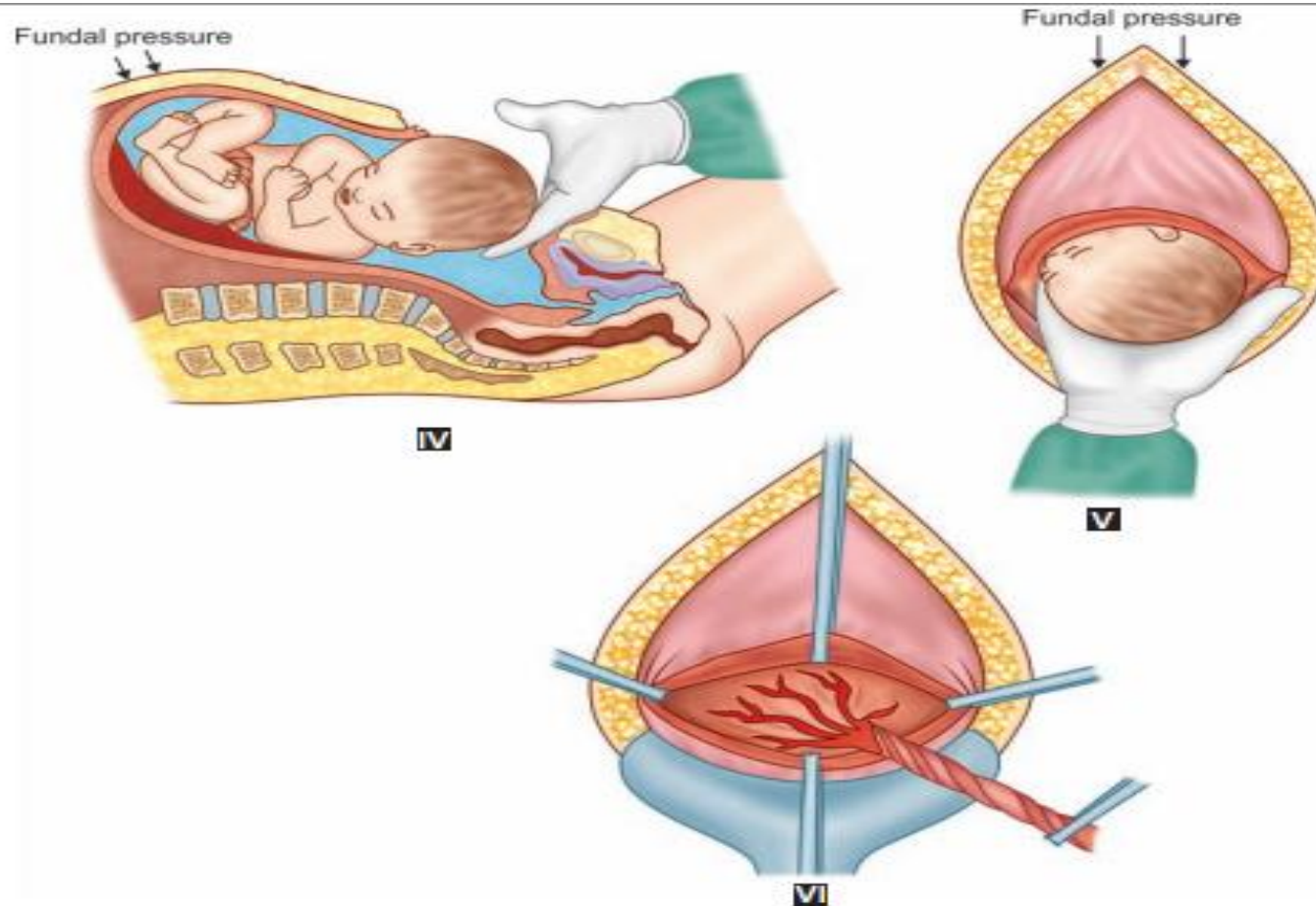


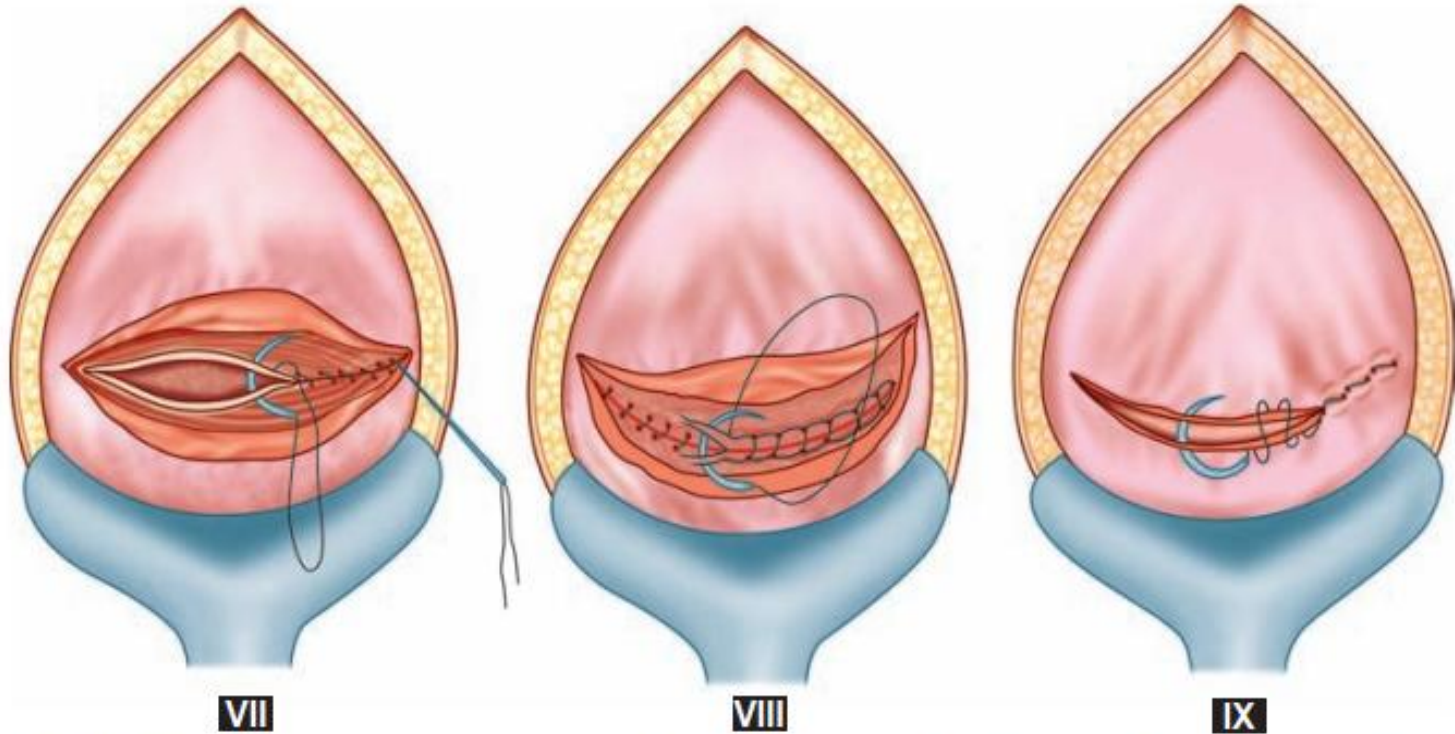
Fig. 7.3B: Extension of the uterine incision with help of scissors

Types of uterine incision for CS...



Figs 36.23IV to VI: Steps of LSCS contd. : (IV) Sagittal section showing insinuation of the fingers between the lower uterine flap and the fetal head until the posterior surface is reached; (V) Methods of delivery of the head; (VI) Placenta is being delivered.

Types of uterine incision for CS...



Figs 36.24VII to IX: Steps of LSCS contd. : (VII) Inserting the continuous catgut (No. '0') suture taking deeper muscles excluding the decidua (VIII) Similar method of continuous suture taking superficial muscles and fascia down to the first layer of suture (IX) Continuous peritoneal catgut suture.

Types of uterine incision for CS

	<i>Lower segment transverse</i>	<i>Lower segment vertical</i>
♦ Extension of incision	- may occur to involve the uterine vessels	- may occur to involve the upper segment or downward the bladder
♦ Bladder dissection	- minimal	- more when extends inferiorly
♦ Uterine closure	- easy	- difficult
♦ Muscle apposition	- good	- often imperfect
♦ Reperitonization	- complete	- often imperfect
♦ Intraoperative bleeding	- less	- more
♦ Subsequent adhesions	- less	- more

Postoperative care

- Vital signs every 15 min until awake, then every hour for 4 hours
- Check vaginal bleeding and uterus is contracted.
- Check and record urine output every 4 - 6 hours.
- Provide analgesics as required.
- Initiate breast-feeding soon as the mother is awake.
- Start sips of fluid (conscious and bowel sounds)
- Discontinue IV fluids once fluid diet is started
- Ambulate early

Induction of labor

- Iatrogenic stimulation of uterine contractions before the onset of spontaneous labor to accomplish vaginal delivery.
- Indicated when the benefits to either mother or fetus outweigh those of pregnancy continuation.
- It can be
 1. Planned(elective)
 2. Emergency.

Augmentation of labor

- Enhancement of spontaneous contractions that are considered inadequate because of failed cervical dilation and fetal descent.
- Increasing the frequency and improving the intensity of existing uterine contractions in a patient who is in labor and not progressing adequately, in order to accomplish vaginal delivery.

Indication of induction of labor

ACCEPTED ABSOLUTE INDICATIONS

Hypertensive disorders

- Preeclampsia/eclampsia

Maternal medical conditions

- Diabetes mellitus
- Renal disease
- Chronic pulmonary disease

Prelabor rupture of membranes
Chorioamnionitis

Fetal compromise

- Fetal growth restriction
- Isoimmunization
- Nonreassuring antepartum fetal testing
- Oligohydramnios

Fetal demise

Postdates pregnancy (>42 weeks)

RELATIVE INDICATIONS

Hypertensive disorders

- Chronic hypertension

Maternal medical conditions

- Systemic lupus erythematosus
- Gestational diabetes
- Hypercoagulable disorders
- Cholestasis of pregnancy

Polyhydramnios

Fetal anomalies requiring
specialized neonatal care

Logistic factors

- Risk of rapid labor
- Distance from hospital
- Psychosocial indications
- Advanced cervical dilation

Previous stillbirth

Postterm pregnancy (>41 weeks)

Contraindications to labor induction

Absolute	Relative
• Placenta previa • Vasa previa • Abnormal lie • Malpresentations • Previous uterine scar (myomectomy, CS) • Contracted pelvis • Macrosomia • Twin pregnancy • Invasive cervical cancer • Active genital herpes infection • Severe IUGR	• Bad obstetric history • Grand multiparity

Elective induction of labor

- Initiation of labor for convenience in an individual with a term pregnancy who is free of medical or obstetric indications.
- May be appropriate in specific instance
 - ⇒ History of very short labors
 - ⇒ For patients who live a great distance from the hospital

Preconditions

- Get informed consent.
- Document the indication.
- Make sure that there are no contraindications.
- Determine Bishop score (cervix score) and if unfavorable, consider cervical ripening.
- Ascertain availability of labor ward staff and also the capacity to do emergency caesarean section.

Cervical ripening

- Is the use of pharmacological or mechanical means to soften the cervix.
- Bishop score is a pre-labor scoring system to assist in predicting whether induction of labor will be required.
- The cervical ripening agent may also initiate labor.
- If not, further pharmacologic agents (i.e. oxytocin) can be used for induction.

Cervical ripening...

Bishop score				
Score Parameter	0	1	2	3
Dilatation (cm)	Closed	1 -2	3-4	≥ 5
Length (cm)	> 4	3-4	1 -2	<1
Consistency	Firm	Average	Soft	N/A
Position	Posterior	Mid	Anterior	N/A

- Interpretation of the Bishop's score:
 - ⇒ **Score ≥ 6** : Favourable - cervical condition and induction is likely to succeed. There is no need for cervical ripening. Induction using Oxytocin can be planned.
 - ⇒ **Score ≤ 5** : Unfavourable - cervix is unlikely to yield for induction; cervical ripening is needed for success with induction; postpone induction for next week if possible or use cervical ripening and plan induction for next day

Methods of cervical ripening

Pharmacological	Mechanical
• Oxytocin • Prostaglandin ⇒ E1-misoprostol ⇒ E2-dinoprostol • Estrogen • Relaxin • Hyaluronic acid • Progesterone receptor antagonists (mifepristol)	• Membrane stripping • Amniotomy • Mechanical dilators ⇒ Laminaria tent ⇒ Dilapan ⇒ Lamicel • Transcervical balloon catheters with concomitant oxytocin

Methods of cervical ripening...

Techniques	Agent	Route/Dose	Compared with Oxytocin
Pharmacological			
Prostaglandin E ₂	Dinoprostone gel, 0.5 mg (Prepidil) Dinoprostone insert, 10 mg (Cervidil)	Cervical 0.5 mg; repeat in 6 hr; permit 3 doses total Posterior fornix, 10 mg	1. Shorter I-D times with oxytocin infusion than oxytocin alone 1. Insert has shorter I-D times than gel 2. 6–12 hr interval from last insert to oxytocin infusion
Prostaglandin E ₁ ^a	Misoprostol tablet, 100 or 200 µg (Cytotec) ^b	Vaginal, 25 µg; repeat 3–6 hr prn Oral, 50–100 µg; repeat 3–6 hr prn	1. Contractions within 30–60 min 2. Success comparable to oxytocin for ruptured membranes at term and/or favorable cervix 3. Tachysystole common with vaginal doses >25 µg
Mechanical			
Transcervical 36F Foley catheter	30-mL balloon	Transcervical	1. Improves Bishop scores rapidly 2. 80-mL balloon more effective 3. Combined with oxytocin infusion is superior to PGE₁ vaginally 4. With EASI, results improved and possible decreased infection rate
Hygroscopic dilators	Laminaria, hydrogel	Intracervical	1. Rapidly improves Bishop score 2. May not shorten I-D times with oxytocin 3. Uncomfortable, requires speculum and placement on an examination table

Pharmacologic methods

- Prostaglandins

- ⇒ Results in dissolution of collagen bundles and an increase in submucosal water content of the cervix.
- ⇒ Vagina or the endocervix route-fewer side effects.
- ⇒ Side effects include fever, chills, vomiting and diarrhea

Pharmacologic methods...

- Prostaglandin E1

⇒ Vaginal

- 25 mcg if required repeat after 6 hours.
- Do not divide or cut a 200-mcg tablet into smaller pieces.

⇒ Oral

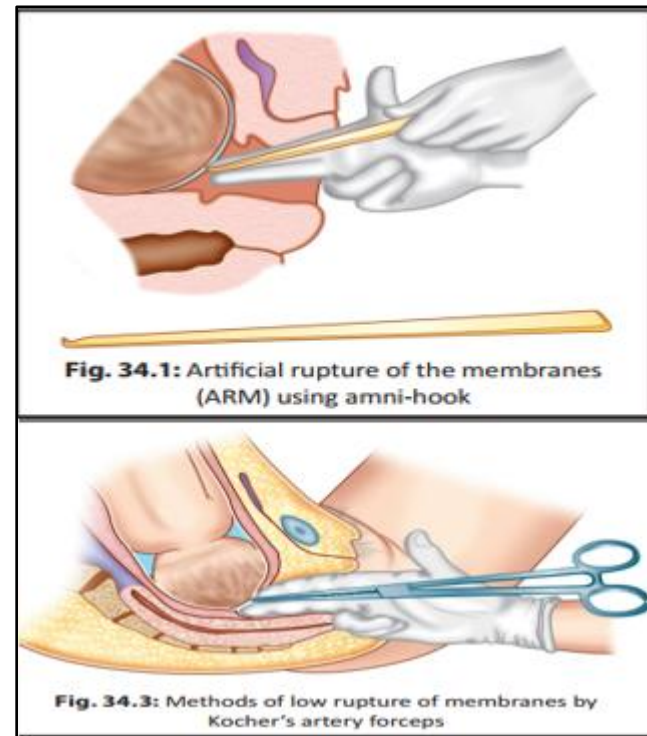
- 25 mcg; if required repeat after 3 hours.
- For patients with PROM

Mechanical agents for cervical ripening

- Mechanical agents work by directly causing cervical dilation, and by releasing endogenous prostaglandins and oxytocin.
 1. Balloon / folly catheter
 2. Osmotic dilators
 3. Amniotomy (artificial rupture of membranes)

Amniotomy

- Involving the rupture of the chorioamniotic membranes.
- Effective method of labor induction or augmentation performed in multiparous women with favorable cervixes



Amniotomy....

- Prerequisite condition

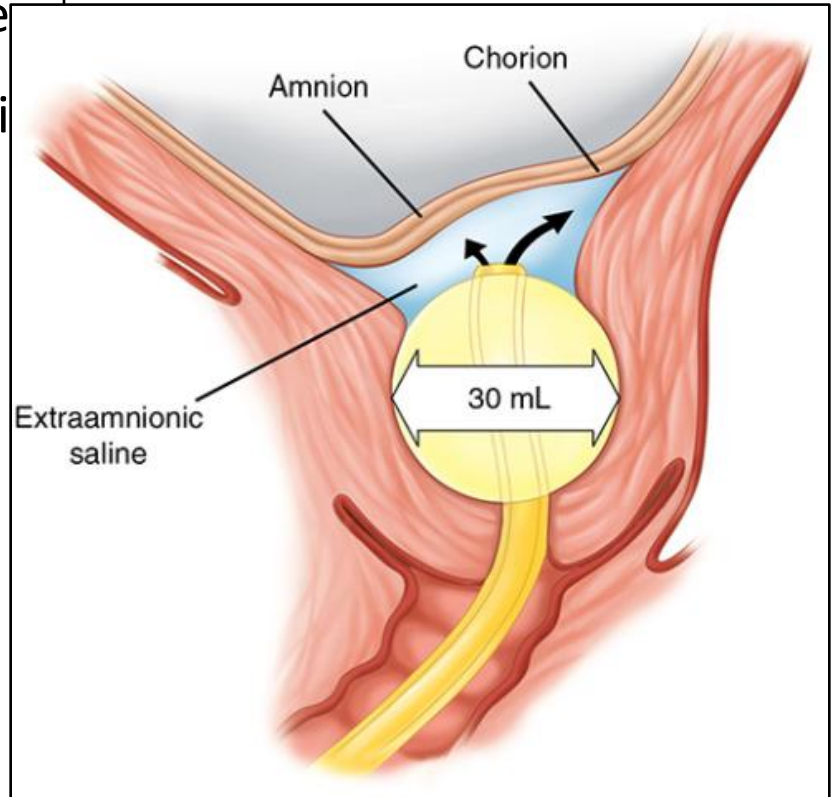
- ⇒ Fetal vertex
- ⇒ Not the umbilical cord
- ⇒ Presenting is well applied to the cervix

- Contraindicated in pregnancy

- ⇒ Unengaged presenting part
- ⇒ Vasa previa
- ⇒ Placenta abruption
- ⇒ RVI pregnant mother
- ⇒ Any contraindications to vaginal delivery

Balloon catheters

- A 16 French 30 ML balloon, can be passed through an undilated cervix into the extra-amniotic space and then inflated.
- Pressure may be applied by attaching the catheter under tension to the patient's inner thigh.



Balloon catheters....

- After insertion, leave the Foley catheter
 - ⇒ Until it is spontaneously expelled or
 - ⇒ keep it in place for at least 12 hours, or
 - ⇒ Until contractions begin

Mechanical methods

Advantages	Disadvantages
• Often less costly • Less hyperstimulation • Easy to store • Fewer side effects for mother and fetus	• Risk of infection • Disruption of a low-lying placenta • Some maternal discomfort on manipulation of the cervix.

Oxytocin induction

- More effective in augmenting labor than in inducing labor, and even less successful as a cervical ripening agent.
- Amniotomy should be performed where feasible before starting infusion.
- Starting dose should be 5IU in primigravida and 2 IU in multigravida in 1000ml NS or RL
- The infusion rate is increased every 30 minutes

Oxytocin induction...

- Aim to maintain strong 3-5 contractions are achieved in 10 min.
- Label the bag and keep timely record of the drops used.
- Monitor mother, fetus and labor according to the labor protocol.
- Record maternal and fetal conditions and progress of labor.
- Continue the oxytocin infusion for at least one hour after delivery.
- Discontinue if there is distress.

Oxytocin induction...

Oxytocin Dose	Oxytocin Concentration (mIU/mL)	Time since induction(hours)	Drip Rate:(Drops/ Minute)	Approximate Dose(mIU / minute)	Total volume infused (ml)
First dose: 2 IU of oxytocin in 1000 ml fluid	2 mIU/mL	0:00	20	2	0
		0:30	40	4	30
		1:00	60	6	90
		1:30	80	8	180
Second dose: Add another 2 IU of oxytocin to the remaining first dose fluid	5 mIU/mL	2:00	50	12	300
		2:30	60	15	375
		3:00	80	20	465
Third dose: Add another 2 IU of oxytocin to the remaining second dose fluid	10 mIU/mL	3:30	50	24	590
		4:00	60	30	665
		4:30	80	40	760
	As above	5:00	As above	As above	880

Failed induction

- Failure to achieve regular(e.g. every 3 minutes) contractions and cervical change after at least 6 - 8 hours of the maintenance dose of oxytocin administration, with artificial rupture of membranes if feasible.

FMOH

Complications of induction

Failed induction

Increased risk of caesarean section

Atonic PPH

Iatrogenic prematurity

Uterine hyper stimulation/ tetanic contractions

Uterine rupture

Fetal distress

Placental abruption

Water intoxication

Unit -III:Family planning (FP)

- Objective

- ⇒ Define family planning and contraceptive
- ⇒ Explain prevalence
- ⇒ Describe WHO tier and medical eligibility criteria
- ⇒ Explain classification of contraceptive

Family planning and contraceptive

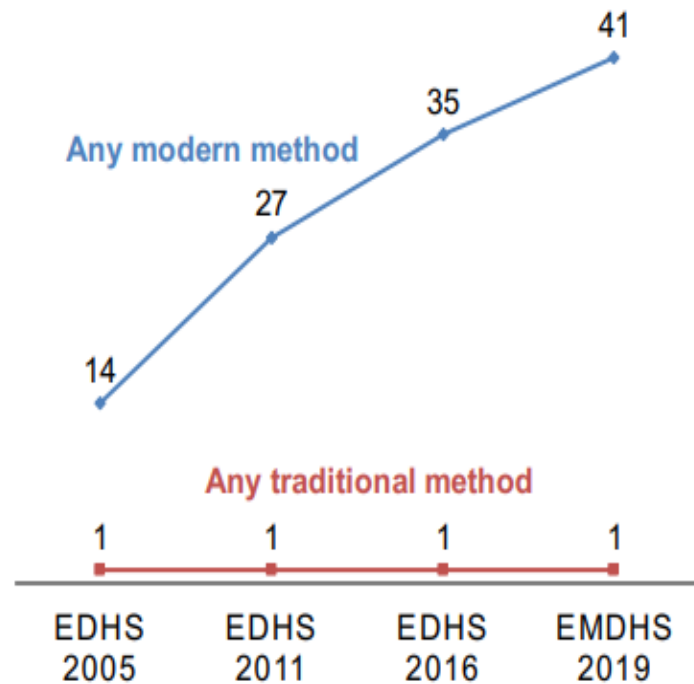
- Family planning
 - ⇒ A conscious effort by couples to anticipate and attain their desired number of children & spacing and timing of their births.
 - ⇒ Achieved through use of contraceptive methods
- Contraceptive
 - ⇒ Is all measures designed to prevent pregnancy due to coital act.

Prevalence

- Overall, 41% of currently married women are using modern methods
- Most popular contraceptive
 1. Injectable (27%)
 2. Implants (9%)
 3. Pill and the IUD (2% each).

Figure 1 Trends in contraceptive use

Percentage of currently married women using contraceptive methods



Criteria of ideal contraceptive

Widely acceptable

Inexpensive

Simple to use

Safe

Highly effective

Requiring minimal motivation maintenance and supervision.

WHO medical eligibility criteria classification categories

Classification	With clinical judgment	With limited clinical judgment
1	• Use method in any circumstances	• Yes • Use the method
2	• Generally use ⇒ Advantages outweigh risks	
3	• Generally do not use ⇒ Risks outweigh advantages	• No • Do not use the method
4	• Method not to be used	

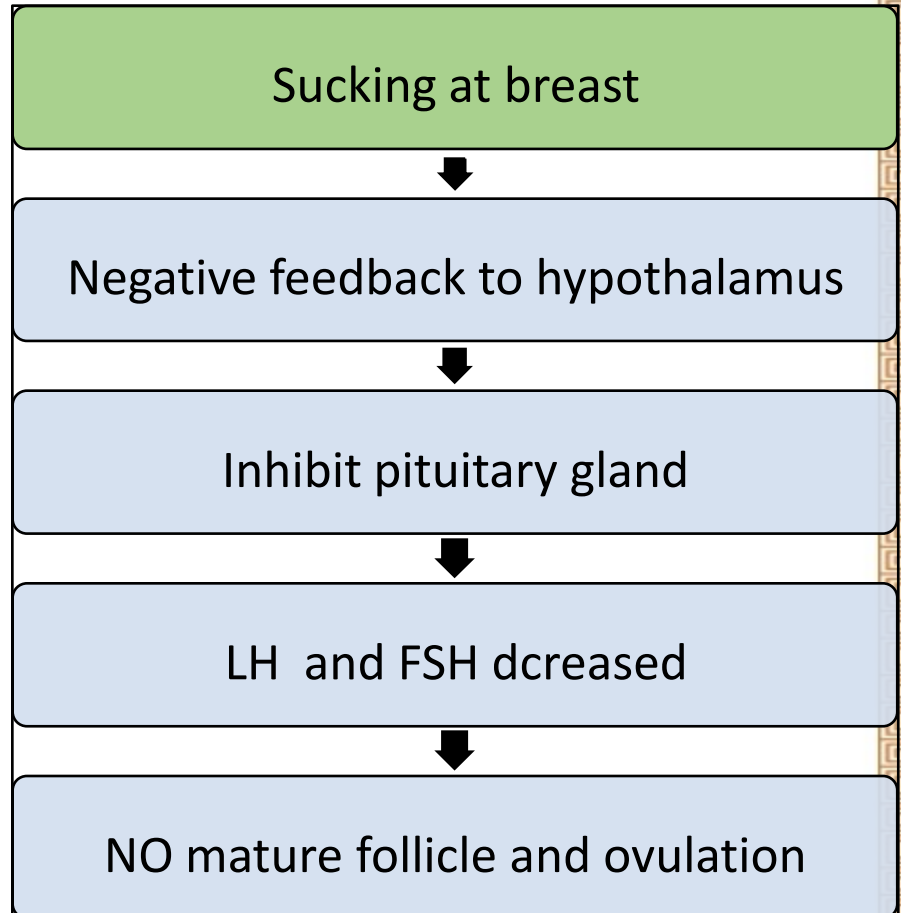
Classification of contraceptive

Criteria	Method	Example
Content	Modern	Sterilisation,IUD, implants, injectables, the pill, condoms, emergency contraception, standard days method (SDM), and lactational amenorrhoea method (LAM).
	Traditional	- Fertility awareness method ⇒ Rhythm /calendar method ⇒ Symptom based methods - Withdrawal method
Hormonal content	Hormonal	Implants, injectables, the pill
	No hormonal	Copper IUCD
Reversibility	Temporary	Hormonal,LAM, Fertility awareness method
	Permanent	Vasectomy and tubal ligation
Duration	Short acting	Oral contraceptive pills, progesterone only pills
	Long acting	Implant ,IUCD

	WHO tier	Method		Perfect use	Typical use
1	Top Tier Most Effective	• Intrauterine devices	• Levonorgestrel system	0.2	0.2
			• T 380A Copper	0.6	0.8
		• Levonorgestrel implants		0.05	0.05
		• Female sterilization		0.5	0.5
		• Male sterilization		0.1	0.15
2	Second Tier: Very Effective	• Combination pill		0.3	9
		• Vaginal ring		0.3	9
		• Patch		0.3	9
		• DMPA		0.2	6
		• Progestin-only pill		0.3	9
3	Third Tier: Effective	• Condom	Male	2	18
			Female	5	12
		• Diaphragm with spermicides		6	12
		• Fertility-awareness			24
4	Fourth Tier: Least Effective	• Spermicides		18	28
		• Sponge	Parous Women	20	24
			Nulliparous women	9	12
	No WHO Category	• Withdrawal		4	22
		• No contraception		85	85

Lactational amenorrhea method(LAM)

- A temporary family planning method based on the natural effect of breastfeeding on fertility
- Effectiveness less than 1 pregnancy per 100 women using LAM in the first 6 months after childbirth.



LAM...

- All 3 conditions must be met:
 1. The mother's monthly bleeding has not returned
 2. The baby is fully breastfed and is fed often, day and night
 3. The baby is less than 6 months old
- Feeding interval shouldn't exceed 4 hours during the day and 6 hours during the night

Start another family planning methods when:

- Her menstrual period return or
- She stop fully or nearly fully breast feeding or
- Her baby is 6 month old.
- She no longer wants to rely on LAM for family planning

Fertility awareness methods

Rhythm method

Standard days method

Cervical mucus method

Basal body temperature methods

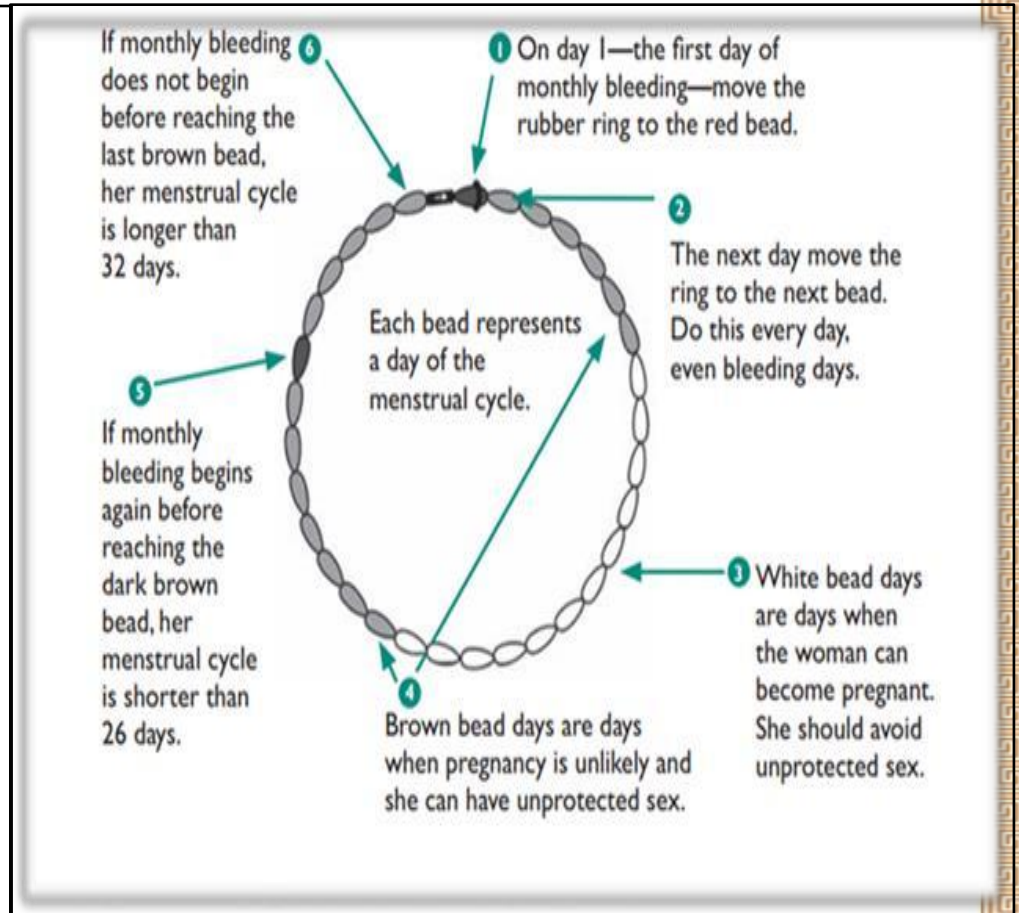
Withdrawal method

Rhythm method

- Is a method by which a woman calculates the fertile days of her menstrual period and the couple avoids vaginal sex, or uses temporary methods during the fertile time.
 - Woman records the number of days in each menstrual cycle for at least 6 months.
 - The first day of monthly bleeding is always counted as day 1.
- ⇒ Shortest cycle -18 = ~first day of her fertile time
- ⇒ Longest cycle -18 = ~last day of her fertile time

Standard days method..

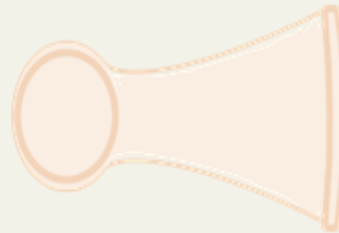
- Is a calendar-based method which can be used by women who have regular cycles that are 26 – 32 days long
- Identifies days 8-19 of the cycle as fertile
- Effectiveness is about 5 pregnancies per 100 women



BARRIER METHODS



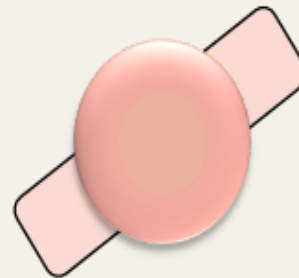
Condom
85-98% Effective



Female Condom
70-90% Effective



**Diaphragm or
Cervical Cap**
71-94% Effective



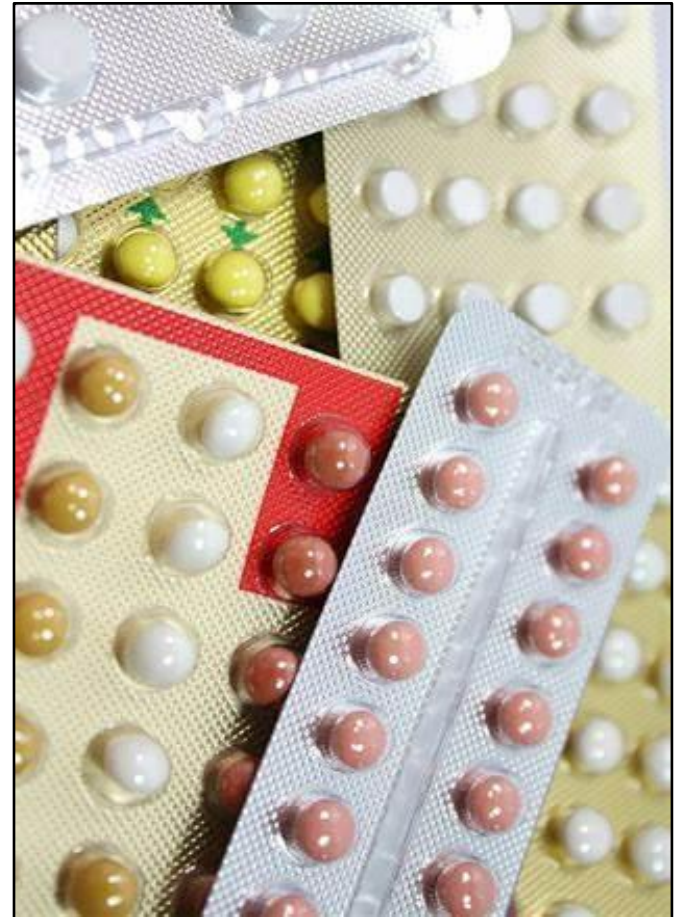
Sponge
76-88% Effective



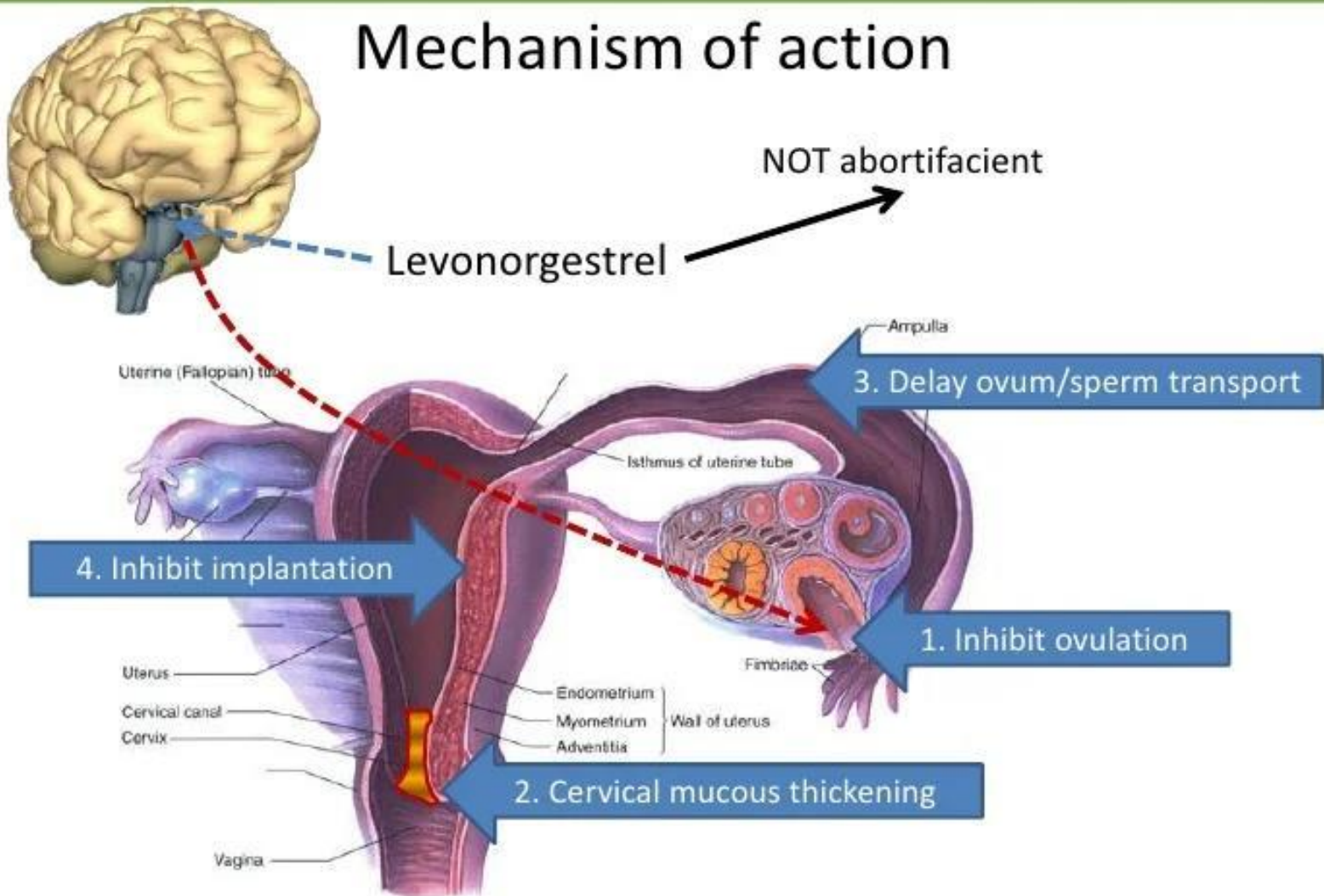
Cervical Cap

Combined oral contraceptives

- Also called “the Pill,” low-dose combined pills, OCPs, & OCs.
- Contain low doses of estrogen and progesterone
- Women who are infected with HIV, have AIDS, or are on ARV therapy can safely use COCs.



Mechanism of action



When to start

Having menstrual cycle	• With in 5 days --- No need of backup • Delayed > 5 days ---- Backup
Switching from IUD	• Start immediately, no backup
From hormonal method	• If use was consistent & correct start at the next schedule & no backup
Fully breast feeding	• Start at 6wks or when supplementation started • >6wks & no menses--- R/O pregnancy + backup
Partial breast feeding	• <3wk---start at 3wkswk • >3wk & no menses ---R/O pregnancy + backup
Non breastfeeding	• <3wks---start anytime b/n 21-28 days • >3wk & no menses---R/O pregnancy + backup
After miscarriage	• <7days ---no backup • >7days----backup
After ECPs	• The day ECPs finished + backup
No monthly bleeding	• R/O pregnancy + backup

Combined oral contraceptives...

How to Take COCs:

Schedule and Missed Pills

Schedule:

- Take one pill every day
- 21-day packs → 7-day break
- 28-day packs → no break between packs

Missed pill:

Missed 1 or 2
active pills



- Take missed pill as soon as remembered
- Keep taking other pills on schedule
- No backup method needed

Source: WHO, 2004.

Combined oral contraceptives...

How to Take COCs: Missed Pills

Miss 3 or more active pills or start pack 3 or more days late

- Take first missed pill as soon as you remember
- Continue daily pill taking as usual and use backup method or abstain for next 7 days
- Count number of active pills remaining in pack

7 or more active pills left in the pack

- Finish active pills
- Take hormone-free break

Fewer than 7 active pills left in the pack

- Finish active pills
- Discard inactive pills
- Start new pack immediately

Source: WHO, 2004.

Non contraceptive benefits of COCs

COCs decrease the risks of

Benign breast disease

Functional ovarian cyst

Anemia

Symptomatic PID

Endometrial & ovarian cancer

Dysmenorrhea

Endometriosis

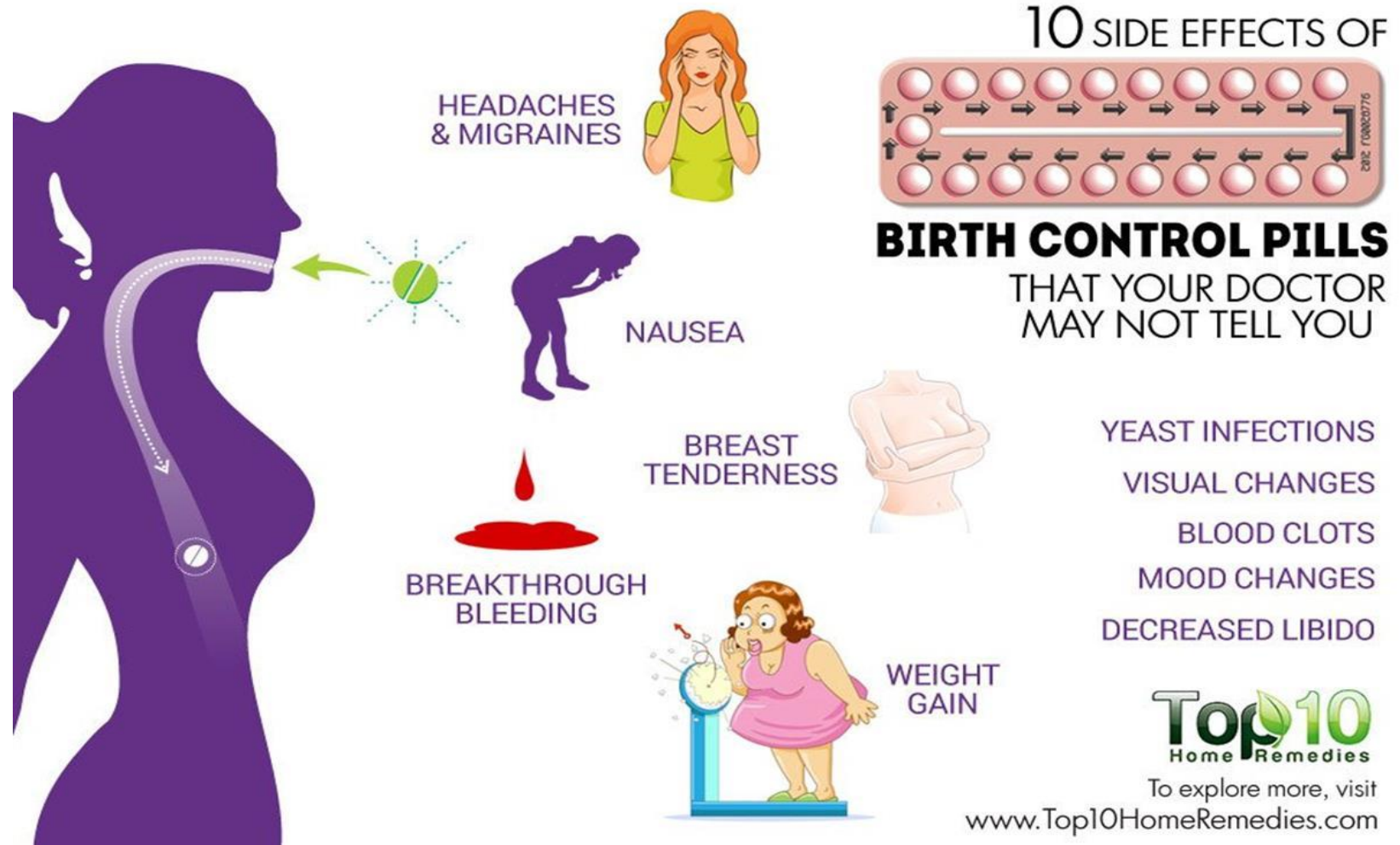
Ectopic pregnancy

Arthritis & osteoporosis

Who should not use cocs

- < 6 weeks post partum if breast feeding(< 3 wks not breast feeding)
- Age ≥ 35 and smoker
- SBP >160 & DBP >100
- DVT or pulmonary embolism
- ↑ risk of cerebrovascular disease
- Cervical and breast cancer
- Liver disease(cancer)
- Migraine head ache
- Woman taking carbamazepine, phenytoin, rifampicin....

Side effects & complications of COCs



Progesterone only pill /POP

- Contain very low doses of a progestin like the natural hormone
- Are also called “minipills” and progestin-only contraceptives
- Doesn't affect breast feeding
- Mechanism of action
 - ⇒ Thickening cervical mucus (blocks sperm from meeting egg)
 - ⇒ Suppress luteinizing hormone (LH) ⇒ suppress ovulation.
 - ⇒ Decrease fallopian tube motility

24

Progesterone only pill....

- Instruction:

- ⇒ 28 pill pack, 1 pill to be taken daily.
- ⇒ Can be started immediately after delivery if not breast feeding.
- ⇒ Pills are taken every day at the same time
- ⇒ For those on breast feeding can be started at 6 weeks

Side effect and advantage of POP

Benefit	Side effect
• Can be given during breast feeding • No estrogen • Decreases ⇒ Dysmenorrhea ⇒ Menstrual blood loss ⇒ Premenstrual syndrome • Fertility doesn't delay after cessation • Doesn't interfere sex • Are controlled by the woman	• Change in bleeding pattern • Needs continuous complains • Minor symptoms • Headaches • Dizziness • Mood changes • Breast tenderness • Abdominal pain • Nausea

Managing missed pill

- Being three or more hours late, take the missed one as soon as possible and then as the schedule
- If women has regular bleeding
 - ⇒ Backup method
 - ⇒ ECPs
- Sever vomiting within 2hrs after taking the pill
 - ⇒ Take another pill from the pack

Injectables preparations

- Progesterone only [can be started immediately after delivery if not breast feeding.
- For breast feeding mother should be started after 6 weeks
 - ⇒ **Depo-provera 150 mg every 90 days IM**
 - ⇒ Noristerat/NET-EN 200 mg every 2 months IM
 - ⇒ Depo-subQ provera 104: SC every 3 months

Injectables preparations...

- Characteristics and side effects:
 - ⇒ Independent of coitus
 - ⇒ Independent of daily activity
 - ⇒ May cause irregular uterine bleeding
 - ⇒ May cause amenorrhea
 - ⇒ Delay in fertility after discontinuation of around 9 months
 - ⇒ May cause weight change, headache, dizziness and fatigue

Depo-Provera (Injectable)

- Injection of 150 mg DMPA every 3 months
- Women of any age and parity can use it
- Who should not use among
 - ⇒ Women with breast cancer
 - ⇒ Women with vaginal bleeding



Depo-Provera (Injectable)....

Health benefits	Side effects
• Helps protect against: ⇒ Risks of pregnancy ⇒ Endometrial cancer ⇒ Uterine fibroids ⇒ Symptomatic PID ⇒ Iron-deficiency anemia ⇒ Sick cell crises ⇒ Symptoms of endometriosis	• Changes in monthly bleeding. • Weight gain • Headaches • Dizziness • Abdominal bloating and discomfort • Mood changes • Less sex drive

Implant

- Small plastic rods, matchstick size, that release a progestin like the natural hormone progesterone in a woman's body.
- Are implanted subdermally under the skin of the upper arm and release hormone over many years

Recently available implants	
Jadelle	• 2 rods and Lasts 5 years • 75 mg of levonorgestrel
Implanon Classic and Implanon NXT	• 1 rod and radiopaque • Lasts 3 year • 68 mg of etonogestrel
Trust implant	• 2 rods • Lasts 4 years
Sinoplast:	• 2 rods & Lasts 5 years • 150 mg of levonorgestrel

Safety

- No delay in return of fertility after removal
- No protection against sexually transmitted infections
- Do not increase frequency of ectopic pregnancy.
- Effectiveness may be reduced when co- administered with antibiotics ,anti-fungals,anticonvulsants,ART(PI)...

Non-contraceptive health benefits

- Non-contraceptive health benefits include:-
 - ⇒ Help prevent ectopic pregnancy
 - ⇒ Help protect against pelvic inflammatory disease
 - ⇒ Help prevent iron deficiency anaemia
 - ⇒ May prevent endometrial cancer
 - ⇒ May reduce sickle crises in women with sickle cell anaemia.

Who can not use Implants ?

Generally avoid in case of

Serious liver disease

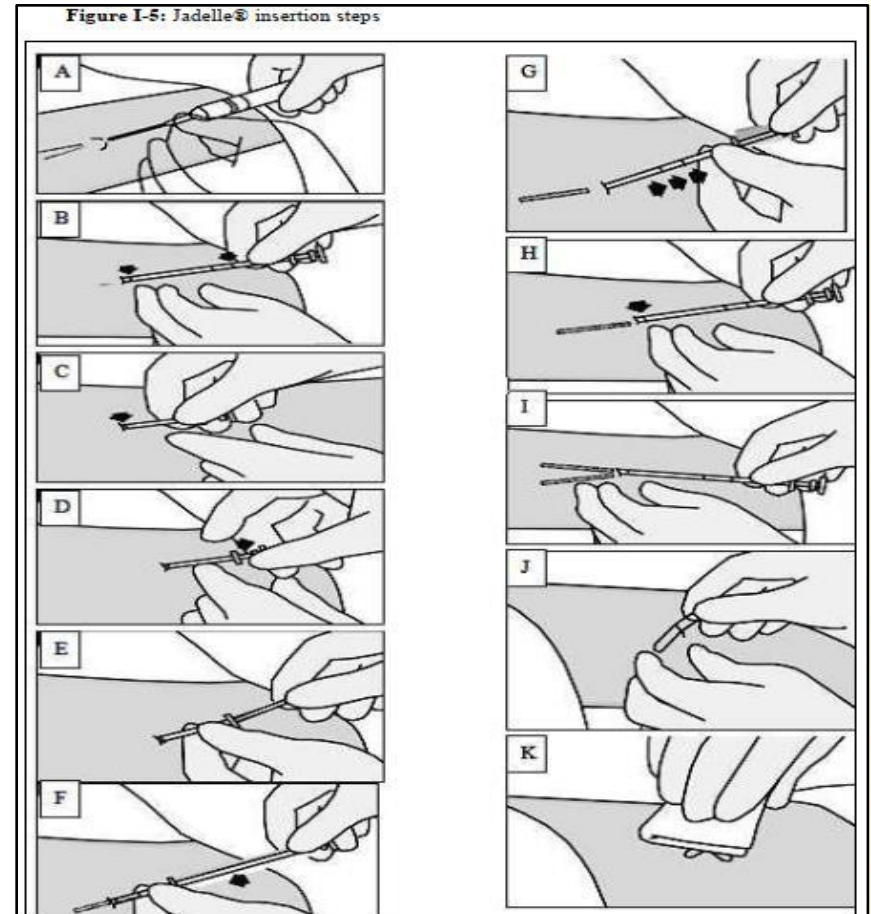
Current DVT

Unexplained vaginal bleeding

Breast cancer, current or History

Timing of insertion

- A woman can start using implants any time she wants if it is reasonably certain she is not pregnant.



Side effects and complications

Side effect	Complications
• Changes in bleeding patterns • Headaches • Breast tenderness • Mood changes • Nausea • All of which usually decrease over time.	• Procedure site problems (early) ⇒ Bleeding / hematoma ⇒ Expulsion ⇒ Infection / cellulites / abscess

Intrauterine contraceptive device

- Small flexible device made of metal and/or plastic, placed in the uterine cavity
- Prevent pregnancy upto 12years
- Type
 - ⇒ Copper – bearing(Cu-T 380A)
 - ⇒ Medicated:levonorgestrel containing (Mirena)



Mechanisms of action



Levonorgestrel-Releasing IUD (LNG-IUS, Mirena®)

- Inhibits fertilization
- Thickens cervical mucous
- Inhibits sperm function
- Thins and suppresses the endometrium

Copper-Releasing IUD (ParaGard® T380A)

- Inhibits fertilization
- Releases copper ions (Cu^{2+}) that reduce sperm motility
- May disrupt the normal division of oocytes and the formation of fertilizable ova

Intrauterine contraceptive device...

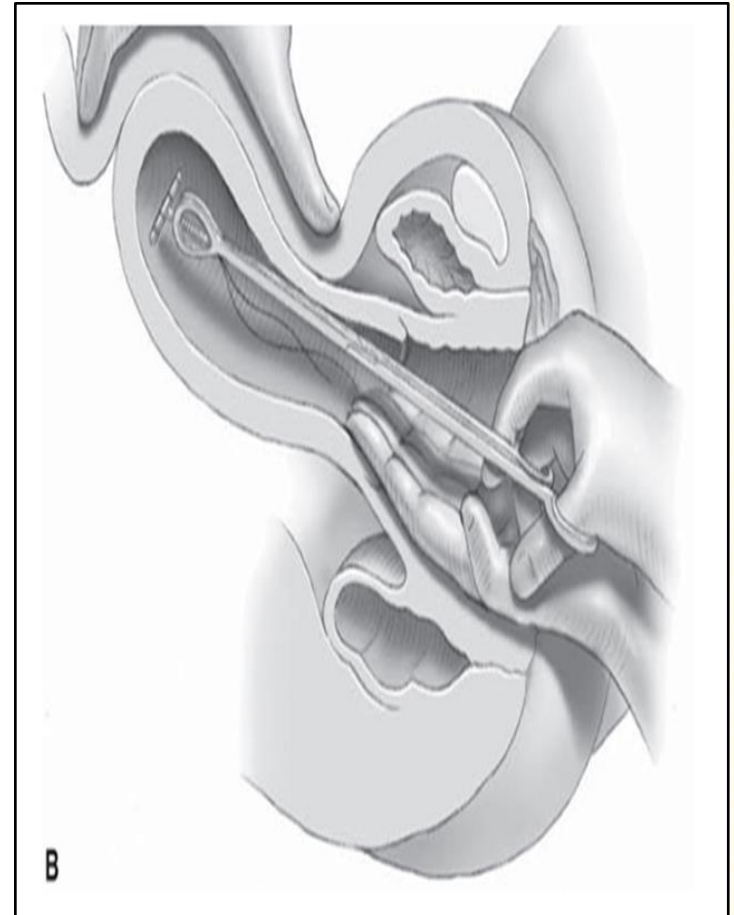
- Allowing women privacy & control over their fertility.
- Does not interact with medications.
- Can be removed whenever the client chooses.
- Does not protect against STIs/HIV.
- Trained provider dependent.

Intrauterine contraceptive device...

❖ Used safely by woman who	❖ Not appropriate for women who
• Healthy reproductive tract • Have precaution(s) for hormonal methods such as COCs. • Is breastfeeding • Is immediately postpartum • Is post abortion. • HIV positive • Are at low or no risk for STI	• Pregnancy (known or suspected) • Unexplained vaginal bleeding • Post partum between 48hrs-4wks • Current pelvic infection (puerperal, post abortal,TB & STI) • GTD or cervical/endometrial cancer • Uterine cavity distortion (myoma or congenital) • AIDS cases (clinically not well)

Timing of insertion

- Any time she wants(if not pregnant).
- Optimal times for insertion are
 - ⇒ Within 12 days of menses
 - ⇒ within 12 days after abortion
 - ⇒ If <48 hours post partum or > 4 wks
- Switching from another method
- For emergency contraception



Timing for IUCD removal

Client requests

?Uterine perforation.

?Pregnancy

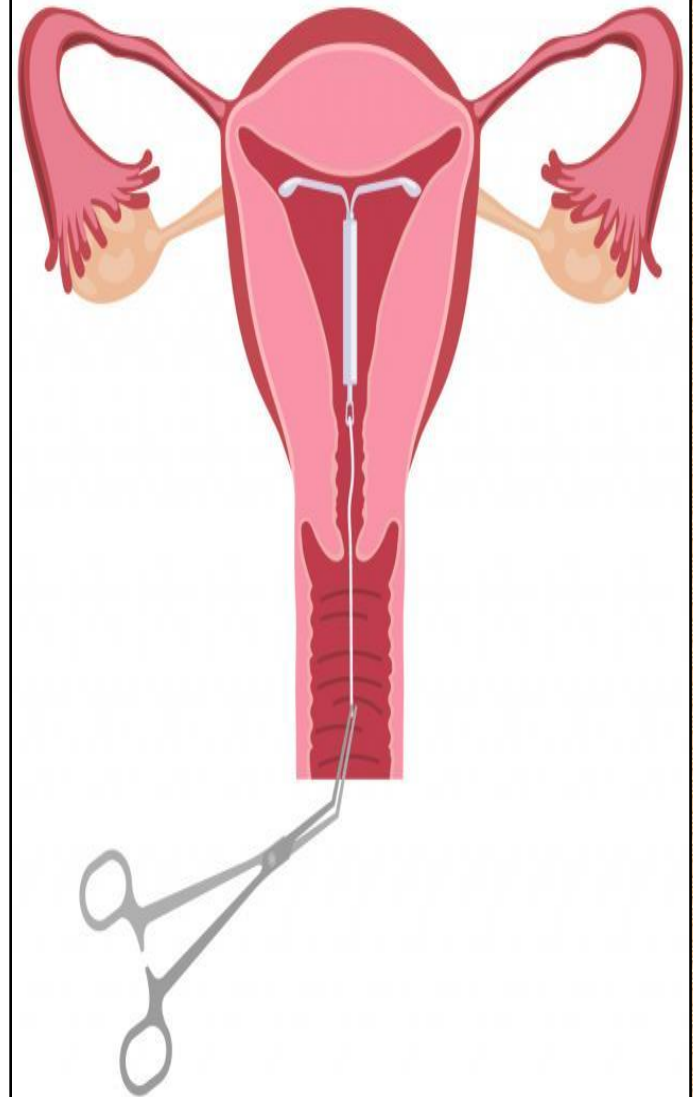
Partial expulsion

Persistent side effects

Severe pain

Sever bleeding

Marked anemia



Side effect and complication

Possible side-effects

*If you choose this method, you may have some side-effects.
They are not usually signs of illness.*

After insertion:

- Some cramps for several days
- Some spotting for a few weeks



Other common side-effects:

- Longer and heavier periods
- Bleeding or spotting between periods
- More cramps or pain during periods



May get less after a few months

*How would you feel
about these side-effects?*



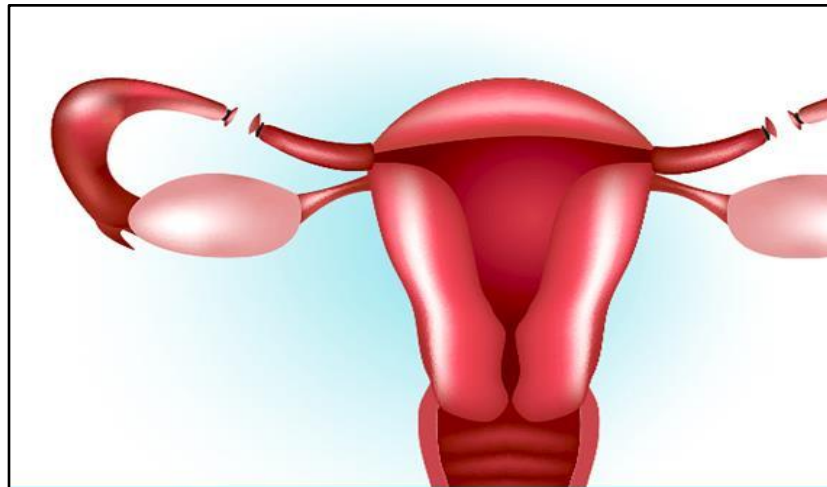
Complications

- Syncope attack (bradycardia, fainting, dizziness...)
- Possible anemia
- Missing strings.
- Expulsion of IUCD
- Pelvic inflammatory disease
- Suspected uterine perforation

Permanent methods

- A surgical method where by the reproductive function of an individual is purposefully & permanently destroyed
- Couple need to be adequately informed before procedure
- Inform on individual procedure in terms of benefit, risks, side effects, failure rate & reversibility
- Can be
 1. Vasectomy in male
 2. Tubal occlusion in female

1. TubaL ligation

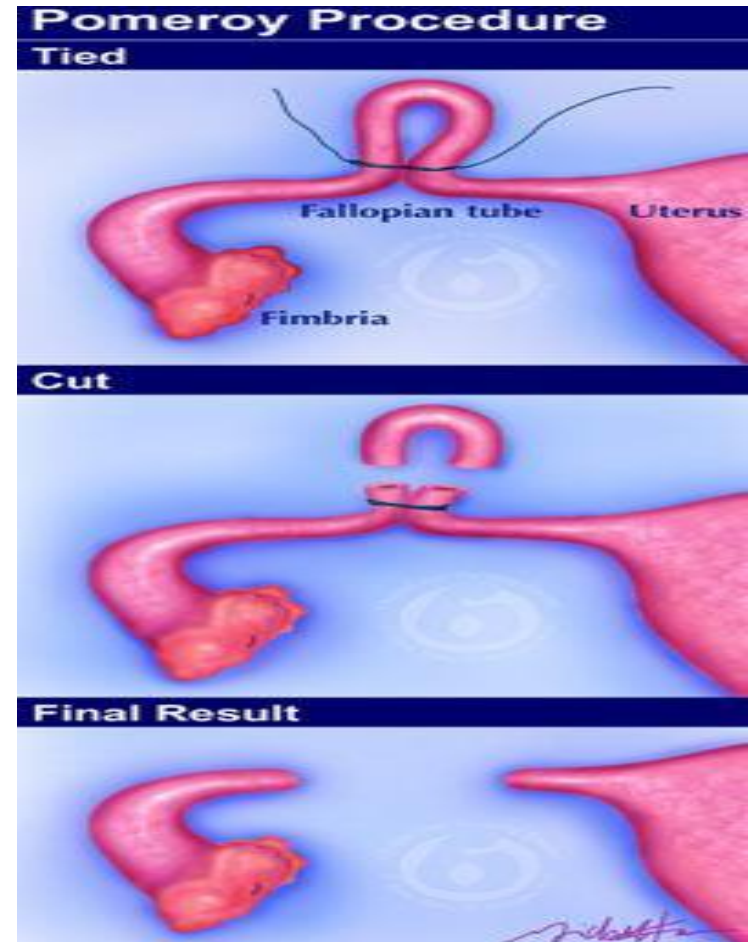


HOW DOES TUBAL LIGATION WORK

Prevents pregnancy by blocking the connection between the ovaries and the uterus

This prevents sperm from reaching and fertilizing the egg

- 0.5 pregnancies per 100 women



2. Vasectomy

- Is the most effective mode of permanent male contraception
- Involves interruption or occlusion of each vas deferens.
- Is highly effective-failure rate of 0.2-0.4%
- Not effective in first three month(need backup method)

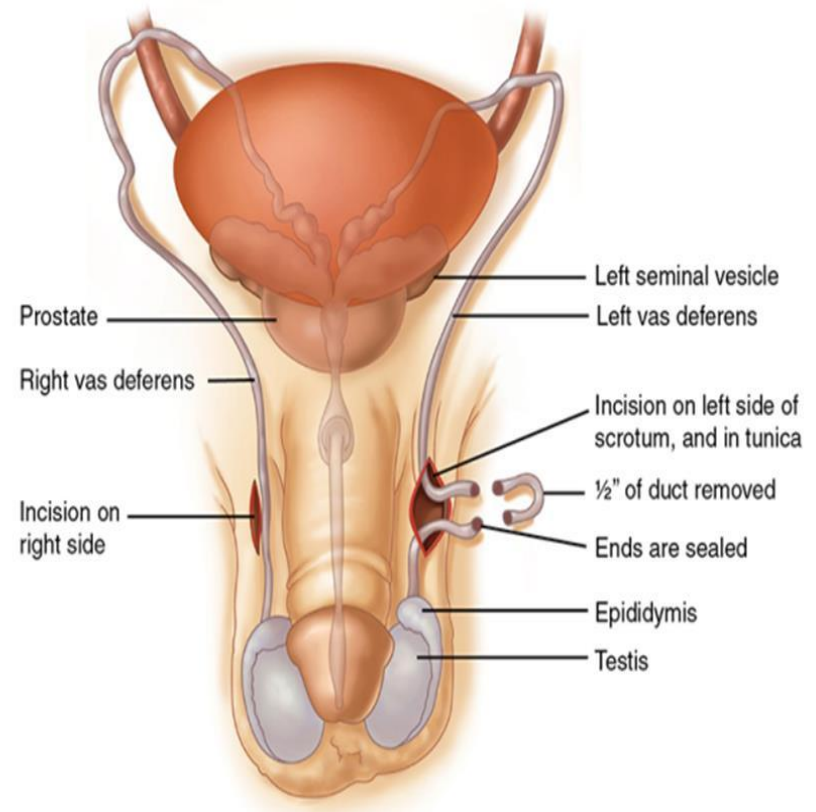


FIGURE 39-5 Anatomy of male reproductive system showing procedure for vasectomy.

Emergency contraception

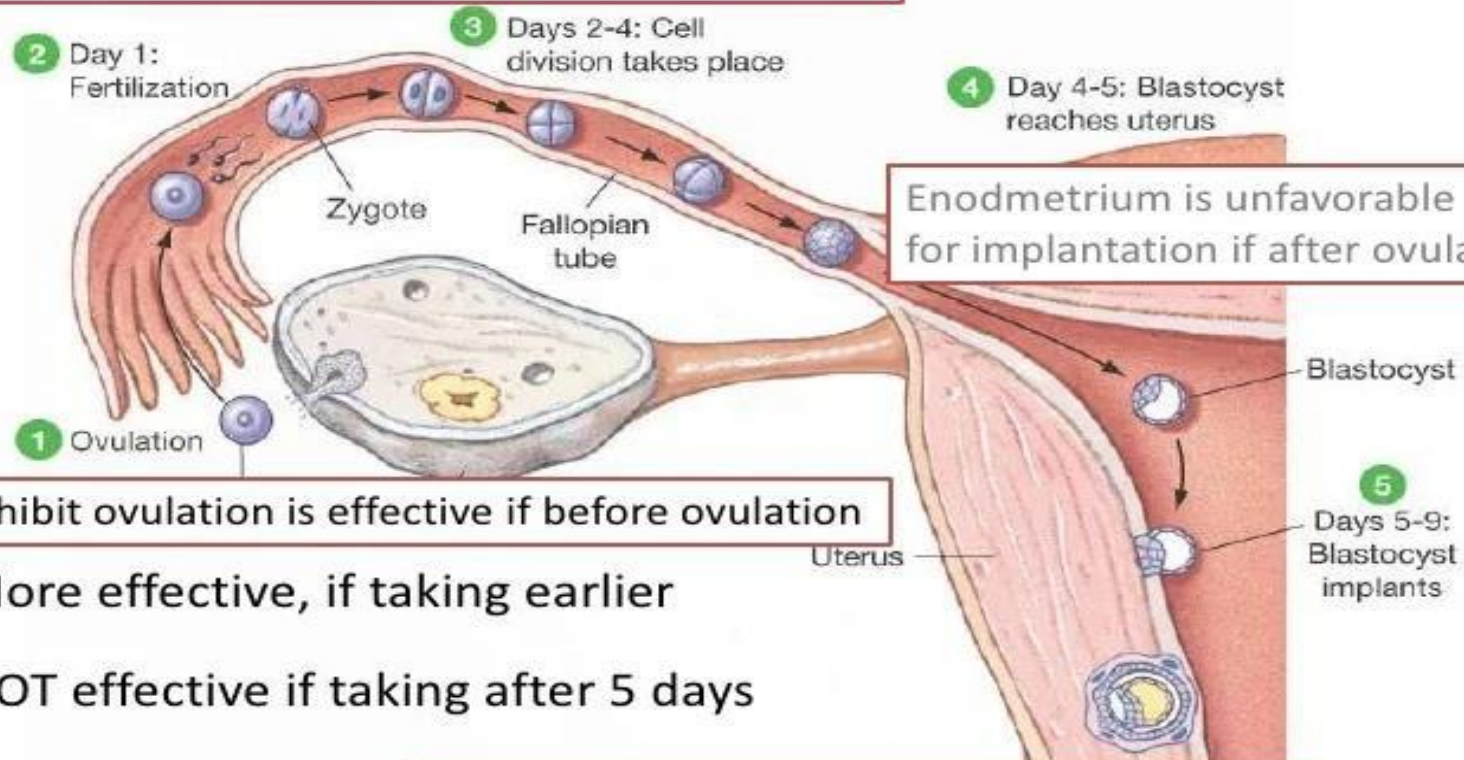
- “Morning after” pills or postcoital contraceptives.
- Taken following unprotected sexual intercourse within 5 days to prevent pregnancy.
- The sooner they are taken, the better.

Type of emergency contraception

		
Type	Copper IUD	Morning-after pill
Efficacy	• Up to 98%	• Up to 95%
When to insert/consume	• Must be inserted within 5 days after sex to have a chance of preventing pregnancy	• Within 24 hours of sex • Effectiveness decreases to 85% when taken between 25 - 48 hours after sex • Effectiveness rate of 58% when taken 48 hours after
Where to get it	• From a clinic or hospital	• From a clinic or hospital

Mechanism of action

Delay ovum/sperm transport if after ovulation



Endometrium is unfavorable for implantation if after ovulation?

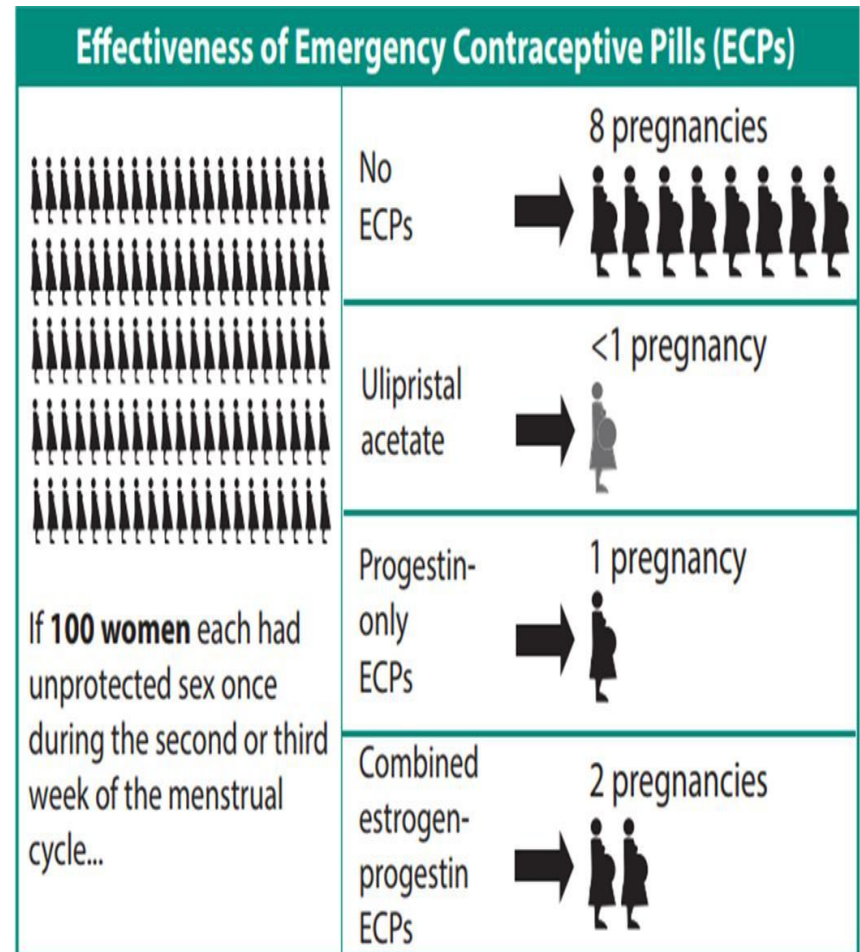
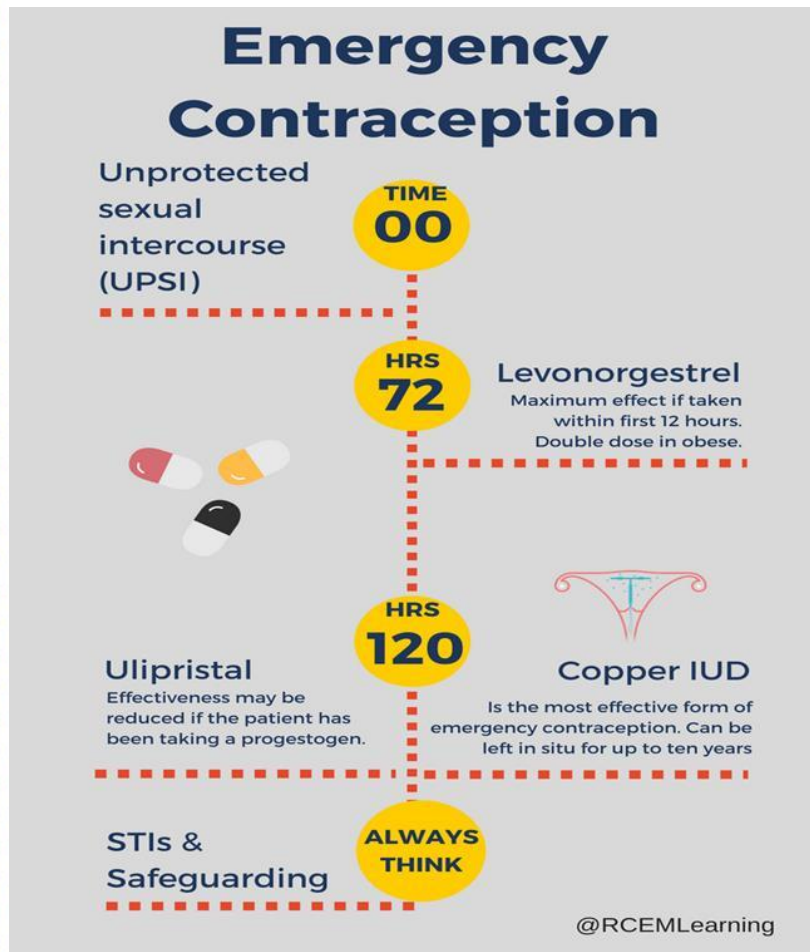
Inhibit ovulation is effective if before ovulation

More effective, if taking earlier

NOT effective if taking after 5 days

Cervical mucous thickening if after ovulation

Emergency contraception...





Side Effects

Some women may experience some side effects including;

- Period that is heavier, lighter, earlier, late
- Nausea
- Abdominal Pain
- Fatigue
- Headache
- Dizziness
- Breast Tenderness
- Vomiting



Unit IV: Abnormal puerperium

1

- Post partum hemorrhage

2

- Puerperal fever

3

- Lactational mastitis

Abnormal puerperium...

- Objective

- ⇒ Define PPH, cause presentation and management
- ⇒ Describe risk ,cause presentation and management of puerperal fever
- ⇒ Discuss on lactational mastitis(risk, cause, presentation and management)

Postpartum hemorrhage

1. A blood loss that exceeds
 - ⇒ 500 ml after a vaginal birth
 - ⇒ 1,000ml after a cesarean birth
 - ⇒ 1500 ml after hysterectomy
2. Hematocrit Change – defined as change $> 10\%$
3. Hemodynamic instability(vital sign deraignment)
4. Need for transfusion – variable practice

Classifications

Primary, early or acute PPH	Secondary or Late PPH
• Delivery - < 24hours PP • 90% PPH cases • Associated with more bleeding	• Affects 1-3 % of all deliveries • Common causes in the first 3 weeks include: ⇒ Sub-involution of the uterus ⇒ Intrauterine Infection (e.g Endometritis) ⇒ Retained pieces of placental tissue(RPOC)

Causes PPH

Early	Late
• Uterine atony • Lower genital tract lacerations • Upper genital tract lacerations • Lower urinary tract lacerations • Retained products of conception • Invasive placentation • Uterine rupture • Uterine inversion • Coagulopathy	• Infection • Retained products of conception • Placental site subinvolution • Coagulopathy

Uterine atony

- Also called uterine inertia
- Inability of the uterine myometrium to contract effectively.
- The most common cause of early postpartum hemorrhage
- The most typical causes in the **1st 4 hours**
- **75-90%** PPH
- Bleeding is from placental implantation site

Risk factor of uterine atony

- Uterine over distention
 - ⇒ Multiple pregnancy
 - ⇒ Macrosomia
 - ⇒ Polyhydramnios
- Weakness of uterine muscle
 - ⇒ Prolonged Labor
 - ⇒ Rapid Labor
 - ⇒ Induced/augmented labor
 - ⇒ Grandmaltiparity
 - ⇒ Advanced age(>35)
 - ⇒ Intrinsic myometrial dysfunction
 - ⇒ General anesthetics- Halothane
- Functional/anatomic distortion of the uterus
 - ⇒ Fibroid uterus(myoma)
 - ⇒ Abnormal placentation
 - ⇒ Uterine anomalies
- Poorly perfused myometrium-hypotension
 - ⇒ Hemorrhage, conduction analgesia
- Couvelaire uterus
- Previous uterine atony

Diagnosis of atony

- Bimanual palpation of the uterus confirms the diagnosis.
 - ⇒ Soft or “ boggy” feel when fundus is located
 - ⇒ Uterine fundus that is located above expected level
 - ⇒ Uterus that becomes firm when massaged but loses its tone
 - ⇒ Uterus not well contracted
 - ⇒ Soft & Lax uterus

Uterine atony management

- Shout for help
- Rapid initial assessment(ABC)
- Left lateral position
- Treat shock if she develop
- Massage uterus to expel blood and tissues
- Give 10 IU oxytocin
- Take 5ml blood for laboratoy
- Catheterization
- Examine the placenta and the membrane
- Treat accordingly the cause

Uterine atony management....

- Fluid replacment
 - ⇒ Use large bare cannula bilaterally
 - ⇒ 1st 1000ml rapidly in 15-20 minute
 - ⇒ At least 2000ml in the 1st hour.
 - ⇒ After she is stable,adjust rate 1000ml/6 hour.
- Aim⇒ Replace blood loss three times

Uterine atony management....

- Massage uterus
- Use uterotonic agent
- Empty bladder and breast feeding

Table 18-6. Uterotonic Therapy

AGENT	DOSE	ROUTE	DOSING FREQUENCY	SIDE EFFECTS	CONTRAINDICATIONS
Oxytocin (Pitocin)	10–80 units in 1,000 ml of crystalloid solution	First line: IV Second line: IM or IU	Continuous	Nausea, emesis, water intoxication	None
Methylergonovine (Methergine)	0.2 mg	First line: IM Second line: IU or PO	Every 2–4 h	Hypertension, hypotension, nausea, emesis	Hypertension, preeclampsia
15-Methyl Prostaglandin F _{2α} (Hemabate)	0.25 mg	First line: IM Second line: IU	Every 15–90 min (8-dose maximum)	Nausea, emesis, diarrhea, flushing, chills	Active cardiac, pulmonary, renal or hepatic disease
Prostaglandin E ₂ (Dinoprostone)	20 mg	PR	Every 2 h	Nausea, emesis, diarrhea, fever, chills, headache	Hypotension
Misoprostol (Cytotec)	600–1000 mcg	First line: PR Second line: PO	Single dose	Tachycardia, fever	None

Conservative management of uterine atony

- If pharmacologic methods fail to use
 - ⇒ Bimanual compression of uterus
 - ⇒ Aortic compression if bleeding continue
 - ⇒ Uterine tamponade
 - ⇒ Selective arterial embolization
 - ⇒ Compression sutures.
 - ⇒ Vessel ligation.
 - ⇒ Surgical intervention.

Uterine atony prevention

1. Active management of the third stage of labor(AMTSL)
2. Spontaneous delivery of the placenta after cesarean delivery
 - Oxytocin
 - Controlled cord traction
 - Massage uterus
3. Avoid risk factor

Trauma

- Is the second most frequent causes 20%
- Clinical finding
 - ⇒ Firm & well contracted Uterus
 - ⇒ Bright red persistent bleeding,oozing,or frank hemorrhage
- Careful inspection of vagina,cervix, perineum
- Treated with repaired

Retained placenta

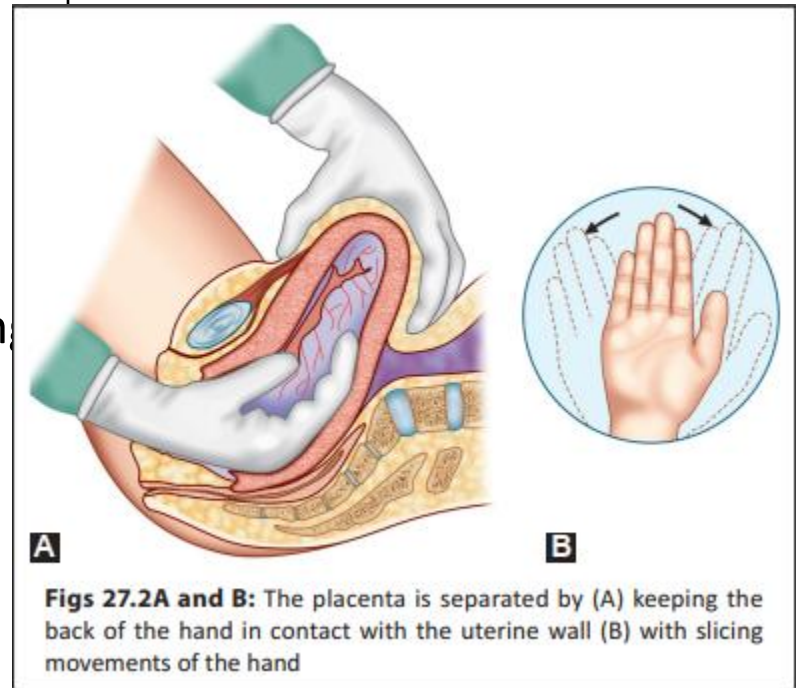
- Cause 5–10% of postpartum hemorrhages.
- Placenta is not expelled 30 minutes after delivery
- Placental tissue and amniotic membranes inhibit uterus from adequate contraction and result in hemorrhage.
- 10% placenta's have fundal implantation

Retained placenta...

- Possible cause
 - ⇒ Mismanagement of the third stage of labor
 - ⇒ Partial separation of a normal placenta
 - ⇒ Cervical constriction
 - ⇒ Failure of the woman to push out
 - ⇒ Closure of the cervix
 - ⇒ Uterine inertia
 - ⇒ Pathological adherence
- Manage with manual separation and removal

Retained placenta management

- Ensure bladder is empty
- Manual removal of placenta
- Pethidine and diazepam
- Ampicillin 2g + Metronidazole 500mg
- Check completeness
- Oxytocin 20 U/1000 ml 60dro/min
- Monitor vital sign



Late postpartum hemorrhage

- Blood loss of more than 500 ml later than 24 hrs after delivery.
- Causes :
 - ⇒ Sub-involution
 - ⇒ Retained placental fragments
 - ⇒ Infection
- Management :
 - ⇒ Uterotonic drug administration
 - ⇒ Curettage
 - ⇒ Broad spectrum antibiotics if there are signs of infection

Puerperal fever

- Also known as postpartum fever is defined as temperature of 38.0°C or higher during the first 10 days postpartum, exclusive of the first 24 hours.
- Fever in the first 24 hours after delivery
 - ⇒ Often resolves spontaneously
 - ⇒ Cannot be explained by an identifiable infection.

Cause of puerperal fever

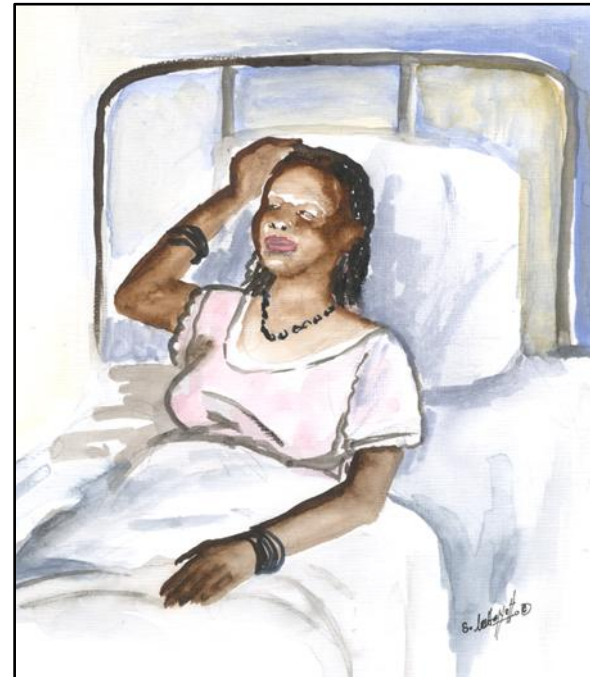
- Endometritis
- Urinary tract infection
- Mastitis
- Genital tract infection.
- Breast engorgement
- Wound infections(episiotomy,incisions or perineal lacerations)
- Respiratory complications after caesarean delivery
- Septic thrombophlebitis.
- Endemic febrile illnesses like malaria

Puerperal fever...

- The most common cause of postpartum fever is endometritis, which occurs after vaginal delivery in about 2% of patients and after cesarean delivery in about 10 % to 15%.

Endomyometritis

- Is inflammation of the endometrium the inner lining of the uterus and myometrium
- The most cause puerperal pyrexia
- Is a major cause of maternal death



Risk factors

- Prolonged and premature ROM
- Prolonged (> 24 hours) labor
- Frequent vaginal examination
- Retained placental fragments or membranes
- Anemia and poor nutrition during pregnancy
- Immunocompromised
- Genital or urinary tract infection prior to delivery
- Cesarean birth (20-fold increased risk)
- Obesity
- Diabetes
- Indwelling urinary catheter.

Cause of endometritis

- The vaginal flora
 - ⇒ Doderlein's bacillus (60–70%)
 - ⇒ Candida albicans (25%)
 - ⇒ Staphylococcus albus or aureus
 - ⇒ Streptococcus—anaerobic common; beta hemolyticus rare
 - ⇒ E. coli and Bacteroides group

Clinical features

- Fever
- Lower abdominal pain
- Offensive lochia
- Subinvoluted uterus
- Lower abdominal tenderness
- Uterine tenderness.
- Less common presentation
 - ⇒ Abdominal distension, nausea and vomiting, diarrhoea, abdominal fluid collection, generalized abdominal tenderness and guarding, adnexal mass

Investigation

- Blood film
- CBC
- C reactive protein
- Urinalysis
- Stool exam
- Abdominopelvic ultrasound

Management

- Supportive care
 - ⇒ Encourage rest.
 - ⇒ Ensure adequate hydration by mouth or IV.
 - ⇒ Use a fan or tepid sponge
 - ⇒ Consider paracetamol 500–1000 mg QID
 - ⇒ Transfusion of blood

Management

- Antibiotics

⇒ Clindamycin phosphate 600 mg IV every eight hours PLUS Gentamicin 5 mg/kg body weight IV every 24 hours.

⇒ Ampicillin 2 g IV every 6 hours PLUS gentamicin 5 mg/kg body weight IV every 24 hours.

⇒ Give 24–48 hours after resolution of clinical signs and symptoms (fever, uterine tenderness, purulent lochia, and leukocytosis).

Prevention

- Avoid risk factors.
- Keep the episiotomy site clean.
- Careful attention to antiseptic procedures during childbirth.
- Administration of prophylactic antibiotics (e.g. cesarean section, manual removal of placenta).

Complication

- Pelvic abscess
- Peritonitis
- Septic shock
- Deep vein thrombosis
- Pulmonary embolism
- Chronic pelvic infection with recurrent pelvic pain and dyspareunia
- Tubal blockage, or infertility

Lactational acute mastitis

- Is inflammation of mastitis
- Usually peripherally situated.
- most frequently in the first 2 to 4 weeks postpartum



Risk of mastitis

- Partial blockage of milk duct
- Pressure on the breast (eg, tight brassiere or car seatbelt)
- Oversupply of milk
- Infrequent feedings
- Nipple excoriation or cracking
- Rapid weaning
- Illness in mother or baby
- Maternal stress or excessive fatigue
- Maternal malnutrition

Etiology

- The most common organisms associated with mastitis are
 - ⇒ *Staphylococcus aureus*
 - ⇒ Methicillin-resistant *S. aureus* (MRSA)
 - ⇒ *S. epidermidis*
 - ⇒ Streptococci and
 - ⇒ Occasionally, gram-negative rods.
- The incidence of sporadic mastitis is 2% to 10% in lactating mothers and less than 1% in nonlactating mothers

Clinical presentation

- Fever $>38.3^{\circ}\text{C}$
- Severe, throbbing type of pain
- Hard, red, tender, swollen area of one breast
- Myalgia, chills, malaise
- Advanced infection
 - ⇒ Large area of breast swelling (erythema)
 - ⇒ Reactive auxiliary lymphadenopathy

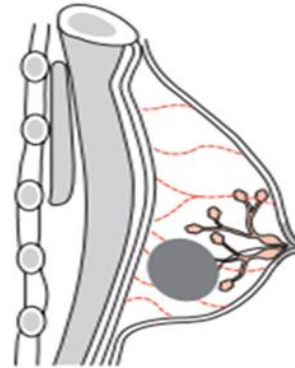


Figure 50.13 Intramammary breast abscess.

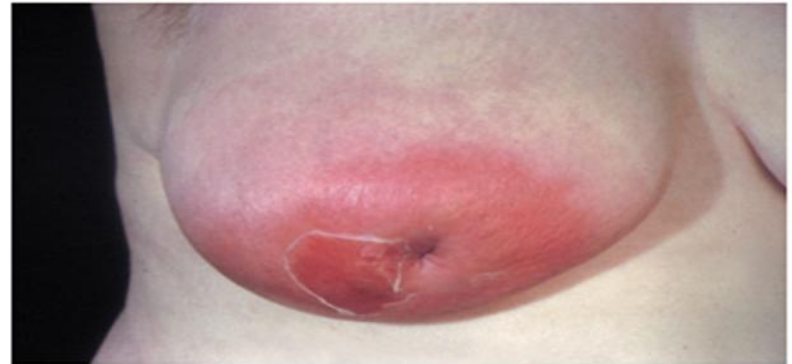


Figure 50.14 Large breast abscess.

Treatment of mastitis

- Fluids
- Encourage the woman to continue breastfeeding,
- Support the breasts with a binder or bra, and
- Apply cold compresses to the breasts between feedings
- Cloxacillin 500 mg PO every six hours for 10 days OR Erythromycin 250 mg every eight hours for 10 days
- Paracetamol 500–1000 mg every six to eight hours as an appropriate alternative (maximum dose 4000 mg in 24 hours).

Complication of acute mastitis

- Abscess
- Skin necrosis
- Toxemia
- Antibiotoma
- Fistula

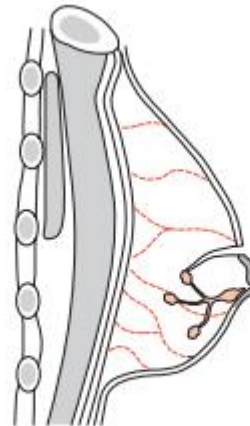


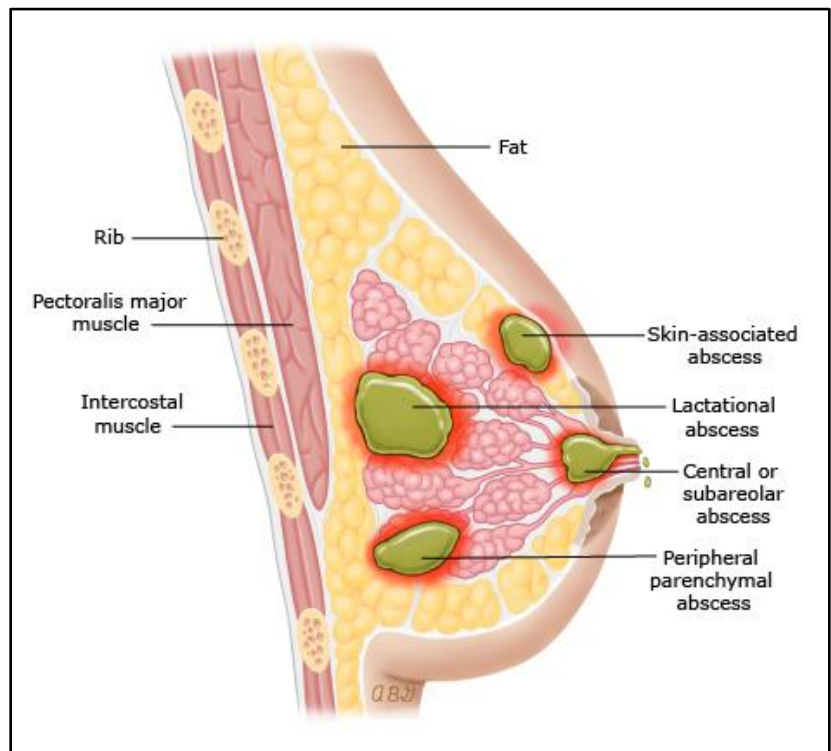
Figure 50.18 Mammary fistula originating in a chronic subareolar abscess.

Breast abscess

- Will occur in about 10% of women who are treated for bacterial mastitis.
- The signs include
 - ⇒ High fever (39° to 40° C)
 - ⇒ Unilateral breast swelling and pain
 - ⇒ Localized and fluctuant mass with erythema of overlying skin
 - ⇒ The patient feels sick, like having the “flu.”
- Abscesses usually occur in the upper outer quadrants

Management of breast abscess

- Cloxacillin 500 mg PO every six hours for 10 days OR
- Erythromycin 250 mg every eight hours for 10 days.
- Drainage of the abscess is indicated
- Breastfeeding should be limited to the uninvolved side during the initial therapy



Unit V: Early pregnancy abnormalities

1

- Ectopic pregnancy

2

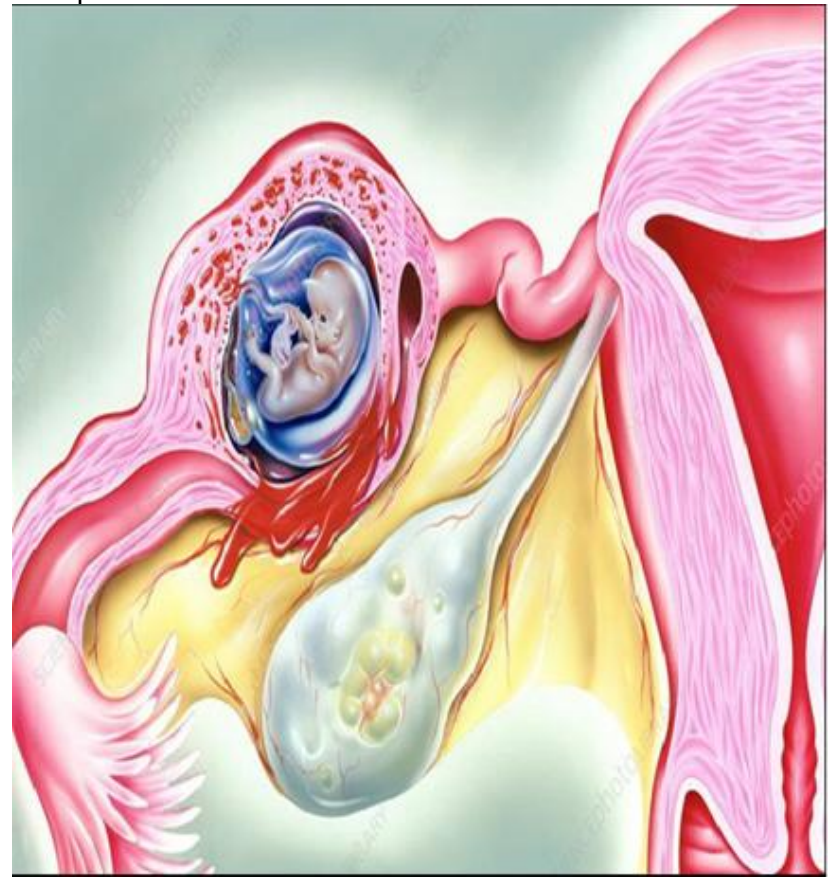
- Abortion

Abnormalities of early pregnancy

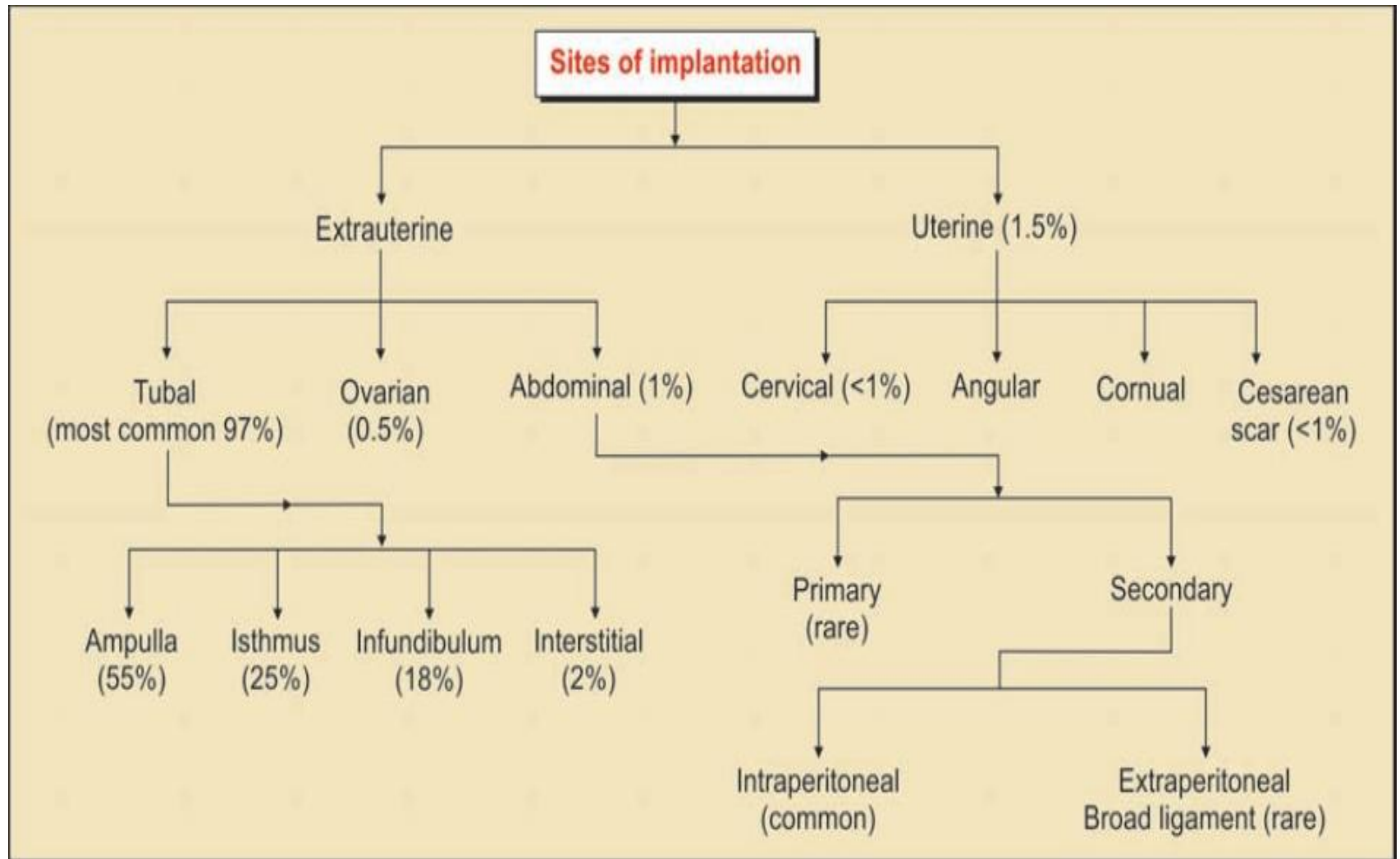
- Objective
 - ⇒ Explain risk factor, presentation and management of ectopic pregnancy
 - ⇒ Define abortion, cause ,classification and management

Ectopic pregnancy

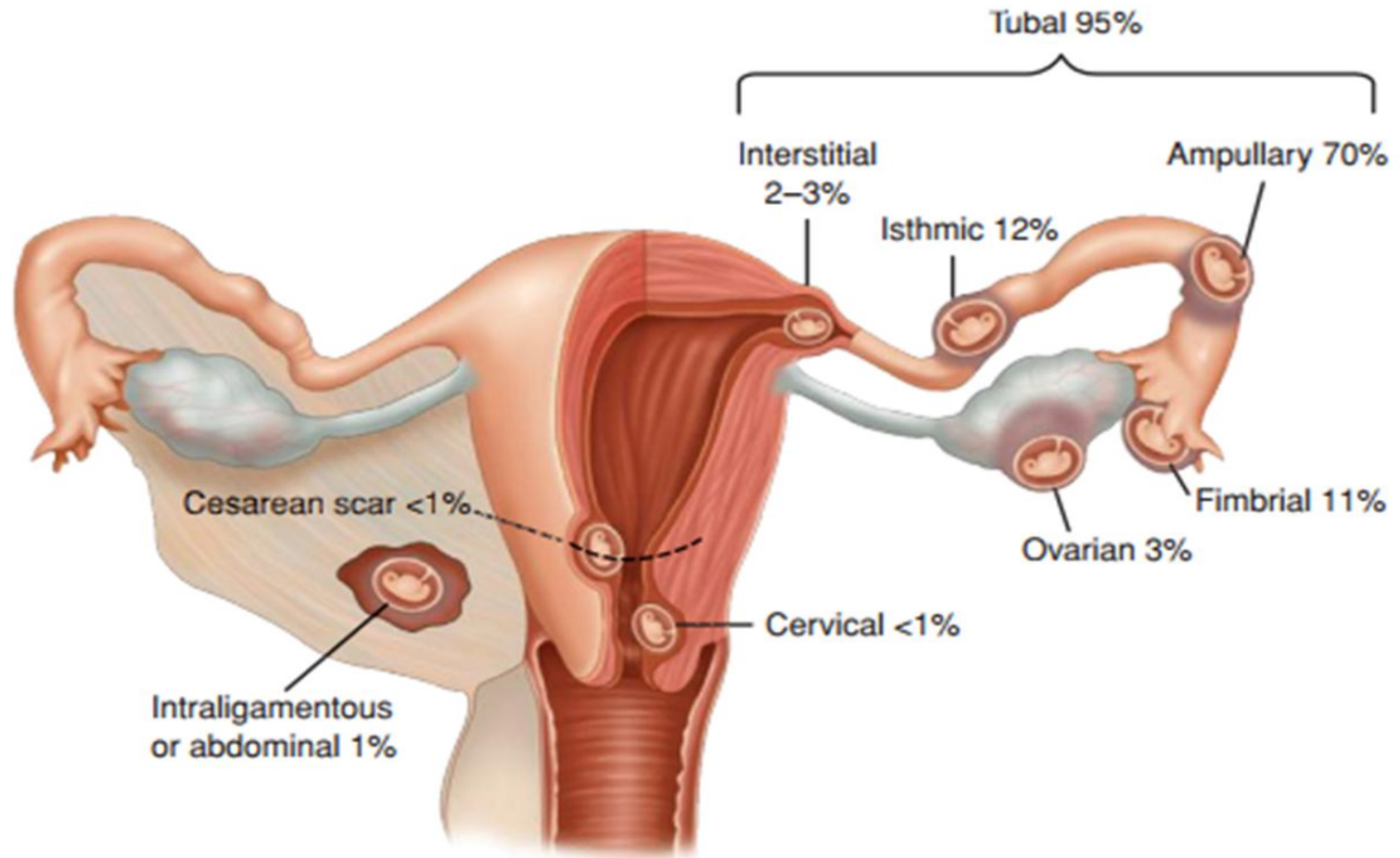
- Is implantation of a fertilized ovum outside uterine cavity.
- Classification
 - ⇒ Fallopian tube
 - ⇒ Extra-tubal
- 2% of pregnancies are ectopic
- 6% of all pregnancy-related deaths.



Site of ectopic pregnancy



Ectopic pregnancy...



19-1 Sites of implantation of 1800 ectopic pregnancies from a 10-year population-based study. (Data from Callen, 2003)

Risk factors

- History of PID (e.g. secondary to STI)
- History of ectopic pregnancy or tubal surgery.
- History of infertility
- Contraceptive failure.
- History of tubal ligation
- IUD use
- Previous induced abortion

Natural history

1. Tubal mole
 - Complete absorption (rare)
 - Abortion → Pelvic hematocele
2. Tubal abortion
 - Complete → Pelvic hematocele
 - Incomplete → Diffuse intraperitoneal hemorrhage
3. Tubal rupture
 - Roof → Diffuse intraperitoneal hemorrhage (p. 180)
 - Floor → Intraligamentary hematoma
4. Tubal perforation
 - Roof → Secondary abdominal pregnancy (rare) (p. 180)
 - Floor → Secondary intraligamentary pregnancy (rare)
5. Continuation of pregnancy—rarest

Tubal abortion beginning

A



Complete abortion (rare)

B



Complete abortion

C



Tubal mole

D

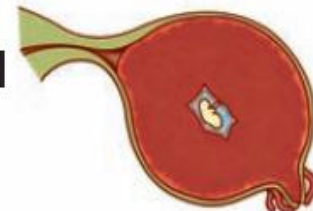


Fig. 15.8: Modes of Termination of Tubal Pregnancy

Natural history....

- Tubal rupture
 - ⇒ Isthmic pregnancies rupture at 6 to 8 weeks
 - ⇒ Ampullary pregnancies rupture at 8–12 weeks.
 - ⇒ Interstitial pregnancies rupture at 12–16 weeks

Clinical features

- Classical triads: present in 50-60%
 1. Abdominal pain: 98%
 2. Missed period: 75%
 3. Vaginal bleeding: 70%
- Early pregnancy symptoms, fainting or syncope
- Features of anemia (e.g. dizziness)
- Pain on defecation and shoulder tip pain

Signs of unruptured and ruptured ectopic pregnancies.

Sign	Unruptured	Ruptured
Vital Signs	• Vital signs can be normal or slightly deranged	• Fast, weak pulse • Hypotension • Pallor
Abdominal examination	• Non-tender abdomen	• Direct abdominal tenderness • Rebound tenderness • Abdominal distension • Bowel sounds are decreased
Pelvic examination	• Minimal vaginal bleeding • Adnexal mass	• Minimal vaginal bleeding • Cervical motion tenderness • Adnexal tenderness

Differential diagnosis

- Abortion
- PID
- Ovarian torsion
- Ruptured ovarian cyst
- Acute appendicitis



Diagnosis

- History and physical findings
- Transvaginal sonography (TVS)
- Serum β -hcG level measurement
- Hematology: CBC, Hgb, Blood group & Rh status, ESR
- Diagnostic surgery
 - ⇒ Uterine curettage
 - ⇒ Laparoscopy
 - ⇒ Laparotomy

Management

- Secure IV line with large bore canula.
- Cross-match blood and prepare for immediate laparotomy.
- Do not wait for blood before performing surgery.
- Surgery
 - ⇒ Salpingectomy
 - ⇒ Salpingostomy
 - ⇒ Salpingotomy

Subsequent management

- Correct anemia
- Anti-D should be administered to all Rh-negatives
- Provide family planning counseling and service.
- Counseling and advice on prognosis of fertility
- Counseled to come as soon as she misses her menses.
- Schedule a follow-up visit.

Abortion

- Is defined as the spontaneous or induced termination of pregnancy before fetal viability.
- ⇒ WHO define abortion as pregnancy termination before 20weeks of gestation or with fetus born weighing < 500g.
- ⇒ Ethiopia and UK the age of viability is 28weeks or <1000gm.
- It is the most common complication of pregnancy

Incidence

- Is difficult to work out
- 10–20% of all clinical pregnancies end in miscarriage
- About 75% miscarriages occur before the 16th week
 - ⇒ Of these 80% occur before 12th week of pregnancy.

Etiology

First trimester	Second trimester
1. Genetic factors (50%) 2. Endocrine disorders • Luteal phase defect • Thyroid abnormalities • Diabetes 3. Immunological disorders 4. Infection 5. Unexplained.	1. Anatomic abnormalities • Cervical incompetence • Mullerian Fusion defects ⇒ Bicornuate uterus ⇒ Septate uterus • Uterine synechiae • Uterine fibroid 2. Maternal medical illness 3. Unexplained

Mechanism of abotion

Hemorrhage



Necrosis and inflammation



Release of inflammatory cells



Uterine contraction



Expulsion of the fetus

Classification of abortion

By occurrence	A. Spontaneous(miscarriage)	
	B. Induced abortion	1. Therapeutic abortion 2. Elective(voluntary) abortion
Based on GA	1. 1st TM vs 2nd TM 2. Early(<12wks) vs Late(12-28wks)	
Based on place of termination	1. Safe 2. Unsafe	
Clinical	1. Threatened 2. Inevitable 3. Incomplete 4. Complete 5. Missed abortion 6. Septic abortion 7. Recurrent/habitual abortion	

Threatened abortion

- The process of miscarriage has started but has not progressed to a state from which recovery is impossible
- Clinical presentation
 - ⇒ Mild vaginal bleeding
 - ⇒ No or mild pain usually following hemorrhage.
 - ⇒ Closed cervix or intact membrane

Threatened abortion...

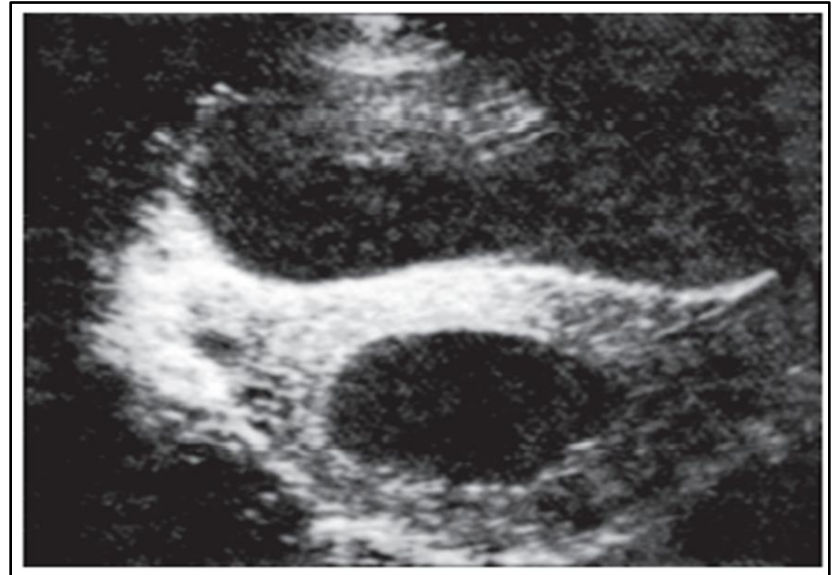
- Investigation

- ⇒ CBC,BG &RH

- ⇒ Urine

- ⇒ hcG test

- ⇒ Confirmed by ultrasound which shows a blighted ovum



Management

- Reassurance (50% probability of continue)
- Decrease physical activity (bed rest), avoid intercourse
- Hormones like Progesterone & hCG
- Treat complication
- Anti- D
- Council on outcome
 - ⇒ Antepartum hemorrhage
 - ⇒ 50 will progress to incomplete abortion
 - ⇒ Are at increased risk of preterm labor

Inevitable abortion

- Continuation of pregnancy is impossible
- Clinical features
 - ⇒ Increased vaginal bleeding.
 - ⇒ Mild to moderate lower abdomen pain
 - ⇒ No passage of products conception
 - ⇒ Dilated internal os with protruded conceptus tissue
 - ⇒ Uterus normal for date
 - ⇒ Leakage of liquor even without open cervix
 - ⇒ The diagnosis is made by clinically.

Management

- CBC , blood grouping , cross match 2 units of blood
- Resuscitation
- AntiD
- Oxytoxic drugs
- Evacuation & curettage.
- Post-abortion management.

Complete abortion

- All products of conception are expelled
- Clinical features
 - ⇒ History of expulsion of a fleshy mass per vagina
 - ⇒ Subsidence of abdominal pain.
 - ⇒ Vaginal bleeding becomes trace or absent.
 - ⇒ Uterus is smaller than the period of amenorrhea
 - ⇒ Cervical os is closed
 - ⇒ Ultrasonography (TVS): reveals empty uterine cavity.

Management

- Post-abortion management
- Ways of confirming completeness
 1. Examine concepts carefully for completeness
 2. Ultrasound to see retained tissue
 3. Documented completeness on referral paper
- If any doubt of completeness evacuate the uterus
- Administer ergometine 0.5mg if bleeding

Incomplete abortion

- Some part of products of conception is expelled and other is not .
- The commonest type
- Clinical features
 - ⇒ Sever lower abdomen.
 - ⇒ Persistence of vaginal bleeding with expulsion of tissue
 - ⇒ Uterus smaller than the period of amenorrhea
 - ⇒ Dilated cervical os often admitting tip of the finger

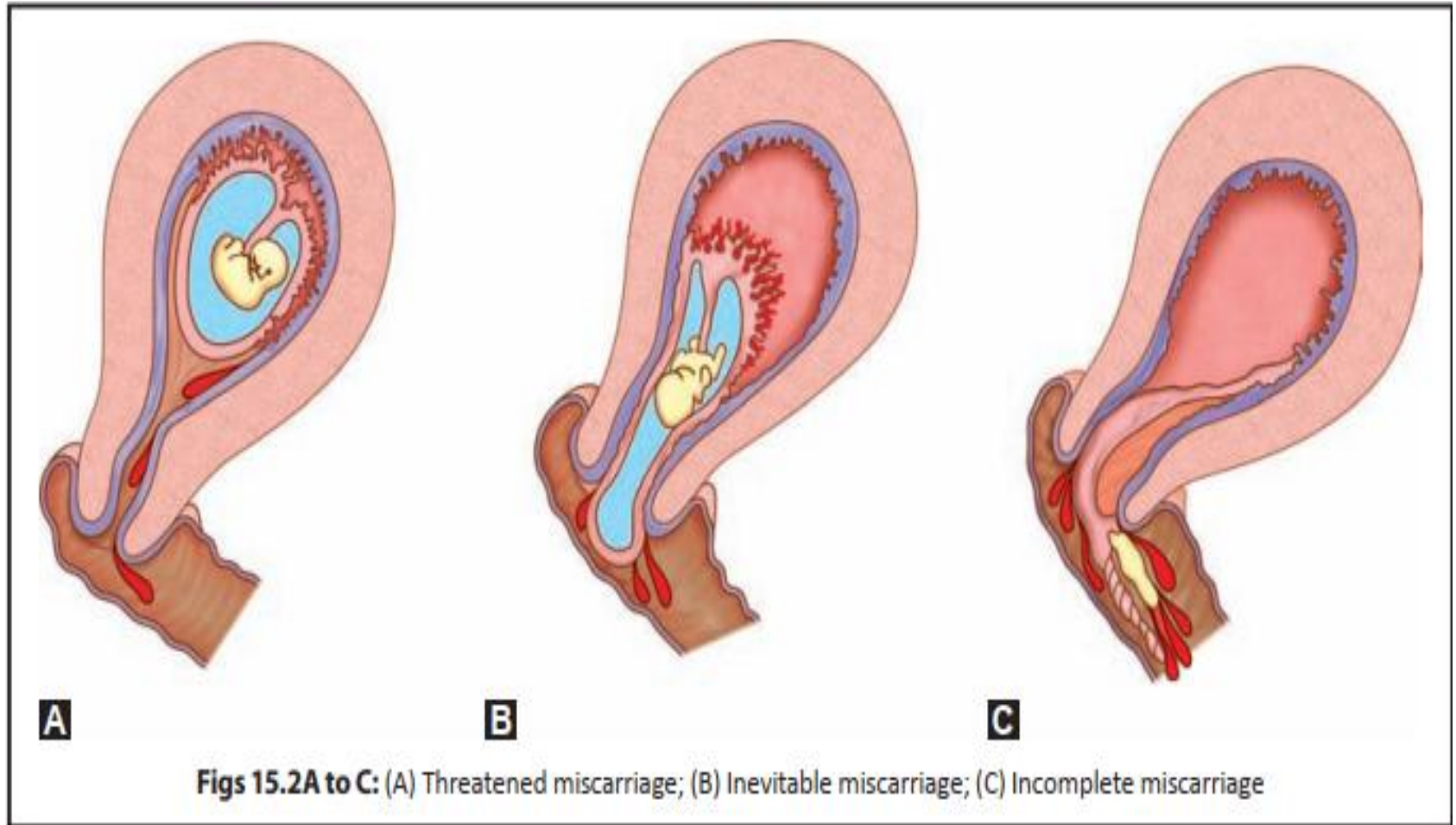
Incomplete abortion...

- Ultrasonography reveals echogenic material in uterus.
- Complications
 - ⇒ Profuse bleeding
 - ⇒ Sepsis
 - ⇒ Polyp

Management

- CBC , blood grouping , cross match 2 units of blood
- Resuscitation
- AntiD
- Uterotonic drugs
- Evacuation & curettage.
- Post-abortion management.

Classification of abortion...



Missed abortion

- Fetus is dead and retained inside uterus for a variable period
- Clinical features
 - ⇒ Persistence of brownish vaginal discharge.
 - ⇒ Subsidence of pregnancy symptoms
 - ⇒ Cessation of uterine growth
 - ⇒ Negative FHB
 - ⇒ Immunological test for pregnancy becomes negative.
 - ⇒ U/S show empty sac, no fetal cardiac motion & fetal movements

Management

- Nearly in 80% of cases spontaneous expulsion during expectant management can occur in three weeks' time.
- If immediate evacuation is decided, one has to use protocol for induced abortion

Septic abortion

- Abortion with clinical evidences of infection of the uterus and its contents.
 - ⇒ Fever at least 100.4°F (38°C) for 24 hours or more
 - ⇒ Offensive or purulent vaginal discharge
 - ⇒ Lower abdominal pain and tenderness.
- About 10% of abortions requiring admission to hospital are septic.
- Majority of are associated with incomplete abortion.

Cause of septic abortion

- Infection is usually due to
 - ⇒ Staphylococcus aureus
 - ⇒ Gram negative bacilli
 - ⇒ some Gram positive cocci.
 - ⇒ Mixed infections
 - ⇒ Anaerobic organisms
 - ⇒ Fungi
- The infection may spread, leading to salpingitis, generalized peritonitis, and septicemia.

Septic abortion...

Symptom	Signs
• Pyrexia • Chills, fever, sweating • Tachycardia • General malaise • Lower abdominal pain • Pelvic tenderness • Prolonged bleeding • Foul smelling vaginal discharge	• Distended abdomen • Tenderness • Tachycardia • Hypotension

Septic abortion...

Clinical grading	
Grade I	• The infection is localized in the uterus. • The commonest • Is usually associated with spontaneous abortion
Grade II	• The infection spreads beyond the uterus to the parametrium, tubes and ovaries or pelvic peritoneum.
Grade III	• Generalized peritonitis and/or endotoxic shock or jaundice or acute renal failure. • Almost always associated with illegal induced abortion.

Management

- Secure IV line and put on intravenous fluids
- Put on broad spectrum antibiotic
 - ⇒ Clindamycin 600 mg IV QID and gentamicin 5 mg/kg IV daily.
 - ⇒ If clindamycin is not available, give ampicillin 2 g IV every six hours and gentamicin 5 mg/kg body weight IV every 24 hours.
- A dilatation and curettage (D&C) or misoprostol
- AntiD if mother is Rh negative blood
- Treat any complication

Recurrent abortion

- A sequence of three or more consecutive spontaneous abortion before 20 weeks.
- It may be primary or secondary (having previous viable birth).
- A woman procuring three consecutive induced abortions is not a habitual aborter.

Etiology

Genetic factors

Endocrine and metabolic

Infection

Anatomic abnormalities

Chronic maternal illness

Unexplained.

Diagnostic

1. History and physical examination
2. Cytogenetic analysis of both partners
3. Ultrasound, HSG, hysteroscopy or laparoscopy
4. Luteal phase endometrial biopsy, mid luteal phase progesterone
5. TSH level, serum prolaction level
6. Lupus anticoagulant activity, Anticardiolipin IgG/IgM, IgM
7. Anti - Nuclear Antibodies
8. CBC, FBS, VDRL
9. Infection screening

Management

- Treatment specifically directed for identified cause
 - ⇒ Use of progesterone
 - ⇒ Anti thrombotic medications
 - ⇒ Repair of anatomic anomalies
 - ⇒ Correction of endocrine abnormalities
 - ⇒ Treatment of infections
 - ⇒ Cervical cerclage
 - ⇒ Counseling & psychological support

Differential diagnosis of abortion

- Ectopic pregnancy in intact tube
- Cervical polyp
- Vaginitis
- Cervical carcinoma
- Gestational trophoblastic disease
- Trauma
- Foreign bod

Induced abortion

- Is surgical or medical termination of a live fetus that has not reached viability.

Indications

- Maternal health problems like:
 - ⇒ Persistent cardiac decompensation
 - ⇒ Pulmonary arterial hypertension
 - ⇒ Advanced hypertensive vascular disease
 - ⇒ Diabetes with end-stage organ failure
 - ⇒ Malignancies
 - ⇒ Maternal mental illness
 - ⇒ Rape or incest
- Non compatible fetal with life
- Age of woman less than 18 years

Techniques for first-trimester

Surgical	Medical
• Dilatation and curettage • Vacuum aspiration • Menstrual aspiration	• Prostaglandins E2, F2α, E1, and analogues • Antiprogesterones—RU-486 (mifepristone). • Methotrexate—intramuscular and oral. • Various combinations of the above.

Complication of abortion

- Hemorrhage, shock, anemia
- Infection
- Intrauterine synechiae (Asherman's syndrome)
- Perforation of the uterus
- Injury to bowel and bladder
- Fistula formation
- Tetanus
- Ectopic pregnancy in later life
- Death

Post abortion care (PAC)

- Emergency treatment of incomplete abortion and potentially life threatening complications
- Post-abortion family planning counseling and services
- Links between post-abortion emergency services and the reproductive health care system.
- Community service provider partnership
- Counseling

Malformations of the female genital tract

Congenital malformations

- Labial fusion
- Imperforate hymen
- Vaginal septum
- Vaginal agenesis.

Imperforate hymen

- Hymen is formed at the junction between urogenital sinus and sinovaginal bulbs.
- An imperforate hymen is formed during fetal development when the sinovaginal bulbs fail to canalize with the rest of the vagina.
- It occurs in 1 in 1,000 female births.

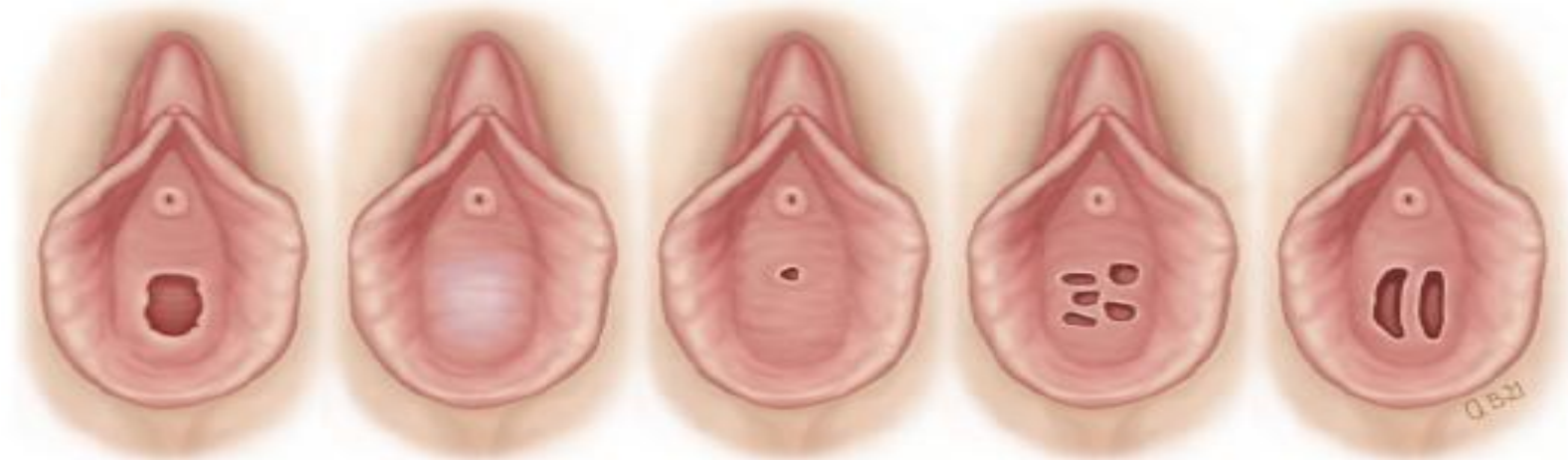
Imperforate hymen...

- An imperforate hymen results in an obstruction to the outflow tract of the reproductive system.
 1. Hydro/ hemato/ mucocolpos
 2. Primary amenorrhea
 3. Cyclic pelvic pain.
 4. Marked distension of the vagina may result in back pain, pain with defecation, or difficulties of urination

Imperforate hymen...

- Physical examination
 1. Absence of an identifiable vaginal lumen
 2. Tense bulging hymen a bluish discoloration
 3. Possibly increasing lower abdominal girth(hematometra).

Imperforate hymen...



[A] Normal

[B] Imperforate

[C] Microperforate

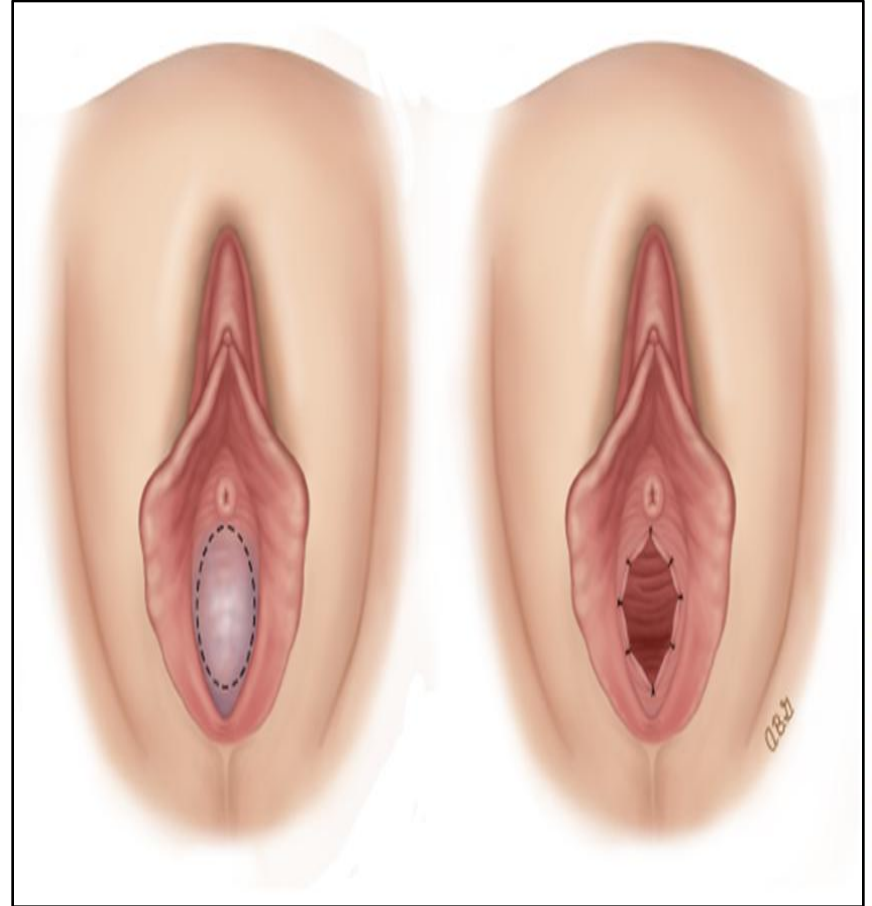
[D] Cribriform

[E] Septate



Management

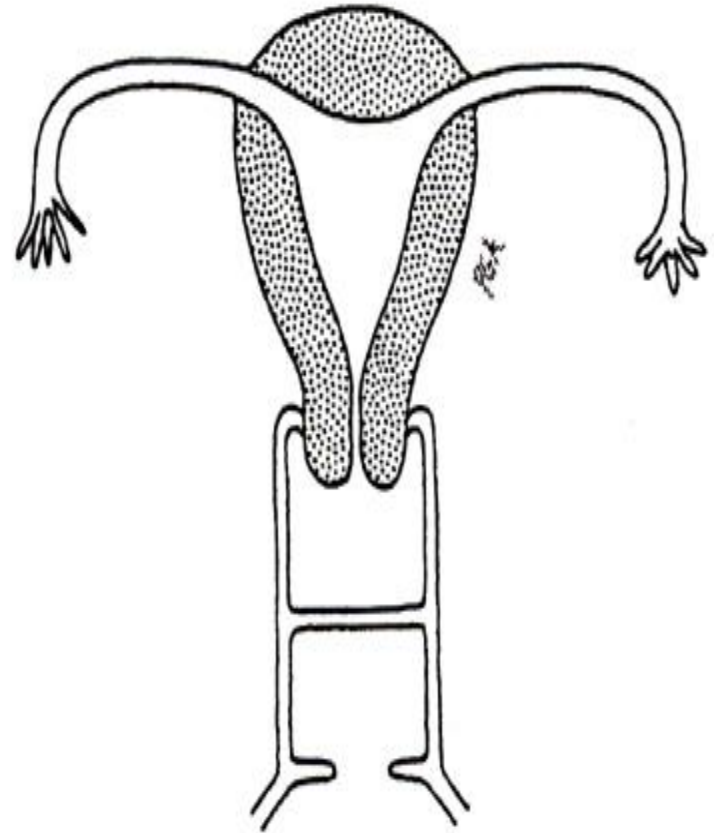
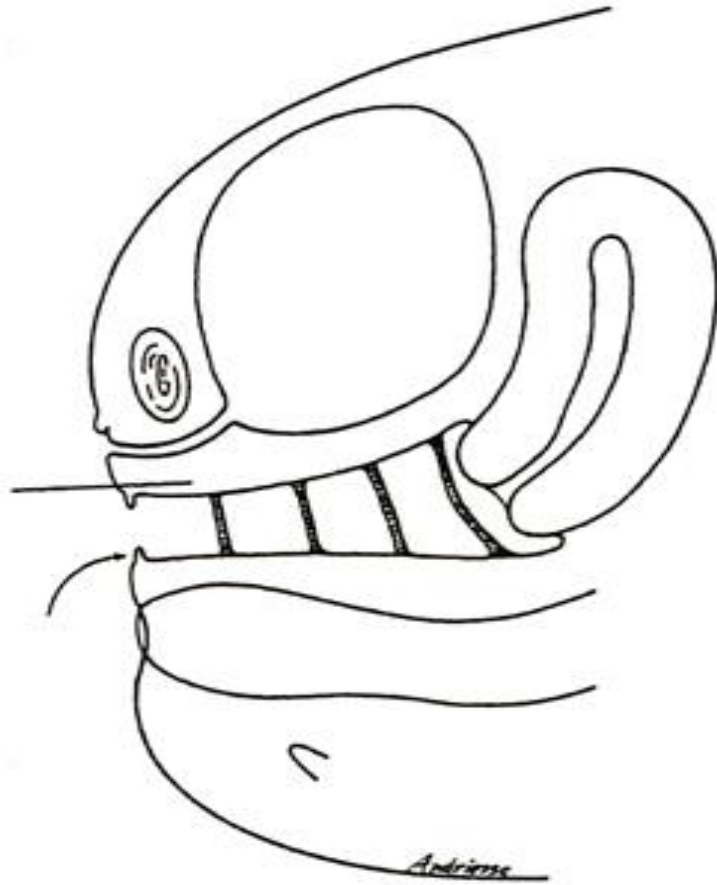
- Before surgical intervention in adolescents, symptoms can be relieved by the
 - ⇒ OCP to suppress period
 - ⇒ NSAIDs to relieve pain.
- Surgery to excise the extra tissue
- Evacuate any obstructed material



Transverse vaginal septum

- Is failure of fusion and/or canalization of the **urogenital sinus and müllerian ducts**.
- Occurs in approximately 1 in 30,000 to 1 in 80,000 women
- The septa are generally less than 1 centimeter in thickness and may have a small central or eccentric perforation
- The majority of transverse vaginal septums have a fenestration and are thus not completely obstructed

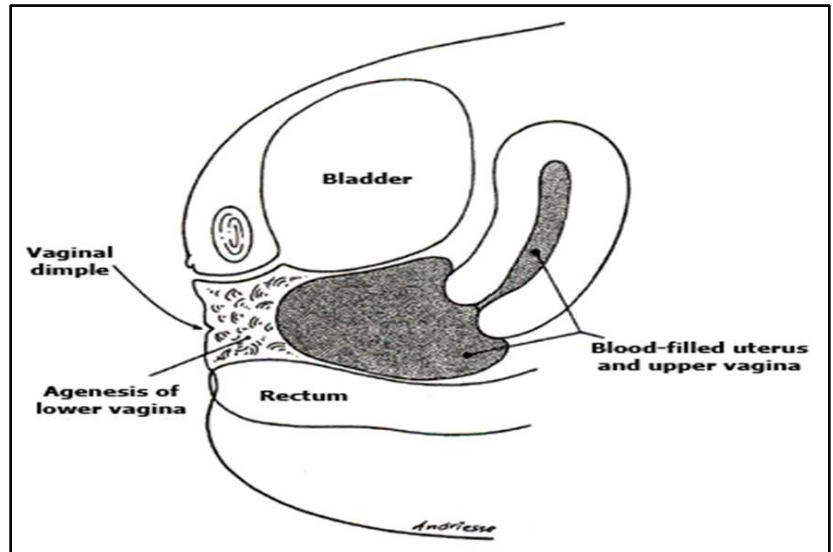
Transverse vaginal septum...



Clinical manifestations

- Normal external genitalia
- Short vagina is shortened
- Children may present with mucocolpos
- Adolescents can develop hematocolpos
- A mass may be palpated

above



Evaluation and treatment

- Diagnosis

- ⇒ Ultrasonographic

- ⇒ Magnetic resonance imaging(location and thickness of the septum)

- Treatment

- ⇒ Resected, followed by an end-to-end anastomosis of the upper and lower vaginal mucosa.

Transverse vaginal septum

- The caudal portion of the Müllerian ducts develops into the
⇒ Uterus and upper vagina.
- The lower vagina is formed as the urogenital sinus evaginates to form the sinovaginal bulbs which then proliferate to form the vaginal plate.
- The vagina is formed as the Müllerian system from above joins the sinovaginal bulb–derived system from below.

Longitudinal vaginal septum

- Are typically associated with uterine anomalies, such as septate uterus and uterus didelphys
- The septum that divides the vagina may be partial or complete.

Clinical manifestations

- The patient may note difficulty inserting tampons, persistent bleeding despite tampon placement, dyspareunia, or she may be asymptomatic.

Evaluation and treatment

- This can be accomplished with
 - ⇒ Two-dimensional ultrasound
 - ⇒ Three-dimensional ultrasound
 - ⇒ Magnetic resonance imaging.
- Treatment involves complete removal of the septum.

Unit -VI: Benign and malignant tumors of genital tract

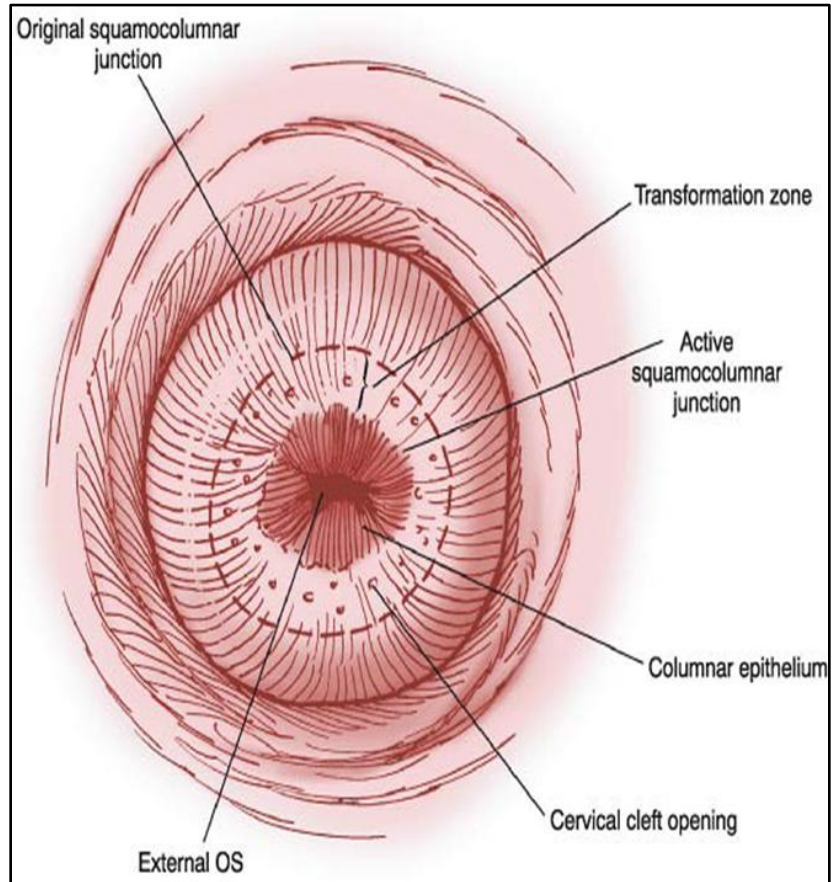
- Objective
 - ⇒ Explain risk, clinical presentation, diagnosis and treatment benign and malignant tumors of the genital tract & breast

Cervical cancer

- Is the second most common cancer in the world
- The third common cause of deaths from cancer in women.
- Diagnosed case each year, an estimated 530,000
 - ❖ More than 270,000 women die from it.
 - ⇒ 85% occur in developing countries.

Origin of cancer

- Metaplasia advances from the
 - ⇒ Original SCJ inward
 - ⇒ Toward the external os
 - ⇒ Over the columnar villi establishes at transformation zone



Risk factors

Human papilloma virus 16/18

Early sexual activities (cohabitarche)

Multiple sexual partners

Partner with multiple sexual partner

Genital hygiene

Smoking

Infection

Immunosuppression (HIV/AIDS)

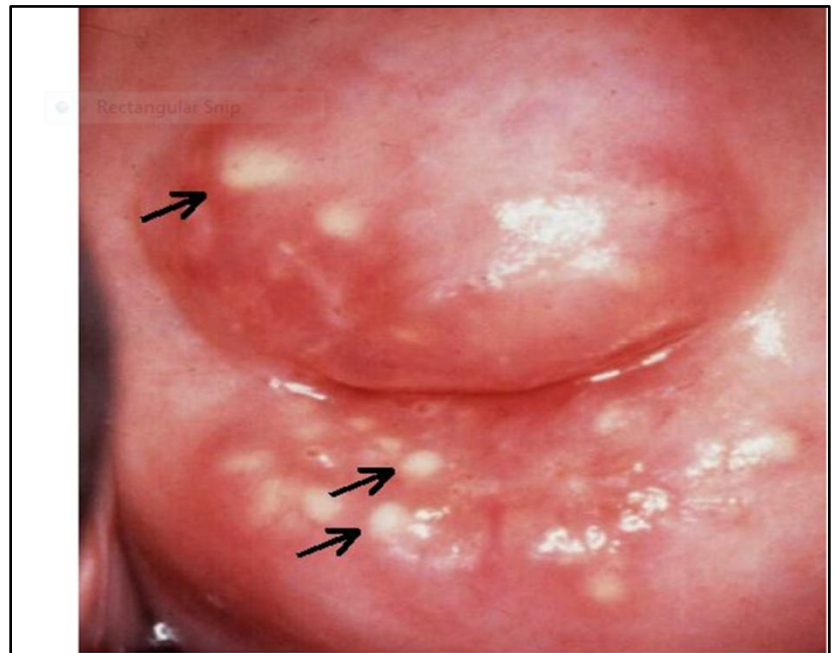
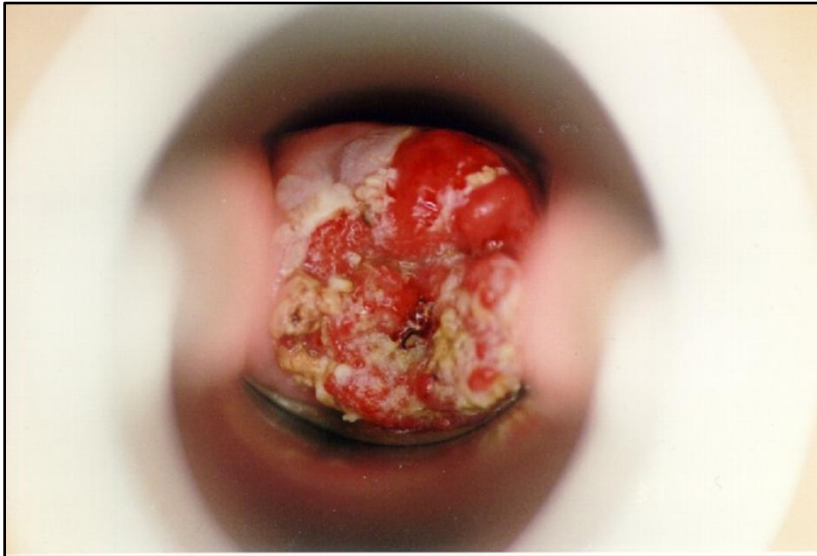
Cervical cancer staging

Stage	Characteristics
0	• Carcinoma in situ, cervical intraepithelial lesion (CIN)
I	• Carcinoma is strictly confined to cervix
II	• Carcinoma extends beyond cervix plus 2/3 upper vaginal • No pelvis side wall
III	• Carcinoma has extended to the pelvic wall • No cancer-free space b/n tumor and pelvic wall(PR) • Tumor involves lower third of vagina • Hydronephrosis or nonfunctioning kidney
IV	• Carcinoma has extended beyond true pelvis • Involved mucosa of bladder or rectum

Clinical presentation of invasive cervical cancer

Stage Ia:	Stage Ib, II	Stage III, IV
• None • Asymptomatic	• Bleeding abnormalities • Vaginal discharge	• Back pain • Lymph edema of the legs • Dysfunction of rectum (bloody stools, subileus) • Hematuria • Dysuria • vagina and vulva are often moist and bloody

Clinical presentation of invasive cervical cancer..



Diagnosis

- Blood (CBC ,ESR ,Serum electrolytes Coagulation status)
- OFT(BUN and Creatinine, SGOT, SGPT, alkaline phosphatase)
- Urinalisys
- Blood glucose level
- Tumor marker (Squamous cell cancer antigen and CA 12-5).
- Papanicolaou smear⇒Histologic evaluation of **cervical biopsy**
- Colposcopy and cervical biopsy

Management principle

- Supportive care
- Operation and medical management
 - ⇒ Operative treatment(Stage Ia,Ib, IIa and IIb)
 - ⇒ Radiotherapy of cervical cancer(FIGO III-IV)
 - ⇒ Chemotherapy of cervical cancer(FIGO III-IV)
- Palliative care

Complication

1. Hemorrhage
2. Infections
3. Peritonitis
4. Anemia
5. Uremia
6. Death

Prevention

Primary prevention	• Girls 9-13 years -HPV vaccination • Health information and warnings about tobacco use. • Sexuality education tailored to age & culture. • Condom promotion/provision for those engaged in sexual activity. • Male circumcision.
Secondary prevention	• Women > 30 years of age • Screening and treatment as needed • “Screen and treat” with low cost technology VIA followed by cryotherapy • HPV testing for high risk HPV types
Tertiary prevention	• Treatment of invasive cancer at any age ⇒ Ablative surgery ⇒ Radiotherapy ⇒ Chemotherapy

Uterine neoplasms

Benign	Malignant
• Polyp • Adenomyosis • Myoma	• Primary ⇒ Endometrial Cancer ⇒ Sarcoma • Secondary arise from ⇒ Ovaries ⇒ Breast ⇒ Bowel

Uterine leiomyomas

- Called myomas, fibroids, fibromyomas
- Are the most common pelvic benign tumor in women
- Are proliferative, well circumscribed, pseudocapsulated benign tumors composed of smooth muscle and connective tissue.
- They arise in reproductive-age women

Uterine leiomyomas

- Are the commonest benign tumor of the uterus
- Are more common in nulliparous or having one child
infertility
- Their size ranges from 1mm to 20cm or more
- May be single or multiple

Uterine liomyomas...

- Can occur out side the uterus

- ⇒ Fallopian tubes

- ⇒ Vagina

- ⇒ Round ligament

- ⇒ Uterosacral ligament

- ⇒ Vulva

- ⇒ Gastrointestine

Predisposing factors

- Risk factors are

- ⇒ Nulliparity

- ⇒ Obesity

- ⇒ Hyperoestrogenic state

- ⇒ Black women

- ⇒ Alcohol

- ⇒ Hypertension

- Protective factor

- ⇒ Multiparity

- ⇒ Dietary vitamin

- ⇒ Smoking reduce the risk.

- ⇒ Progestin-only contraceptives

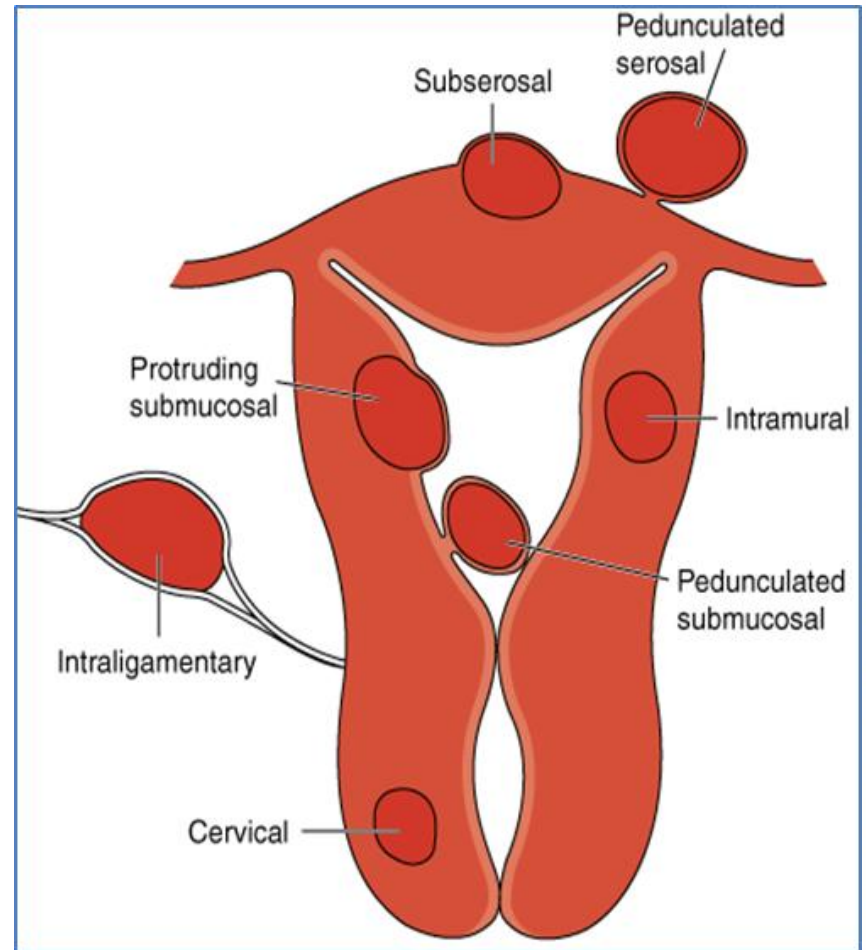
Types (according to location)

Intramural myomas

Subperitoneal or subserous

Submucous myomas

Cervical



Clinical features

Bleeding	Bulk-related symptoms	Reproductive dysfunction
• Heavy or prolonged menstrual bleeding. • 75% are asymptomatic • Most commonly menorrhagia (metrorrhagia)=heavy and prolonged • Menorrhagia-excessive bleeding during menses(more than80ml)	• Feeling of pelvic heaviness • Urinary frequency • Urinary retention • Ureteral obstruction • Constipation • Varicosity or edema • Pelvic pain: chronic, intermittent, dull pressure or pain.	• Ectopic pregnancy • Infertility due to • Distortion and enlongation • Preventing rhythmic uterine contraction • Cornual block • Anovulation • Abortion • Preterm labor

Diagnostic evaluation

- History
- Physical examination
- Laboratory testing(Hgb,CBC,....)
- Pelvic U/S is first-line study used to evaluate for uterine fibroids.

Management

- Conservative for small, diagnosis is certain and no symptoms
 - ⇒ Patient may be reassured but kept under regular supervision
- Surgical
 - ⇒ Myomectomy with is need of further fertility
 - ⇒ Hysterectomy
- GNRH analogues
- Uterine artery embolization(UAE)
- Myolysis
- Immunotherapy

Endometrial cancer

- Most common genital tract malignancy in developed countries
- 50% of all gynecologic malignancies
- 4th most common malignancy in developed world next to breast,CRC,lunge
- The average age at diagnosis is the early 60s

Endometrial cancer risk factor

Factors Influencing Risk	Estimated Relative Risk ^a
Obesity	2–5
Polycystic ovarian syndrome	>5
Long-term use of high-dose menopausal estrogens	10–20
Early age of menarche	1.5–2
Late age of natural menopause	2–3
History of infertility	2–3
Nulliparity	3
Menstrual irregularities	1.5
Residency in North America or northern Europe	3–18
Higher level of education or income	1.5–2
White race	2
Older age	2–3
High cumulative doses of tamoxifen	3–7
History of diabetes, hypertension, or gallbladder disease	1.3–3
• Protective factors ⇒ Long term use of COC. ⇒ Cigarette smoking.	

Endometrial cancer pathogenesis

Type I	Type II
• Estrogens dependent • Associated with endometrial hyperplasia • Good prognosis • These patients tends to have risk factors ⇒ Obesity ⇒ Nullparity ⇒ Estrogen excess ⇒ Diabetes mellitus ⇒ Hypertension • Longer 5 years survival than type II (92% vs. 33%) • Example- Endometrioid cancer	• Unrelated to estrogen or endometrial hyperplasia • Present with high grade tumors • Poor prognostic cells • Usually multiparous , older & thin • Example ⇒ Papillary serous cancer ⇒ Clear cell tumors

Clinical features

- An average age of 60 year .
- Vaginal bleeding or discharge(90%)
- Pelvic pressure or discomfort(enlargement or extrauterine spread)
- Hematometra or pyometra(cervical stenosis)⇒purulent vaginal discharge
- <5% of women diagnosed are asymptomatic.
- Risk like obesity,hypertension

Clinical features...

- Peripheral lymph nodes and breasts should be assessed carefully
- Ascites or hepatic or omental metastases may be palpable.
- Speculum examination may confirm the presence of bleeding
- Atrophic vaginitis
- Enlarged, fixed and immobile uterus from parametrial extension.

Diagnosis

- CBC
- Endometrial biopsy is 90% to 98% accuracy
- Imaging
 - ⇒ Ultrasound
 - ⇒ Computer tomography
 - ⇒ Magnetic resonance imaging (MRI)

Histologic classification

Endometrioid Adenocarcinoma

Secretory carcinoma

Mucinous carcinoma

Papillary serous carcinoma

Clear cell carcinoma

Squamous cell carcinoma

Endometrial cancer clinical stage

Stage	Characteristic	Distribution
I	limited to endometrium	70
II	Extends to the glandular epithelium of the endocervix	13
III	Involves serosa and/or adnexa	14
IV	Invades bladder mucosa and/or bowel mucosa	3

Pretreatment evaluation

- A complete history
- Physical examination
- Chest radiography should be performed to exclude **pulmonary metastasis**
- CBC, BG and Rh, OFT,urinalysis,CA125
- Ultrasonography and MRI to assess myometrial invasion

Endometrial ca treatment

Surgery	• Abdominal Hysterectomy & BSO with cytology(>IB LND) • Primary operative procedures for carcinoma of endometrium. • Remove adnexa-the site of microscopic metastasis.
Radiation therapy	• Progestin (because most endometrial cancers have both E and P receptors)
Chemotherapy	• Adjuvant cytotoxic chemotherapy has been studied in a few trials • No benefit to adjuvant chemotherapy could be found

Benign ovarian mass

- Due to hormonal dysfunction related to ovulation.
- Due to accumulation of intrafollicular fluids
- Causes include
 - ⇒ Follicular cysts
 - ⇒ Corpus luteum (Granulosa Lutein) cyst
 - ⇒ Theca Lutein cysts
 - ⇒ Endometriomas
 - ⇒ Infectious
- Except the endometrial cyst, all are functional & called as “cystic ovaries

Benign ovarian mass ...

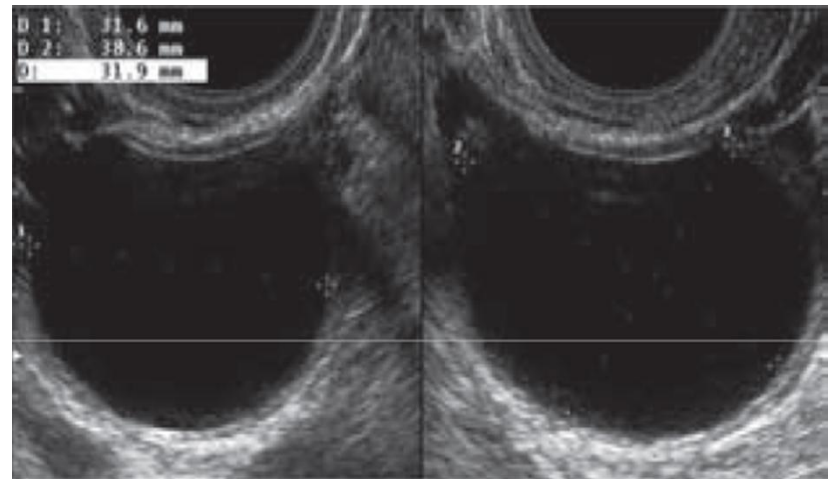
- Generally have similar symptoms and management
- **Cystic ovaries** are different from neoplasia by
 1. Not exceeding 5cm
 2. Unilocular (with only one cell or cavity)
 3. Content is clear fluid
 4. Lining epithelium is corresponding functional epithelium
 5. Spontaneous regression

Symptoms of benign ovarian mass

- Most women with ovarian cysts are asymptomatic.
- Pain and vague pressure or achiness sensations
- Cyclic pain with menstruation(endometriosis, endometrioma).
- Intermittent pain may reflect early torsion.
- Severe persistent acute pain-torsion due to ovarian ischemia.

Diagnosis

- History
- Physical examination(masses are mobile, cystic, nontender, and found lateral to uterus(
- Ultrasound
- Tumor marker
 - 1.HCG
 - 2.Tumor markers



Treatment and complication

- Treatment

- ⇒ Treat symptom and complication
- ⇒ Surgical

- Complications

- ⇒ Torsion of the pedicle
- ⇒ Intracystic hemorrhage
- ⇒ Rupture
- ⇒ Infection
- ⇒ Pseudomyxoma peritonei
- ⇒ Malignancy

Ovarian cancer

- Is the 7th most common cancer in women in USA while the 6th in East Africa accounting for 6% death from cancer in women.
- Worldwide each year, 204,000 women are diagnosed, and 60-75% of women die from this disease.
- A woman's risk at birth of having ovarian cancer sometime in her life is 1% to 1.5%, and that of dying from ovarian cancer is almost 0.5%.

Risk factors

- Low parity/nulliparity & infertility
- Early menarche & late menopause
- Galactose consumption
- Family history
- Increasing age
- White race
- Mutation of tumor suppressor genes, BRCA1 & BRCA2.
- Lynch II (HNPCC) syndrome

Histogenetic WHO classification of ovarian tumors

Classic tumor	Sub classification	%
Neoplasms derived from surface (coelomic) epithelium and stroma:	• Serous tumor • Mucinous tumor • Endometrioid tumor • Clear cell (mesonephroid) tumor • Transitional cell tumor • Malignant mullerian mixed tumor • Endometroid stromal sarcoma	65%–70%
Neoplasms derived from germ cells	• Teratoma • Dysgerminoma • Embryonal carcinoma • Yolk sac tumor • Choriocarcinoma • Gonadoblastoma	15%–20%
Neoplasmas derived from specialized gonadal stroma	• Granulosa-thecal cell tumors • Sertoli-Leydig tumors	5%–10%
Neoplasms derived from nonspecific mesenchyme:	• Fibroma, hemangioma, leiomyoma, lipoma • Lymphoma • Sarcoma	

Patient evaluation

History	Physical examination	Investigation
• May be asymptomatic • Abdominal pain • Abdominal swelling • GI complain/constipation • Weight loss • Abnormal bleeding • Urinary frequency • Pelvic pressure • Back pain • Palpable mass	• Chronically sick looking • Evidences of anemia • Supraclavicular lymphadenopathy • Pleural effusion • Presence of upper abdominal mass • Solid, irregular, fixed, pelvic mass • Ascites • Bimanual Pelvic and rectovaginal examination • Lower extremity swelling	• Complete blood count • Blood group & R • Urinalysis • Abdominal and pelvic Ultrasound • Renal and Liver function tests • Serum CA125 • Chest X-ray • Pap test

Patient evaluation...

Clinical findings	Benign	Malignant
Unilateral	+++	+
Bilateral	+	+++
Cystic	+++	+
Solid	+	+++
Mobile	+++	++
Fixed	+	+++
Irregular	+	+++
Smooth	+++	+
Ascites	+	+++
Cul-de-sac nodules	-	+++
Rapid growth rate	-	+++

FIGO Stage grouping for primary carcinoma of ovary

Stage	Features
I	• Growth limited to the ovaries.
II	• Growth involving one or both ovaries with pelvic extension
III	• Tumor involving one or both ovaries with peritoneal implants outside pelvis
IV	• Distant metastasis

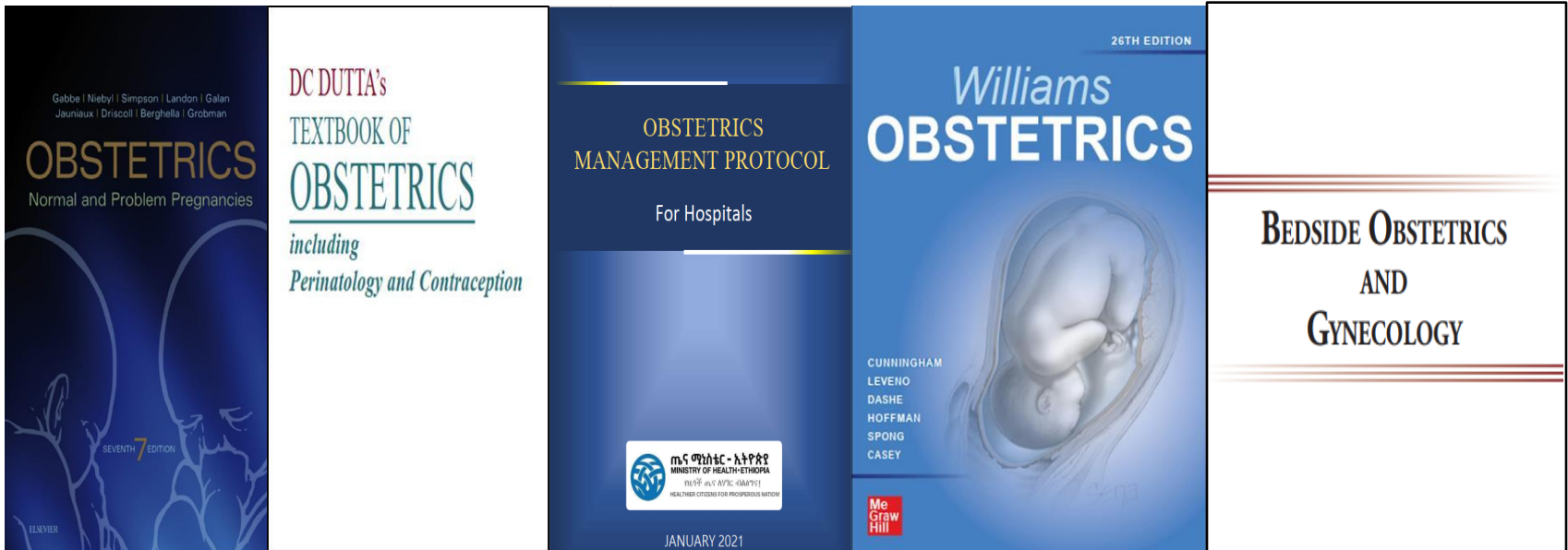
Management

Clinical stage	Management
Early stage (stage I-II)	• Total abdominal hysterectomy (TAH) and bilateral salpingo-oophorectomy (BSO)
All stage II	• Surgery(TAH, BSO & surgical staging) • Chemotherapy:
Advanced-stage (III-IV) Epithelial Ovarian Cancer	• Surgery • Chemotherapy: • Chemotherapy for advanced EOC

Prevention of ovarian neoplasm

1. Factor reducing the number of lifetime ovulatory cycles.
 - Full-term pregnancies
 - Lactational periods
 - Use of OCP
2. Tubal ligation
3. Hysterectomy
4. Prophylactic oophorectomy (hereditary predisposition)
5. Lifestyle modification

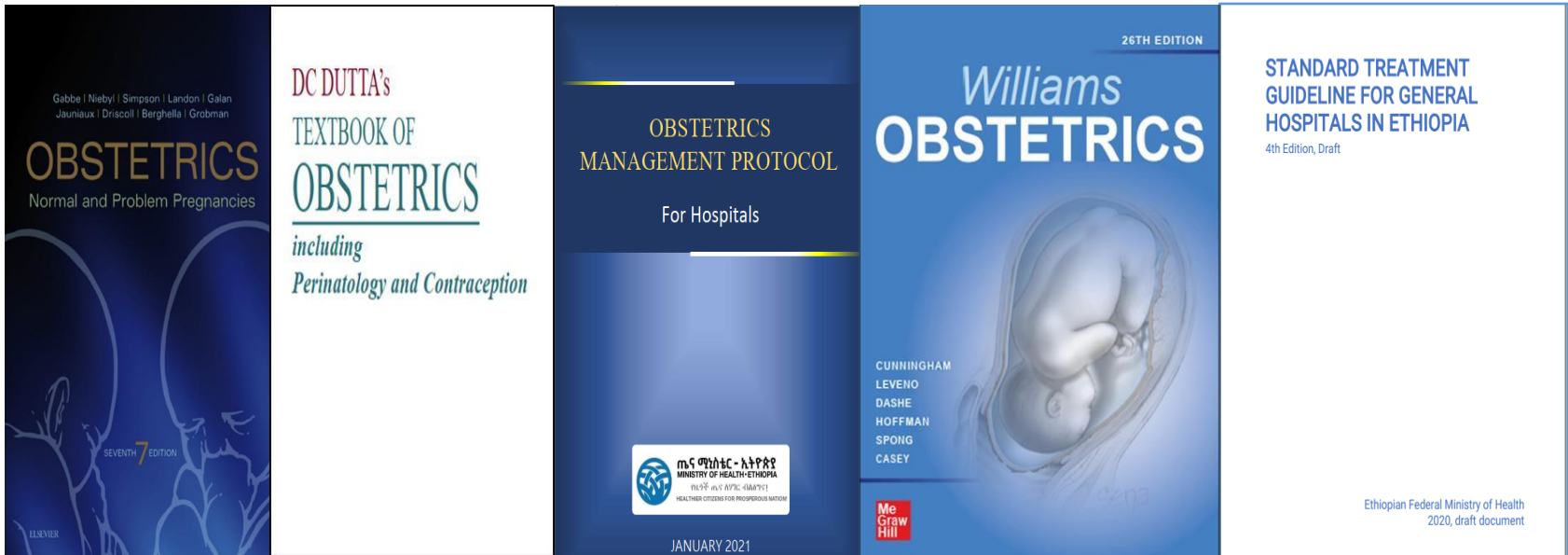
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